

HALO Frontend

Complete Specification, Every Page, Every Element, Every Data Binding

Built from	The full HALO blueprint (all 70 sections, read directly from source text) and the backend's API contract, every route's response read field by field from the code, and the fields the finished backend added. Part D is that contract in one place
Why it exists	The blueprint describes what every page must do in detail, but (1) it never maps a page element to the actual API field that feeds it, (2) its functional rules are interleaved sentence-by-sentence with visual values that are now being redesigned, (3) one page's content is scattered across up to eight different sections, and (4) it was written before the backend existed, so it doesn't know the fields the backend actually uses. This document fixes all four.
What's locked	Every page's purpose, content, states, buttons, copy, interactions, and data rules, from the blueprint, unchanged
Design foundation	Locked: colors (brand and severity, light and dark), typefaces, and the tokens already built in <code>app/onboarding.css</code> (sections A11 and P6)
What's open	Page-level visual choices: composition, spacing, motion values. Listed per page as decisions to make. Each page is wireframed, approved by Yogi, frozen, then built (Part P8 and A13). Shared components are decided once, in Phase 0, so every page looks like the same app
Which backend	The finished backend, merged into <code>main</code> and specified in Part D. Pages bind to the live API from the first commit
Companions	HALO_Codex_Context.md (the whole-app briefing) and HALO_Wireframe_to_Build.md (the per-page workflow Codex follows). Read Part P of this document first, then Part A, then your own pages
As of	September 29, 2026 (first written September 27). Checked against <code>main @ e3f2147</code> : backend merged, Onboarding built and approved. Reviewed page by page against the routes' code by three independent readers

0. How to read this document

Every item is tagged so there's never a question of where a requirement came from or how firm it is:

Tag	Meaning	What to do with it
LOCKED	Functional or content requirement from the blueprint	Build exactly. Not up for redesign
BIND	The exact API field that feeds this element (full contract in Part D)	Bind to this field name, verbatim. Never compute the value yourself
FRONTEND RULE	A rule the frontend must follow because of how the API behaves, usually a trap that would silently show wrong information	Build it into the shared data layer, not per page
PROPOSED	Content or copy that isn't in the blueprint, usually extra richness from data the API already provides	Build it, every page should hold all the content it can, and list each PROPOSED item you built when you hand the page back, so Yogi can keep or cut it
DESIGN	A visual decision still to make	Decided in the page's design pass: Codex proposes from Yogi's reference inspiration and A11-A13; Yogi approves before the page is finished

THE ONE RULE THAT GOVERNS EVERYTHING BELOW: the server decides every severity level, every score, and every household-specific sentence. The frontend displays what it's given. It never compares a raw number against a threshold, never picks a severity word, never writes a personalized sentence of its own. Two screens computing the same threshold independently will eventually disagree about the same household (§6.1). Where this document says the frontend combines server values (e.g., "worst of four severity words"), it's choosing among words the server already produced, never deriving new ones.

THE BACKEND IS LIVE. Everything here targets the merged backend. Bind to the real API from the first commit; mock data exists only to force states in previews and tests. A field that is absent still never crashes a page: it shows the words No data or a hidden element (Part D, D0).

THREE PEOPLE BUILD IN PARALLEL. Part P says who owns the shared layer (the shell, shared components, data client, formatting, tokens), who builds which page, and the signatures every page codes against. If a sentence elsewhere in this document says "the first page that needs it builds it" or "Codex does everything", Part P wins.

ON "EVERY PAGE SHOULD HAVE ALL THE CONTENT IT CAN": for Today, HomeGuard, Journal, Act, Map, and the overlays, this document adds **PROPOSED** content wherever the API provides data the blueprint didn't use. **Onboarding is the deliberate exception:** the blueprint's own rule (§10.1) is that every extra question costs completions, and a household that abandons onboarding never sees any other page. Onboarding stays minimal on purpose; the richness goes into the pages after it.

Contents

- Part P** Foundation (Phase 0), owners and lanes, shared-code contracts, dependencies between pages, tokens, backend asks, open decisions
- Part A** Cross-cutting systems, shell, navigation, data layer, severity, wording, formatting, copy, forms, accessibility, shared components, signature visuals, the locked design foundation (A11), priorities, performance, testing and limits (A12), the design process and the proposed base visual system (A13)
- Part B** Pages, in build order: B1 Onboarding · B2 Today · B3 HomeGuard · B4 Journal · B5 Act · B6 Settings · B7 Map and District panel
- Part C** Overlays: C1 Learn · C2 Assistant · C3 Alerts
- Part D** API reference, every endpoint and field the frontend binds to
- Part E** Definition of done, per page, and the build order
- Part F** Decision log, every product, design, and privacy decision made so far, and what's still open

Each page in Part B follows the same structure: purpose → address and entry → data calls in order → layout top to bottom → element-by-element table (content, binding, states) → page-level states → proposed content → visual decisions still to make.

Part P: Foundation (Phase 0), Owners, and Parallel Building

This part exists so that three people, each working in their own Codex account on their own device, can build different pages at the same time and end up with one app. **It overrides any older sentence in Parts A to F that says otherwise** (for example "the first page that needs a component builds it" or "Codex does all frontend work"). Read this part first, then Part A, then your own pages.

THE RULE THAT MAKES PARALLEL BUILDING SAFE. Nobody edits a shared file except its owner. A page builder **imports** shared code and never changes it. If a page needs something the shared layer lacks, the page builder asks the owner (a one-line message: file, what, why) and keeps building against the agreed signature in this part. Two people editing `api.ts`, `copy.ts`, `types.ts`, `storage.ts` or `onboarding.css` in different branches is the single most likely way this project breaks in the last week.

P1. Who builds what

Lane	Owner	Builds	Depends on
Phase 0 (Foundation)	One person, named by Yogi (OPEN D1). Recommended: whoever has the most Codex quota, reviewed by Yogi	Everything in P2 to P6: the app shell, shared components, the data layer additions, the formatting and text modules, the token additions, empty copy files and preview scaffolding. Merged to main before any page branch starts.	Onboarding as built (already on main)
Lane 1	Yogi	Today (B2), Journal (B4), Settings (B6), and the two Onboarding follow-ups in P7	Phase 0
Lane 2	Naggi	HomeGuard (B3) + Learn sheet (C1), then Act (B5) + Alerts sheet (C3)	Phase 0
Lane 3	Vibhav	Map + District panel (B7), then Assistant sheet (C2)	Phase 0. Backend wrap-up items first

Why this split: Today, Journal and Settings share the daily-reading and profile data. HomeGuard and Learn share the home-reading data and the "What does this mean?" link. Act and Alerts mostly compose data other pages already load. Map and Assistant are the two large, self-contained builds, and Vibhav knows both backends. If someone is slower than planned, the blueprint's own cut order applies: Assistant, then Settings, then Act, then the district panel.

P2. What Phase 0 must deliver, with owners and files

Everything below is built once, by the Phase 0 owner, and then frozen. **The look of every shared component is decided in Phase 0** (Yogi approves it with the same wireframe, freeze, build, prove workflow used for Onboarding), so no page can restyle it later. This replaces A9's "the first page that uses it decides."

The shell **FRONTEND RULE**

File	What it does
<code>app/(app)/layout.tsx</code>	A Next.js route group for the five tabs plus <code>/settings</code> and <code>/map</code> (URLs do not change: the folder name is not part of the address). It renders once: <code><div className="halo-app" data-wide={isMap}></code> containing <code>Header</code> , <code>OfflineBanner</code> , <code><main>{children}</main></code> , <code>AssistantButton</code> , <code>TabBar</code> , <code>SheetHost</code> , <code>ToastHost</code> . Pages render only their own content. They never add <code>.halo-app</code>, a header, a tab bar or an assistant button. Onboarding stays outside the group (full screen, no shell)
<code>components/shell/AppGate.tsx</code>	Runs on every route in the group. Calls <code>ensureAnonSession()</code> , then <code>GET /api/profile</code> , then routes exactly like A1 (uses <code>hasCompletedOnboarding()</code> and <code>markCompleted(profile)</code> from <code>lib/frontend/storage.ts</code>). Also routes to onboarding when a page reports <code>no_location</code> (daily-score) or <code>no_county</code> (home-guard). A deep link that is interrupted is kept as <code>?next=</code> and restored after onboarding
<code>components/shell/Header.tsx</code>	Reads the route. Title, variants (Today: bell + gear; other tabs: gear; Settings: back arrow; Map: over the map). The bell is <code><BellBadge/></code> , exported by the Alerts owner; until it merges the shell renders nothing in that slot. Back arrow on Settings: <code>router.back()</code> when the previous history entry is an in-app route, otherwise <code>/today</code>
<code>components/shell/TabBar.tsx</code>	Five destinations, plain links with no query string. Tapping a tab pushes a history entry (so Back returns to the previous tab, as A1 says). Each tab remembers its scroll position through

	<code>useScrollMemory(pathname)</code> (module memory, per session)
<code>components/shell/OfflineBanner.tsx</code>	Uses <code>copy.common.offlineCached</code> (already in <code>copy.ts</code> : "You're offline. Showing your last readings.")
<code>components/shell/AssistantButton.tsx</code>	Fixed, hidden while any sheet is open. Opens the Assistant through <code>openAssistant(page)</code> . Read a build-time switch <code>NEXT_PUBLIC_HALO_ASSISTANT</code> (<code>on</code> by default, <code>off</code> removes the button). This is how the Assistant is "switched off without a dead button"
<code>components/shell/SheetHost.tsx</code> + <code>lib/frontend/sheets.ts</code>	The one place sheets open. Full contract in P3
<code>components/shell/ToastHost.tsx</code>	One toast at a time above the tab bar. One duration for every toast: 8 seconds (long enough to reach Undo). Contract in P3
<code>app/not-found.tsx</code>	The \$27.5 message and a button back to Today. Never a raw error page

TOKENS AND PORTALS. Today `--halo-*` tokens are defined under `.halo-app` only, so a sheet or toast rendered into `document.body` would lose every color. Phase 0 changes the selector to `:root, .halo-app` (the values stay exactly as they are) **and** renders `SheetHost` and `ToastHost` inside the layout's `.halo-app`. Do both.

Shared components **FRONTEND RULE**

All in `components/ui`, all with the states listed in A9. These are the components used by two or more pages, so Phase 0 builds them. Components used by one page stay in that page's folder and belong to that page's owner (P7).

Component	File	Props (minimum)	Used by
BottomSheet	<code>BottomSheet.tsx</code>	<code>{ id, title, height: 'standard' 'tall' 'content', onClose, children }</code> . Registers with <code>useSheets()</code> . Drag, backdrop, Escape and close button all call <code>closeSheet()</code>	Learn, Alerts, Assistant, Act, Journal day detail, Map, Settings delete
Toast	<code>useToast()</code> in <code>lib/frontend/sheets.ts</code>	<code>show(message, { actionLabel, onAction })</code>	Journal, Alerts, Act, Settings, Map
Card / ExpandableCard	<code>Card.tsx</code> , <code>ExpandableCard.tsx</code>	<code>{ title, summary, severity, defaultOpen, open, onOpenChange, children }</code> . Controlled and uncontrolled. Never a history entry	Today, HomeGuard
ScoreRing	<code>ScoreRing.tsx</code>	<code>{ score: number null, severity, partial }</code> . No-data state: no arc and the words "No data" in the centre	Today, HomeGuard, health card
ContributionBar	<code>ContributionBar.tsx</code>	<code>{ segments: { key, label, share, severity 'no_data' } [], onSelect }</code> . Rules in B2 and B3	Today, HomeGuard
Callout	<code>Callout.tsx</code>	<code>{ tone: 'info' 'caution' 'notice' 'error', children }</code> . The existing <code>.halo-callout</code> is red only; Phase 0 adds the other tones	Today, HomeGuard, Journal, Map, Assistant
Skeleton	<code>Skeleton.tsx</code>	<code>{ shape: 'block' 'text', width, height }</code> . Fill uses a new token (see P6) so it is visible in dark mode	Every page
EmptyState	<code>EmptyState.tsx</code>	<code>{ icon, heading, body, action? }</code>	Journal, Alerts, Act, Map
SegmentedControl	<code>SegmentedControl.tsx</code>	<code>{ options, value, onChange, label }</code> . Arrow keys move between options	Journal, Act letter, Settings
ToggleChip	<code>ToggleChip.tsx</code>	<code>{ pressed, onToggle, children }</code> . Extract the chip already built inside <code>OnboardingScreen.tsx</code> so Onboarding and Journal use the same one	Onboarding, Journal, Settings
Switch, AccordionRow	<code>Switch.tsx</code> , <code>AccordionRow.tsx</code>	Per A9	Settings (and Onboarding if reused)
SwipeRow	<code>SwipeRow.tsx</code>	<code>{ actionLabel, onAction, children }</code> . Always also renders a visible 44px button for the same action , so it works by keyboard and without gestures	Journal, Alerts
ProvenancePill, ConfidencePill	<code>ProvenancePill.tsx</code> , <code>ConfidencePill.tsx</code>	Pill words from A3 only	Today, HomeGuard, Map
Already exist	<code>Button</code> , <code>DataState</code> , <code>ErrorCallout</code> , <code>FieldError</code> , <code>SeverityPill</code>	Reuse. Do not rebuild	Every page

The formatting and text modules **FRONTEND RULE**

File	Contents
lib/frontend/text.ts	Move (and re-export from the old place so Onboarding keeps working) <code>displayText</code> , <code>contaminantLabel</code> , <code>normalizeSeverity</code> , <code>isOutsideNC</code> from <code>lib/frontend/onboarding.ts</code> . <code>displayText</code> replaces U+2014 with ", ". Every server-provided string passes through it before it is rendered (see A6)
lib/frontend/format.ts	The A5 module: <code>formatAqi</code> , <code>formatUv</code> , <code>formatPpt</code> , <code>formatRatio</code> , <code>formatScore</code> , <code>formatPopulation</code> , <code>formatCost</code> , <code>formatPercent</code> , <code>formatRelative(iso)</code> , <code>formatSampleDate(iso)</code> ("July 2025"), <code>formatMonthDay</code> , <code>parseLocalDate("YYYY-MM-DD")</code> (splits into numbers, never <code>new Date(string)</code>), <code>todayET()</code> (uses <code>Intl.DateTimeFormat('en-CA', { timeZone: 'America/New_York' })</code> , the zone the server uses), <code>weekdayInitial</code> , <code>formatUtilityName</code> (title case, keeps acronyms such as WTP, and state abbreviations, in capitals; has tests). Missing values return the words "No data" from <code>copy.common.noData</code>
Lint test	A test that scans <code>app/</code> , <code>components/</code> and <code>lib/frontend/</code> for U+2014 and fails. Also each page's browser test asserts the rendered body text contains no U+2014 in every state

Copy, types, storage, and per-page files FRONTEND RULE

- **Copy.** All strings live in `lib/frontend/copy/<page>.ts`, composed into the existing `copy` object by Phase 0 under these namespaces: `today`, `homeGuard`, `journal`, `act`, `settings`, `alerts`, `learn`, `map`, `assistant`, `shell`. Phase 0 creates each file (empty) so every owner edits only their own file. The existing `copy.home` is Onboarding's step 4: never reuse the name. Parameterized strings are functions. A string that is not in a copy file is not in the app.
- **Types and adapters.** Each page owns `lib/frontend/<page>.ts` containing its response types, a small API interface (`TodayApi`, `HomeGuardApi` ...), `createLive<Page>Api()` and `createMock<Page>Api(scenario)`. This is the same pattern as `OnboardingApi` and `lib/frontend/mock.ts`. The fully typed `DailyResponse` is Today's; `HomeResponse` is HomeGuard's (they currently type most fields as `unknown`; the owners tighten them in their own files and Phase 0 re-exports).
- **Preview routes.** Each page gets a development-only `/<page>/preview` with a scenario selector, hidden in production unless `HALO_ENABLE_FRONTEND_PREVIEW=1`, exactly like `/onboarding/preview`. Scenarios force loading, error, empty, populated, partial, offline and every page-specific state.
- **CSS.** Colors only through `--halo-*` tokens (they carry both light and dark through `light-dark()`). New page styles go in `app/<page>/<page>.css`, scoped under `.halo-app`, with class names prefixed per feature (`.halo-today-*`, `.halo-hg-*`, `.halo-journal-*`, `.halo-act-*`, `.halo-settings-*`, `.halo-learn-*`, `.halo-map-*`, `.halo-assistant-*`, `.halo-alerts-*`). Phase 0 renames `app/onboarding.css` to hold only the foundation (`app/halo-foundation.css`) and moves the onboarding-only rules to `app/onboarding/onboarding.css`. The repository uses hand-written CSS; Tailwind is installed but the foundation does not depend on it.
- **Type sizes.** Use `rem` (or `calc(Npx / 16 * 1rem * var(--halo-text-scale))`), not `px`, for every font size in new CSS, or the Settings text-size control cannot work (see B6).
- **Storage.** Two namespaces, no others. `halo.frontend.v1.<identity>.*` for anything about the household (readings, dismissed prompts, last-used journal group, cached Learn pieces, map data); `clearReadings()` clears it, and so does Change address and an identity change. `halo.device.*` for device settings (text size, contrast, motion): survives Change address and is removed only by Delete everything. `ReadingKind` widens from `"daily" | "home"` to add `"history"`. `storeReading` strips `lat` and `lng` from what it stores: never use it for map data (a separate store, IndexedDB, holds that).
- **Fonts, icons, meta.** Phase 0 adds `public/fonts/ibm-plex-mono-latin.woff2` with `next/font/local` and the variable `--font-plex` (used only for water system IDs and certifications), and the icon set A5 lists (32, 96, 192, 512 and the 1200 by 630 preview) plus a web app manifest. Today the layout has only `/halo-icon.png`.

P3. The contracts every page codes against

These signatures are fixed now. Phase 0 may add optional fields; it may not rename or remove any.

The data layer BIND

```
// lib/frontend/api.ts (Phase 0 owner)
export type HaloMethod = 'GET' | 'POST' | 'PUT' | 'PATCH' | 'DELETE';
export function haloRequest<T>(path: string, opts?: {
  method?: HaloMethod; body?: unknown; signal?: AbortSignal; timeoutMs?: number;
}): Promise<T>;
// Same behavior as the private request() inside createLiveOnboardingApi today:
// authedFetch, identity check before and after (session changed), deadline,
// status to FrontendError code mapping, never surfaces a server error string.
// The 429 mapping keeps retry_after_seconds from the body on the error object.

// lib/frontend/useHaloData.ts (Phase 0 owner)
export function useHaloData<T extends object>(endpoint: string | null, opts: {
  loader: (signal?: AbortSignal) => Promise<T>;
  enabled?: boolean;
  cache?: ReadingKind; // 'daily' | 'home' | 'history'; omit for previews and uncached data
  isEmpty?: (data: T) => boolean;
}): { data: T | null; status: 'loading' | 'error' | 'empty' | 'success';
  error: ErrorCode | null; retrievedAt: string | null;
  fromCache: boolean; refreshing: boolean; refresh: (o?: { fresh?: boolean }) => Promise<void> };
```

- The four universal states in A2 map to the hook's statuses: loading, error, empty, and **success** (called "populated" in prose).
- When a refresh fails and data is already on screen, the hook keeps the data, keeps `status` at "success" and sets `error`. That combination (`error !== null && data !== null`) is how a page knows to show "Showing your last readings from [time]. We couldn't reach the data source."
- `refresh({ fresh: true })` passes `fresh` to the loader for exactly that one call (pull to refresh on Today). Auto-refresh after 15 minutes away calls `refresh()` without it, so it cannot trip the 5-minute limiter.
- New `ErrorCode` values Phase 0 adds: `not_found` (404), `unavailable` (503 that is not an identity problem, and a platform 502 or 504), `invalid_request` (400 with a body the client did not expect), `unsupported` (501). Existing codes stay.

Client timeouts BIND

Request	Client timeout
<code>ensureAnonSession</code> , <code>POST /api/onboard</code> , <code>GET /api/daily-score</code> , <code>GET /api/home-guard</code>	15 seconds
<code>GET /api/map</code> , <code>GET /api/district</code> (a cold rebuild can be slow; the server allows up to 60 seconds)	20 seconds
<code>POST /api/assistant</code>	25 seconds (a timeout is treated like <code>reason: "unavailable"</code>)
Everything else	8 seconds

Sheets, toasts, and the back button BIND

```
// lib/frontend/sheets.ts (Phase 0 owner; pages depend only on this file)
export type LearnTopic = 'pfas' | 'radon' | 'lead' | 'air' | 'pollen' | 'uv' | 'mold';
export function openLearn(req: {
  topic: LearnTopic;
  reading?: { severity: Severity; valueText?: string }; // the "Your reading" pill
  context?: Record<string, string | number | null>; // query params for /api/learn
}): void;
export function openAlerts(): void;
export function openAssistant(page?: 'today' | 'homeguard' | 'journal' | 'map' | 'act'): void;
export function closeSheet(): void;
export function useSheets(): { active: 'learn' | 'alerts' | 'assistant' | 'page' | null };
export function useToast(): { show(message: string, o?: { actionLabel: string; onAction(): void }): void };
// <BottomSheet id title height onClose> (components/ui) registers itself here,
// which is how the assistant button hides and how "only one sheet" holds.
```

- **Opening.** A sheet that has an address (`?learn=` and `?system=`) opens with `router.push`; the others use `history.pushState`. Either way exactly one history entry is added.
- **Closing.** Every close path calls `closeSheet()`. If this app pushed the entry, close calls `history.back()`. If the sheet came from a pasted deep link (nothing to go back to), close uses `router.replace` to the same URL without the sheet parameter, so closing never leaves the app.
- **Sheet from a sheet** (a map water sheet opening "Learn about PFAS"): the second opens with `replaceState`, so Back closes to the page beneath and deselects the map feature. A button inside a sheet that navigates to another page (home pin "Go to Today") closes the sheet by replace, then pushes the route.
- **Deep links to Learn.** `?learn=<topic>` is accepted on every tabbed page. Today accepts `air` | `uv` | `pollen` | `mold`; HomeGuard accepts `pfas` | `radon` | `lead`. On a deep link the page rebuilds `reading` and `context` from its own loaded data if the topic belongs to it. Otherwise the sheet opens with no "Your reading" pill and without section 2 (see C1). Any other value is ignored (no error page).

- `valueText` **per topic** (the pill's text): air "AQI 37"; uv "5.2"; pollen "Grass 3"; mold "Estimate"; pfas "8.3 ppt · 2.1×"; radon "Zone 3"; lead "Built 1975" or "Year unknown".
- **Toast strings.** Each page's owner lists its toast strings in its own copy file. Toasts are for confirmation and Undo only. A failure that needs action stays on the page as an inline error.

Shared profile and household saves BIND

```
// lib/frontend/profile.ts (Phase 0 owner)
export function useProfile(): DataResult<ProfileResponse> & { refresh(): void };
// One GET /api/profile per app session, shared by every page and the shell.

// lib/frontend/household.ts (Settings owner; HomeGuard and Onboarding import it)
export function saveHousehold(next: {
  household: Household; renter_mode?: boolean; locale?: 'en' | 'es';
}): Promise<void>;
// ALWAYS sends all seven booleans. Reads the current values from useProfile(),
// never from the PUT response. Serializes overlapping calls so the last tap wins
// in arrival order. Controls that use it stay disabled until GET /api/profile
// has succeeded, otherwise a save from an empty local copy would wipe every group.

// lib/frontend/alerts.ts (Alerts owner; the shell imports BellBadge)
export function useAlerts(): { unread: number; alerts: Alert[]; status; refresh(); markRead(id); markAll(); dismiss(id);
undo(id) };
export function BellBadge(): JSX.Element; // unread count, "9+" cap, hidden at zero
export function AlertsSheet(props: { open: boolean; onClose(): void }): JSX.Element;
```

Alerts badge behavior (decided here). The badge is `GET /api/alerts` → `unread`. Opening the sheet does not clear it; reading rows does (this keeps "which alerts are new" visible). It refetches on app load, on window focus after 15 minutes away, and on sheet close. Optimistic updates revert with a toast if the call fails.

Change notice between pages FRONTEND RULE

After any successful Settings save that changes wording (household, renter or owner, water source, build year, language), Settings calls `clearReadings()` and dispatches the window event `halo:settings-changed`. Today, HomeGuard and Act listen and refetch. Act's sheets always fetch fresh on open rather than reading the stored copy.

P4. Rules that apply to every page

Rule	Detail
Bind to the live API from the first commit	The backend is merged into <code>main</code> and running. Pages call the real API through the live adapter. The mock adapter exists only to force states (loading, error, no data, partial, offline) in previews and tests. There is no "fixture first, wire later" step.
Never test destructive calls against production	The local <code>.env.local</code> points at the production Supabase project. <code>DELETE /api/account</code> , <code>DELETE /api/journal?all=true</code> and <code>PUT /api/household</code> change real rows. Build and test with mock adapters; run live checks only against a separate Supabase project or a throwaway anonymous session you are willing to lose.
Server strings	Every server-provided string passes through <code>displayText()</code> (A6). The server also emits some text that must never be shown (raw error strings, dataset names): D0 and D3 list them.
Missing values	The words "No data" (<code>copy.common.noData</code>). Never 0, never blank, never a dash. In a card that has both a value slot and a severity pill, "No data" appears in only one of them (DESIGN: the pill).
No em dashes	None in app-facing text (this spec's own examples follow the rule). Titles use the template <code>HALO Page</code> .
Appearance	There is no default mode. Every color is a <code>light-dark()</code> token and the phone's setting decides. Print output is always light. See A11.
Next.js	Version 16.2.10 has breaking changes from what most tooling assumes (<code>AGENTS.md</code>) and known security advisories; the fix is 16.3.6, outside the pinned range. Do not upgrade inside a page branch. Read <code>node_modules/next/dist/docs/</code> before writing routing code.

P5. How the pages depend on each other

Dependency	Provided by	Used by	Until it exists
<code>openLearn</code>	Phase 0 (contract), Learn owner (the sheet)	Today, HomeGuard, Map	SheetHost shows a labeled placeholder in development only
<code>openAlerts</code> , <code>BellBadge</code>	Phase 0 (contract),	Header on Today	Slot renders nothing

	Alerts owner		
<code>openAssistant</code>	Phase 0 (contract), Assistant owner	Shell button	<code>NEXT_PUBLIC_HALO_ASSISTANT=off</code>
<code>saveHousehold</code>	Settings owner	Settings, HomeGuard (renter control), Onboarding (its own PUT is fine as is)	HomeGuard hides the renter control
<code>halo:settings-changed</code> and <code>clearReadings</code>	Settings owner	Today, HomeGuard, Act	Pages ignore an event that never fires
Stored <code>daily</code> and <code>home</code> readings	Today and HomeGuard owners	Act (health card, volunteer causes), Map (home pin sheet)	Act and Map show "No data" for the missing part and never trigger a provider fetch themselves
<code>/settings?open=home renting household</code>	Settings owner	HomeGuard ("add the year your home was built", "Change" renter)	Plain link to <code>/settings</code>
<code>/map?panel=district</code>	Map owner	Act's district card	Plain link to <code>/map</code>
<code>/onboarding?change=1</code>	Onboarding (built)	Settings (Change address), HomeGuard (400 no county)	Works today

P6. Design tokens

These are the tokens that exist on `main` (defined with `light-dark(light, dark)`). Every page uses these names; the A11 tables give the meaning.

Token	Light	Dark	Use
<code>--halo-page</code>	<code>#eef8f7</code>	<code>#081725</code>	Page background
<code>--halo-card</code>	<code>ffffff</code>	<code>#0f2436</code>	Cards and sheets
<code>--halo-ink</code>	<code>#0b1f33</code>	<code>#e8f3f5</code>	Body text
<code>--halo-secondary</code>	<code>#4a6275</code>	<code>#9bb4c2</code>	Timestamps, sources, captions
<code>--halo-primary</code> , <code>--halo-on-primary</code>	<code>#0e5e6f</code> , <code>ffffff</code>	<code>#2cc4bd</code> , <code>#081725</code>	Primary button, active tab, links, focus ring, and the text on them
<code>--halo-border</code>	<code>#d3e4e6</code>	<code>#1d3a50</code>	Dividers, outlines
<code>--halo-tint</code>	<code>#d9f0ef</code>	<code>#1d4f5c</code>	Selected chips, soft tints (this is the approved value; it replaces the older A11 light aqua)
<code>--halo-error</code> , <code>--halo-error-tint</code>	<code>#c0262d</code> , <code>fff6f6</code>	<code>#f0605f</code> , <code>#301d2b</code>	Field errors and the error callout
<code>--halo-good</code> , <code>-moderate</code> , <code>-elevated</code> , <code>-high</code> , <code>-severe</code> , <code>-no-data</code>	A11 severity table	A11 severity table	Severity fills; <code>--halo-good-ink</code> is the pill text on a filled level
<code>--halo-navy</code> , <code>--halo-teal</code>	<code>#0b1f33</code> , <code>#0e5e6f</code>	same	Welcome band only
<code>--halo-gutter</code> , <code>--halo-radius</code> , <code>--halo-heading</code>	20px, 14px, 26px	same	Page gutter, card and input radius, step and page heading size

Tokens Phase 0 adds (each with a light and a dark value, decided in Phase 0's design pass and frozen): `--halo-skeleton` (visible in both modes), `--halo-callout-info`, `-caution`, `-notice` (background and text), `--halo-limit-marker` (the federal limit line on charts), `--halo-house-ground`, `--halo-house-roof`, the four Today hero gradient stop pairs (see B2), and a high-contrast override set. Text contrast is checked for every hero state in every time-of-day and both modes.

P7. The two Onboarding follow-ups FRONTEND RULE

Onboarding is built and approved, and it is the reference for everything above. Two small changes are still needed so Settings can use it. Owner: Yogi, before Settings is built.

1. **Change address returns properly.** In `/onboarding?change=1`, the Back arrow on the address step returns to `/settings` (today it lands on the Welcome screen, which makes no sense for someone changing an address). Completing the flow ends at `/today` (already true).
2. **The Household step in change mode.** Recommended (`OPEN D2`): keep it, prefilled with the current groups, because the built flow already does this and it lets someone who moved change who lives there. Settings' Your home row must say so; do not describe a flow that skips it.

P8. Building each page: wireframe, freeze, build, prove

Every page follows the workflow in **HALO_Wireframe_to_Build.md** (the file given to Codex), which is Onboarding's proven process:

1. **Wireframe**: a real, final-looking preview route with mock scenarios for every state, in light and dark, at 375px. Yogi reviews screenshots.
2. **Freeze**: approved screenshots become Playwright baselines (`docs/<page>/approved/`) with a README listing every element. Approved screenshots are never regenerated to make a test pass.
3. **Build**: the same screen component, now with the live adapter, bound to Part D.
4. **Prove**: zero visual difference against the frozen screenshots, a live check against the real API, a flow check into and out of the page, and the element checklist from this spec.

The element tables in Part B are the checklist. The pages' wireframes may differ in composition and spacing; they may not drop an element, a state or a string. Shared components look identical everywhere because Phase 0 froze them first.

P9. Asks for Vibhav (backend, all additive)

Nothing here blocks the frontend. Each item has a stated interim rule in the page that needs it, so builders can proceed. Tell Vibhav which he wants to do; do not change backend files from a frontend branch.

#	Ask	Interim frontend rule
1	Add <code>severity</code> to each entry of <code>water.scored_contaminants[]</code> (the value <code>waterRiskSeverity(risk)</code> already computes)	HomeGuard threshold rows take the water box severity; rows that do not exceed a limit use the neutral bar (B3)
2	Add <code>sends_household_context</code> (boolean) to <code>GET /api/assistant</code> , so the sheet and the privacy statement can say which mode is live. Also make the default safe: today the code turns household data ON unless the variable is exactly the string <code>false</code> (unset, <code>off</code> or <code>0</code> all mean on). Change it so only <code>true</code> turns it on	Assume false (public sources only) and use the false-state privacy line (C2, B6). Until the code changes, Vibhav sets the Vercel variable to exactly <code>false</code>
3	Add <code>export const maxDuration = 30</code> to the assistant route (the client waits 25 seconds; no route limit is set)	None needed
4	Add <code>history[].backfilled</code> (and <code>is_partial</code>) to <code>GET /api/history</code>	Compare severity words only, and show the "change since yesterday" line only when the rule in B2 allows
5	Remove em dashes from server-written text at the source (<code>lib/explain.js</code> , <code>learnContent.js</code> , <code>learnWhyYours.js</code> , <code>wellTestData.js</code> , <code>leadRisk.js</code> , <code>actionPlan.js</code> , <code>alertRules.js</code> , the assistant refusal, home-guard messages)	<code>displayText()</code> handles it either way
6	Replace <code>{ error: err.message }</code> on the routes that still return raw database text (journal, history, alerts, notifications, push, volunteer, learn, sources, map, district, assistant, account, health)	The frontend never displays a server error string
7	Change the map suggestion chip "What does 'not tested' mean on the map?" (the map says "No detection reported")	Chip text is server-supplied; note it in testing
8	Soften <code>score.reason</code> wording "is capped for that reason" when <code>score.capped</code> is false	Gate any "capped" wording on <code>score.capped === true</code>
9	Never store an empty district or water payload over a good one. When the water geography lookup fails, the server currently stores <code>total_systems: 0</code> , <code>systems_over_limit: 0</code> and <code>affected_population: 0</code> , which reads as an all-clear	Treat <code>total_systems</code> of 0, or <code>unlocated_systems</code> near <code>total_systems</code> , as no data (B7)

P10. Decisions still open, with the default a builder uses meanwhile

ID	Decision (Yogi unless noted)	Default until decided
D1	Who owns Phase 0	Recommended: whoever has the most Codex quota, reviewed by Yogi
D2	Change address: keep the prefilled Household step, or skip it	Keep it (as built)
D3	Dark mode: follow the phone (built and approved in both), or dark only	Follow the phone. A dark-only switch is a one-line change to <code>color-scheme</code> plus a review of the light screenshots
D4	Send readings and symptom names and dates to the Assistant's model (<code>ASSISTANT_SEND_HOUSEHOLD_CONTEXT</code>)	Off. The Assistant answers from public sources only (the Vercel variable must be exactly <code>false</code> : any other value, including unset, turns household data on)
D5	Who writes the missing content (Act's help actions and letters, About text, radon and	Builders use the PROPOSED strings in this spec and list each one used, so nothing invented ships unnoticed

	pollen notes, named local district facts) and by when (Aadit unless changed)	
D6	Season summary alert target	Show the summary callout at the top of both Log and Trends, keep the alert opening <code>/journal?mode=trends</code>
D7	Map library and tiles	Mapbox GL JS with the existing <code>NEXT_PUBLIC_MAPBOX_TOKEN</code> and Mapbox's muted light and dark styles; NC county shapes from <code>lib/data/ncCounties.json</code> (100 counties, names identical to the radon keys, about 46 KB compressed) copied to <code>public/data/nc-counties.geojson</code>
D8	Device notifications (service worker) in this build	Build after the core pages; the Settings switch is hidden until the service worker ships (B6)
D9	Volunteer organizations: all 13 have <code>last_verified</code> empty	Rows say "Not yet verified" until a person confirms each one and sets the date (Vibhav's wrap-up item)

Part A: Cross-Cutting Systems

Everything in this part applies to every page. Built once, in the shared layer, before any page, the blueprint's own first build step (§26.1) and the thing that decides whether dark mode and consistency are cheap or impossible later.

A1. Platform, shell, and navigation

Platform LOCKED

- A web application with a mobile-shaped layout, running in a mobile browser. Not native iOS or Android (§3).
- Primary viewport 375–430px. Every screen verified at 375px. Content column caps at 430px and centers above that; the Map is the one exception and widens to 600px (§53.2). Under 375px content compresses; nothing hides; no horizontal scroll ever.
- Portrait is the target. Landscape isn't blocked, but has no special layout (§53.4).
- Header respects the top safe-area inset, tab bar the bottom inset, sheets the bottom inset (§53.3).
- Stack: Next.js 16.2.10 App Router, React 19.2.4, Tailwind v4, `lucide-react` as the single icon family. The repo's `AGENTS.md` warns that this Next.js version has breaking changes from what most tooling assumes, whoever builds reads `node_modules/next/dist/docs/` before writing routing code. Version 16.2.10 also has published security advisories (fix: 16.3.6, outside the pinned range); do not upgrade inside a page branch (P4).
- Styling is the repository's hand-written CSS with `--halo-*` tokens (P2, P6). Tailwind is installed but the foundation does not depend on it.

Pages and overlays LOCKED

Page	Address	Tab label	Header title	Reached by
Entry	<code>/</code>	none	none	Redirects to <code>/onboarding?entry=1</code> , which decides where the visitor goes
Onboarding	<code>/onboarding</code>	n/a	none (full screen)	Through <code>/</code> on first visit; <code>?change=1</code> from Settings; <code>/onboarding/preview</code> in development only
Today	<code>/today</code>	Today	Today	Tab 1, default landing
HomeGuard	<code>/home</code>	Home	Your home	Tab 2
Map	<code>/map</code>	Map	Map	Tab 3
Journal	<code>/journal</code>	Journal	Journal	Tab 4
Act	<code>/act</code>	Act	Act	Tab 5
Settings	<code>/settings</code>	n/a	Settings (back arrow, not gear)	Gear icon in any header

Three overlays, Learn, Assistant, Alerts, are bottom sheets, never pages, never tabs. Learn opens only from a specific reading's "What does this mean?" link (no index, no browsing). Assistant opens from a floating button on all five tabbed pages. Alerts opens from the bell on Today's header only (§7.2).

Header bar LOCKED

Fixed top on every tabbed page. Page title left. Today: alerts bell (unread badge, capped at "9+", hidden at zero) and settings gear on the right. Every other tabbed page: gear only. Never hides on scroll (§7.3, §58.1). On Map, the header sits over the map (§58.1). `BIND` badge count = `GET /api/alerts` → `unread`. **The shell renders the header, the bell and the assistant button** from the route (Part P2); pages render none of them, and no page fetches `/api/alerts` for the badge.

Bottom tab bar LOCKED

Fixed bottom, five equal-width destinations, icon above label, active destination visibly distinct from inactive, always visible, never scrolls away (§7.4). Icons (§32.4): Today = sun over horizon, Home = house, Map = folded map, Journal = notebook, Act = outstretched hand. Tab changes are instant, no transition animation; content fades in as it arrives (§28.2). Each tab remembers its scroll position for the session; a newly visited tab starts at the top (§27.4).

Floating assistant button LOCKED

Lower right, above the tab bar, on all five tabbed pages. Hides while any sheet is open. Carries a chat-bubble icon (§32.4). The blueprint also has it fade out on scroll down and back on scroll up (§58.3). **Tier 3: default is a fixed button** (§65.3). Opens the Assistant sheet carrying the current page as context.

First-visit routing LOCKED FRONTEND RULE

Blueprint (§7.5): on every load, check for a stored location. Have one → Today. Don't → onboarding. If the check itself fails → onboarding, never an error screen.

```

/                                redirects to /onboarding?entry=1
/onboarding?entry=1 (OnboardingScreen bootstrap):
  await ensureAnonSession()      // lib/auth.js, also run app-wide by AuthInitializer
  GET /api/profile
    onboarding_complete === true  -> markCompleted(profile), go to /today (or ?next=)
    request failed, hasCompletedOnboarding() true
                                  -> /today?verify=1 (ProfileRecheck retries in the background)
    anything else                 -> stay on onboarding
Every route in the app shell (AppGate, Part P2):
  no session or no completed profile -> /onboarding?entry=1&next=<current address>
  a page reports no_location / no_county -> /onboarding?entry=1

```

FOUR RULES. (1) Route on `onboarding_complete`, not `onboarded`. `onboarded` turns true as soon as coordinates are saved, at the address step, so someone who closes the app on step 3 or 4 would be sent to Today with no water source, and a well household inside a utility boundary would be shown that utility's numbers. A household that did not finish restarts at the address step (the address was never kept). (2) The local marker is `halo.onboarded`, a JSON record `{ version: 1, identity, complete: true }` written by `markCompleted(profile)` and read by `hasCompletedOnboarding()` (both in `lib/frontend/storage.ts`). It belongs to one session identity: a marker from another identity, or the old bare-boolean form, is ignored. It is set only after step 4 succeeds. (3) When `GET /api/profile` succeeds with an incomplete profile, clear the marker and every stored reading first (after a genuine 401 reset creates a new anonymous user, the old marker and readings would otherwise linger). (4) When the profile request *fails* and the marker is valid for this identity, go to `/today?verify=1`. Today must render `<ProfileRecheck />` when `verify=1`; it retries, redirects an incomplete profile to onboarding, and offers Retry if verification stays unavailable. It never loops and never resets the identity. A finished household is never sent back through onboarding because of a failed request.

Addresses and state in the URL LOCKED

Address	Restores
<code>/map?layer=radon</code>	Map with radon layer active
<code>/map?</code> <code>layer=water&contaminant=pfos</code>	Map colored by PFOS
<code>/map?system=NC0112010</code>	Map with that system's sheet open
<code>/journal?mode=trends</code>	Journal in Trends mode
<code>/home?risk=water</code>	HomeGuard with the water card expanded (also <code>radon</code> , <code>lead</code>)
<code>/today?learn=air</code>	Today with that Learn sheet open (Today accepts <code>air</code> , <code>uv</code> , <code>pollen</code> , <code>mold</code> ; HomeGuard accepts <code>pfas</code> , <code>radon</code> , <code>lead</code>)
<code>/today?verify=1</code>	Today plus the background profile recheck (onboarding handoff, above)
<code>/onboarding?entry=1</code> · <code>?</code> <code>change=1</code>	Normal entry · replace the address (from Settings and HomeGuard)
<code>/settings?open=home</code> (also <code>renting</code> , <code>household</code>)	Settings with that section open. One-shot: opened on load, then removed from the address with <code>router.replace</code> . The only exception to "the open section is not in the URL"
<code>/map?panel=district</code>	Map with the district panel open

Rules for every address parameter: unrecognized or invalid values are ignored and the default is used (never an error page). Expanding a card, changing the Map layer, contaminant or quarter, and switching the Journal mode use `router.replace` (no history entry); opening a sheet adds one entry. Only one card is open at a time on Today and HomeGuard. Tab links carry no query string, so a saved viewport or scroll position applies only when the address has no parameters (URL parameters win).

Not carried in the URL: scroll position, assistant conversation, unsaved journal input, which Settings accordion is open (§27.2). `/map?layer=water&quarter=2024Q3` applies only when the time slider exists (the map reports available `quarters`; see B7). Any unrecognized address shows a short message and a button back to Today, never a raw error page, never a silent redirect (§27.5).

The back-button contract LOCKED

- Opening any sheet (Learn, Assistant, Alerts, a map detail, any Act feature, Journal day detail, delete confirmation) pushes a history entry.
- Back with a sheet open closes the sheet and stays on the page beneath.
- Back with no sheet open goes to the previous tab, or exits.
- Closing a sheet any other way (drag down, tap backdrop, Escape, close control) also removes its history entry, so Back doesn't reopen it.
- Only one sheet at a time. Opening a second replaces the first (using `replaceState`, so Back closes to the page beneath). The exact open and close algorithm, including deep links, is in Part P3.

- Expanding a card is not a history entry. Back never collapses a card.
- Each onboarding step pushes an entry; Back moves between steps without losing entered values.
- Opening a sheet locks page scroll beneath it; closing restores the exact prior position. Expanding a card scrolls just enough to reveal its new content (§27.3–27.4).

Layering LOCKED

Back to front: page content → fixed header → fixed tab bar → floating assistant button → sheet backdrop → sheet → toast → offline banner. Nothing else creates a layer (§53.1).

A2. The data layer

Authentication BIND

- `ensureAnonSession()` (lib/auth.js), silent anonymous session, runs app-wide from the root layout. Throws with `.code` `'captcha_failed'` or `'signin_failed'`; the onboarding screen shows *"Verification could not be completed. Check your connection and try again."* or *"Could not start a session. Please reload and try again."*
- `authedFetch(url, init)`, attaches `Authorization: Bearer <token>`. Pass `init.headers` as a plain object (a `Headers` instance gets dropped by the spread).
- Every data route requires the bearer token. No route accepts identity from the body or query.
- 401 vs 503, the rule that protects a household's data.** `401` means the session is definitively invalid (the account no longer exists): discard it and start a new anonymous session, silently (§31.4). `503` means identity couldn't be verified right now: show the generic error with retry and **never** reset the session. Resetting on a temporary failure would silently orphan the household's profile, history, and journal under an abandoned account. If the silent retry after a 401 also fails, show *"Could not start a session. Please reload and try again."*
- The built layer also protects against mixing sessions: the identity is compared before and after every request, and a changed identity rejects the old response with `session_changed` instead of showing another household's data. A browser-wide observer clears frontend caches and emits the window event `halo:identity-changed`. It does not create or reset a session.

One data hook, one state wrapper FRONTEND RULE

Every page gets data the same way: `useHaloData(endpoint, { loader, enabled, cache, isEmpty })` (built in `lib/frontend/useHaloData.ts`, made generic in Phase 0, signature in P3) and a single `<DataState>` wrapper that renders the right one of the four universal states. The hook returns `{ data, status, error, retrievedAt, fromCache, refreshing, refresh }`; its statuses are `loading`, `error`, `empty`, `success`. Pages never write their own loading or error logic. Requests go through the one `haloRequest` function (P3), which owns authentication, timeouts, identity checks and error mapping. Pages never call `fetch` directly. This is how "every screen implements all four states" (§24) holds across ten screens and three builders.

The four universal states LOCKED

State	When	Shows
Loading	Request in flight, nothing cached	Skeleton shaped exactly like the content it replaces, same dimensions, so nothing shifts when data arrives. Never a bare spinner (Map is the one exception, §46.9)
Error	Request failed	What failed + that it isn't a finding about their environment + Retry
Empty	Succeeded, nothing to show	Why there's nothing, and what would make content appear
Populated	Data present	The content

Error wording and empty wording are never the same text (§24). Every screen tolerates a missing water utility ID without breaking.

Error responses: what the frontend shows BIND

Response	Show
401, any message	Nothing (silent re-auth, §31.4), subject to the guard above
503 "Could not verify your session right now..."	Generic error + retry
429 (any route)	<i>"You've made a lot of requests. Please try again later."</i> (Assistant has its own 429 text, see C2. A 429 on Today's daily-score has its own handling, next row.)
429 on <code>/api/daily-score</code> (with <code>?fresh=1</code> , or a plain call that misses the server's cache and would reach the providers) with body <code>{ error, retry_after_seconds: 300 }</code>	When readings are already on screen or stored: not an error state. The readings stay, the freshness line is unchanged, and a toast shows PROPOSED <i>"Your readings were just updated. Try again in a few minutes."</i> When nothing is stored: the generic 429 copy above. The server allows one provider refresh per household per 5 minutes
400 on <code>/api/daily-score</code> , "No location yet..."	The household has no stored location: treat it exactly like <code>onboarding_complete: false</code> and route to onboarding (A1)
400 on <code>/api/home-</code>	Rare (the geocoder returned no county). HomeGuard shows its error state with a <i>Change address</i> action (B6);

<code>guard, "No county on file..."</code>	on the results reveal, the water and radon lines read <i>Couldn't check right now</i>
500, or network failure	<i>"Something went wrong on our end, not yours. Try again in a moment."</i>
Timeout (client-side, 8s default)	<i>"That took too long. The data source may be slow right now."</i> Longer deadlines (the table in P3): <code>POST /api/onboard</code> , <code>GET /api/daily-score</code> and <code>GET /api/home-guard</code> use 15s ; <code>GET /api/map</code> and <code>/api/district</code> use 20s (cold rebuilds are slow; map layer providers have their own 20s cap on the server); <code>POST /api/assistant</code> uses 25s (its timeout shows the Assistant's own "couldn't reach" message with Retry, C2). The daily, HomeGuard and onboarding providers have no server-side cap, so a cold call can run long. When <code>POST /api/onboard</code> times out, the client re-checks <code>GET /api/profile</code> before showing an error: each address submission carries a fresh <code>request_id</code> , and the profile's <code>onboard_request_id</code> equal to it confirms that exact write (Onboarding already does this)
Offline (<code>navigator.onLine</code> false)	Offline banner + last stored readings (below)

TWO BLUEPRINT STRINGS DELIBERATELY NOT USED: §31.4's "[Source name] isn't responding right now. The rest of your readings are current." is dropped because §39.1 forbids naming services a household won't recognize, a missing reading shows its card's own no-data state instead (B2). §31.4's "You appear to be offline. We're showing the last readings we have." is replaced by §58.4's banner text, "You're offline. Showing your last readings." (`copy.common.offlineCached`). Don't reintroduce either.

NEVER DISPLAY A SERVER'S `error` STRING DIRECTLY: most routes return raw database or JavaScript messages in 500s. Map every error to the copy above. The one exception is onboarding's specific address errors (B1), which are mapped by status code.

Stored readings, offline, and refresh LOCKED

- The last successful Today and HomeGuard responses are stored locally with the time they were received and shown immediately on open, while fresh data loads (§24, §36). Cached reading display target: under 200ms (§35.1).
- Fresh request fails and stored data exists → keep it on screen with *"Showing your last readings from [time]. We couldn't reach the data source."*
- Offline → thin banner under the header: *"You're offline. Showing your last readings."* Pushes content down, never covers it (§58.4).
- Learn: cache only the parts that do not depend on the household, `what_it_is`, `protect[]` and `sources[]`, keyed by topic and locale. **Never cache** `why_yours` **or** `household_note` (they are built from the reading and the household, which can change), and refetch every time the sheet opens. Offline: show the cached parts and omit sections 2 and 3. Map data is cached after first load (§36) in its own store (P2), never through `storeReading`, which strips coordinates.
- Pull to refresh on Today and HomeGuard only. On Today, the page calls `refresh({ fresh: true })` for exactly that one call, which sends `/api/daily-score?fresh=1` and bypasses the server's one-hour cache. HomeGuard simply re-requests `GET /api/home-guard`, which has no server cache and no `fresh` parameter. Disabled while a request is in flight. Automatic refresh when the app regains focus after 15+ minutes away (§28.1) is a plain request without `fresh`, so it cannot trip the 5-minute limiter. A refresh that returns cached data still updates the freshness line.
- Freshness lines use the server's `retrieved_at` (or `assembled_at`), never the time the browser received the response, a cached reading can be up to an hour old.
- Degradation order: fresh fails → stored data with its time → a single reading unavailable while others show → all fail: explanatory empty state with retry → server unreachable: stored data + offline banner (§39.2).

Journal writes offline TIER 3

The blueprint queues entries written offline and sends them on reconnect, with a small pending marker (§36). It's Tier 3 (§65.3): **the default is a clear failure instead**, keep everything typed and show PROPOSED *"Couldn't save that entry. It's still here. Try again."* Build the queue only if time allows.

A3. Severity and confidence

The six severity words LOCKED BIND

The backend emits exactly these words: `good`, `moderate`, `elevated`, `high`, `severe`, `no_data`. The frontend holds one mapping (word → label, icon, color token), ideally importing the word list from `lib/severity.js` so it can't drift. Labels: Good, Moderate, Elevated, High, Severe, No data. Any unrecognized word renders as No data (§62.4).

- Every severity appears as **color + icon + word together**, never color alone (§6.1, §25.1). This is an accessibility requirement, not a style preference.
- No data is visually distinct from Good everywhere. Missing data is never shown as zero or green, the single most dangerous failure the blueprint names (§65.1).
- Pill labels capitalized; the same word lowercase inside a sentence (§29.3).

Where to read severity, per endpoint BIND

Screen element	Read	Never read
Today's four cards	<code>air.severity</code> , <code>uv.severity</code> , <code>pollen.severity</code> , <code>mold.severity</code>	<code>*.status</code> , older field with different thresholds (AQI 120 is status "unhealthy" but severity "elevated")
Today trend lines, Journal grid	<code>/api/history</code> → <code>history[].air</code> / <code>uv</code> / <code>pollen</code> / <code>mold</code> (plain words). Note <code>history[].pollen</code> is the worst pollen category that day, while a pollen trend plots the value of today's dominant category, so a non-dominant day's point takes its color from the display-only edges in A10	n/a
HomeGuard cards & house diagram	<code>breakdown[].severity</code> , matched by <code>breakdown[].key</code> (<code>water</code> , <code>radon</code> , <code>lead</code>), never by position	<code>breakdown[].status</code> , older field with its own thresholds
HomeGuard threshold-chart rows	No per-row severity exists yet (ask 1 in P9). Interim: every bar takes the Drinking-water box severity; a row with <code>exceeds_limit</code> false uses the neutral bar color	Never derive a color from <code>risk</code> or the ratio
Both heroes (score ring, headline)	<code>score.severity</code>	n/a
HomeGuard action plan	<code>action_plan[].severity</code>	n/a
Lead card	<code>lead.level</code> (canonical)	n/a
Map	Water: <code>features[].overall_severity</code> , <code>features[].contaminants[X].severity</code> , <code>features[].hazard_index.severity</code> . Radon, facilities, air: <code>features[].severity</code>	n/a
District panel	<code>counties[].worst_water_severity</code> , <code>counties[].radon_severity</code>	n/a
Alerts	<code>alerts[].severity</code> (set for alerts that carry a level; else null)	n/a

Hero severity: what colors the score ring BIND

Both heroes read `score.severity`, the server decides the composite's level (§11.6). On HomeGuard it reflects water and radon only; **lead never colors the ring**, because it isn't scored. When the score is null (a private well, or nothing available), the ring shows its no-data state.

Confidence: a separate signal LOCKED BIND

value	Shown	Meaning
<code>full</code>	nothing	Everything evaluated, current
<code>limited</code>	quiet pill: <i>partial</i>	Some detections couldn't be evaluated
<code>stale</code>	quiet caution pill: <i>3+ yrs old</i>	Evaluated, but the sample is old
<code>none</code>	quiet pill: <i>not evaluated</i>	Nothing could be evaluated

HomeGuard's `breakdown[].confidence` carries this. A reading can be low severity and low confidence at once, never merge the two into one element (§6.2).

Provenance pills LOCKED

Three words: *measured*, *estimate*, *modeled* (§31.2). Air: `is_measured === true` → *measured*; `false` → *modeled*; `null` → no provenance pill (unknown isn't the same as modeled). UV and pollen: no provenance pill (the API exposes none). Today's cards show no confidence pill (`/api/daily-score` has no confidence field). Mold: always *estimate* (`is_proxy: true`). Radon: always *estimate* (county zone, not a home measurement). Lead: *estimate* when `lead.is_estimate` .

A4. Household-aware wording

LOCKED Seven yes/no facts per household, no ages, names, or medical records: `has_toddler` , `has_child` , `has_teen` , `has_adult` , `has_senior` , `has_pregnant` , `has_respiratory` (§8). Composition never changes a measurement, a severity, or a score, only the sentence under a reading and the order of actions (§8.2).

BIND The sentence is already written server-side: `air.sentence` , `uv.sentence` , `pollen.sentence` , `mold.sentence` (Today), `lead.sentence` (HomeGuard), `household_note` (Learn). Display them as-is. The frontend never composes a household sentence.

When several groups apply, the server uses the most specific one that has wording for that reading, in the blueprint's priority order: respiratory condition → toddler → pregnancy → senior → child → general (§8.5).

A5. Formatting rules LOCKED

One formatting module, used everywhere. The same quantity formatted two ways on two screens reads as unfinished (§29).

Quantity	Format	Example	Source field
Air quality index	Whole number	37	<code>air.aqi</code> (may be a decimal from the modeled source, round)
UV index	One decimal	5.2	<code>uv.index</code>
Pollen	Whole number with its category	Grass 3	<code>pollen.dominant</code> + <code>pollen[dominant]</code>
Concentration	One decimal, unit after a space	8.3 ppt	<code>value_ppt</code> (sent unrounded)
Ratio to limit	One decimal + ×	2.1× the limit	<code>value_ppt / limit_ppt</code> , arithmetic on server numbers, not a threshold
Composite score	Whole number, no unit, no %	78	<code>score.display_score</code> , never <code>score.score</code>
Radon zone	"Zone" + numeral	Zone 2	<code>radon.zone</code>
Population	Commas, whole number, no rounding and no "+" sign. Where the server marks a figure as incomplete (<code>affected_population_complete</code> <code>false</code> , <code>over_population_unknown</code>), write "at least" before it	12,400 · at least 120,000	<code>map population</code> , <code>district affected_population</code> , <code>population_over</code> , letter <code>population_served</code>
Costs	Range, en dash, no decimals	\$40-90	<code>action plan cost</code> (already formatted); <code>test plan estimated_cost.low/high</code>
Percent	Whole number + %. Journal <code>elevated_rate</code> and <code>other_rate</code> are already 0 to 100 integers: do not scale them again. District <code>district_share_over</code> and <code>state_share_over</code> are fractions from 0 to 1: multiply by 100	78%	<code>journal</code> , <code>district</code>
Lithium	Micrograms per liter, never ppt: value = <code>value_ppt / 1000</code> , formatted one decimal	25.0 µg/L	<code>excluded_from_score[] .value_ppt</code> ; <code>reference level health_reference_level_ppt / 1000</code> . The raw name <code>lithium</code> displays as "Lithium"
Missing value	The words "No data" (<code>copy.common.noData</code>). Never 0, never blank, never a dash	No data	any <code>null</code>

- Units spelled out the first time in a view, abbreviated after: "parts per trillion (ppt)" then "ppt."
- Measured values use tabular digits (Instrument Sans with `font-variant-numeric: tabular-nums`) so digits align, not a monospaced face. Monospace (IBM Plex Mono) is for codes only (A11).
- Never more precision than the source provides.

Dates LOCKED FRONTEND RULE

Situation	Format	Example
Under 1 hour	Relative minutes	updated 14 minutes ago
Under 24 hours	Relative hours	updated 6 hours ago
Yesterday	The word	yesterday
Last week	Weekday	Tuesday
This year	Month and day	March 14
Older	Month and year	October 2024
Lab sample dates	Always month + year, never relative	sampled October 2024
Data age warnings	Rounded years	this sample is about 3 years old

WHICH DATE FIELDS: water sample dates: bind the `*_iso` fields (`latest_sample_iso`, `earliest_sample_iso`, `date_iso`). The same objects also carry older `M/D/YYYY` strings (`latest_sample_date`, `date`) kept for compatibility, don't parse those; `new Date("7/1/2025")` is locale-dependent. Every other date is ISO.

ONE DATE CONVENTION: daily readings (`history[].date`) and journal entries (`entry_date`) are both the household's **local** calendar date. Join them by the same `YYYY-MM-DD` string. The server computes every date in **America/New_York** (`lib/localDate.js`). The client must use the same zone to join with history and entries: `todayET()` in `lib/frontend/format.ts` (P2), or the `to` field of `/api/history`. Never use the device's own date or a UTC timestamp. Parse `YYYY-MM-DD` by splitting into numbers, never with `new Date("2026-09-28")` (that parses as UTC and returns the previous weekday in US zones).

Text conventions **LOCKED**

- Sentence case everywhere, buttons and headings included. No title case, no all caps.
- PFOA, PFOS, PFHxS, PFNA, GenX keep their conventional capitals. The backend key `HFPO-DA` displays as "GenX (HFPO-DA)".
- Numbers under ten as words in prose, numerals in data. "County" always included. Locked strings that use numerals in a sentence ("3 days selected.", "Based on 3 of 4 readings.") are data strings and keep them.
- **No em dashes (U+2014) anywhere in app-facing text:** screens, buttons, errors, alerts, accessibility labels, page titles, and server-provided strings. Use a period, comma, colon or parentheses. A test enforces this (P2). This is a team decision that supersedes older punctuation in the blueprint.
- HALO always in capitals, except the approved Welcome subtitle "To Halo". Tagline where needed: "*Know what's around your home.*" Browser title template: `HALO | [page name]` (built in `app/layout.tsx`). Link preview description: "*See what's in the air, water, and ground around your home, and what to do about it.*" (§32).

Identity and icons **LOCKED** (§32)

- The approved logo is `public/halo-logo-mark.png`, a transparent PNG used as a monochrome mask and drawn **white on the navy Welcome band** (a contrasting foreground on light surfaces). The blue tile, `finallogo.png` and every earlier concept are superseded. It is used as the Welcome mark; the browser tab icon (32, 96, 192px); the home-screen icon (192 and 512px, with a web app manifest); and the link-preview image (1200×630, the mark and the name over the primary color). Only `/halo-icon.png` exists today; Phase 0 produces the full icon set (P2).
- The browser `theme-color` is the Primary token, `#0e5e6f` in light mode, `#2cc4bd` in dark (two `<meta name="theme-color">` tags with `media` queries).
- One icon family (lucide-react), one line weight, no mixed styles. Concept icons: Today, sun over horizon · Home, house · Map, folded map · Journal, notebook · Act, outstretched hand · Air quality, wind lines · UV, sun · Pollen, flower · Mold, droplet · Water, tap · Radon, radiation trefoil · Lead, pipe · Alerts, bell · Settings, gear · Assistant, chat bubble. Where lucide has no exact match (e.g. a trefoil), pick the closest line icon at the same weight and keep it consistent everywhere.

A6. Voice and copy **LOCKED**

Plain, specific, steady. Never minimize a real problem; never frighten someone who can't act tonight (§30).

- Never "safe" or "unsafe" as a bare verdict. Never exclamation marks. Never fear words: toxic, poison, dangerous, alarming, shocking.
- Never apologize for data that doesn't exist. Never congratulate a good reading. Never "simply," "just," or "easy" for something that costs money or effort. Never imply a medical outcome.
- Second person ("your water"). HALO calls itself HALO, never "we" as an organization making promises.
- Reading level about sixth grade.
- Anything not listed in the content inventory (§31) and not derived from data doesn't appear in the interface. Where this document proposes new copy, it's tagged **PROPOSED** and needs approval.

BACKEND MESSAGES THAT BREAK THE VOICE RULES, USE BLUEPRINT COPY INSTEAD: some server `message` strings name services a household won't recognize (§39.1). Where the blueprint has specific copy for a state, use the blueprint's. Example: HomeGuard's `no_data_yet`, show the blueprint's "*Testing results for your utility haven't been published yet. This is normal. Federal testing is still rolling out nationally.*" instead. Each page section notes where the server's text is used and where the blueprint's replaces it.

DATA IN CAPITALS AND SERVER TEXT THAT BREAKS VOICE: utility names arrive as EPA and state records spell them, often in capitals (e.g. "CONCORD, CITY OF" in `lead.utility_inventory.utility`; `water.pws_name` likely the same). **PROPOSED**: display them in title case ("Concord, City of"), keeping acronyms such as WTP and state abbreviations in capitals. Don't build fixtures around the blueprint's "Concord, Hillgrove WTP", which is a treatment-plant name, not a utility record. A few server-written strings break A6 and are flagged for rewording by the content owner (Aadit) or Vibhav, for example a well test's `why`, "This is the one that can make your family sick this week, not in twenty years.", and alert messages like "Air quality dropped to Elevated today." (a capitalized severity mid-sentence). **Display server text unchanged, except that every server-provided string passes through** `displayText()` **first** (`lib/frontend/text.ts`), which replaces U+2014 with " ". That covers every `sentence`, `reason`, `message`, `title`, `action`, `why`, `where`, `text`, `protect[]`, `household_note`, `why_yours`, `what_it_is`, `source` labels, `regulatory_notice`, `utility` and organization names, alert text, and the Assistant's answer and refusal. Do not otherwise rewrite it in the frontend. Server fields that must **never** be shown at all are listed in D3.

Universal copy **LOCKED**

Controls: Close, Back, Retry, Save, Cancel, Skip, Continue, Done, See more, Show less. Severity: Good, Moderate, Elevated, High, Severe, No data. Confidence: partial, 3+ yrs old, not evaluated. Provenance: measured, estimate, modeled (§31.1–31.2).

Confirmations (§31.5): changing address, "*This will replace your current results. Continue?*"; delete all, "*This permanently deletes everything HALO knows about your household. This cannot be undone.*" + typing a confirmation word; clear journal, "*This deletes all [n] of your entries permanently.*"; delete one journal entry, immediate, with undo, no dialog.

A7. Forms and validation LOCKED

- Validate on blur, not on every keystroke. Error text directly beneath the field, field border also marked. An error clears the moment the input becomes valid.
- The submit button is never disabled for validation, let people submit and tell them what's wrong. (Different from disabling during an in-flight request, which is required.)
- Every field keeps its value after a failed submission. Every input has a visible label, not just a placeholder. Text inputs at 16px or larger (smaller makes mobile browsers zoom on focus). Address requests device autofill; build year opens the numeric keypad and accepts digits only (§34).

Field	Rule	Error text
Address	Required, 5+ characters	"Enter an address or ZIP code."
Address not found (server 404)	n/a	"We couldn't find that address. Try adding your city or ZIP code."
Build year	Optional; if present, four digits, 1700 to current year	"Enter a four-digit year, or leave this blank."
Build year in the future	Rejected	"That year hasn't happened yet."
Water source	Required, from the list	"Choose where your water comes from."
Household groups	Never required	n/a
Journal note	Optional, max 500 characters	Character counter appears past 400
Assistant question	Required, max 1000 characters	"Ask a question to get started."

BUILD YEAR: validate 1700 to the current year on the client (the rules above), and send it as a number, or `null` when blank. The server rejects values outside that range too, but the client check is what gives the person the right message.

A8. Accessibility, motion, text handling

Non-negotiable LOCKED

- Severity never by color alone. Every text/background pair meets standard contrast, verified separately in light and dark.
- Every interactive element at least 44×44px regardless of icon size.
- Everything reachable and operable by keyboard, in visual order, with a visible focus indicator. Escape closes any sheet. Enter/Space activates. Arrow keys move between segmented options and map layer chips; the map's time slider changes quarter with the arrow keys and the district handle is a button. Calendars and grids (the Journal's month calendar and heat grid) use a roving tabindex: arrows move by day and week, Space toggles or opens the day, and a visually hidden list of the same days is the text equivalent.
- Any gesture-only action (swipe to delete or dismiss) also has a visible 44px control for the same action (`SwipeRow` , P2).
- Sheets trap focus while open and return focus to whatever opened them.
- Icons and images have text alternatives; decorative ones are marked decorative. Every chart has a text equivalent stating the same finding.
- Body text never below the body size floor and respects the user's own text size setting.
- Hover states exist for pointer devices but never shift layout, and hover is never the only way to reveal something (§25.1, §28.6).

Motion: functional rules only LOCKED

Motion is limited to: sheet entry/exit, card expansion, content fade-in, press feedback, score ring sweep on first render, chart bar growth, onboarding step transitions, map color changes, toasts. Nothing longer than 400ms. Everything becomes instant under the system's reduced-motion setting, except opacity fades (kept very short so content doesn't pop). Exact durations and easing are DESIGN .

Touch and request feedback LOCKED

Every tappable element visibly responds on press within 100ms. Any button that triggers a request shows an in-place spinner and disables until it resolves (no double submits). Toggle chips flip instantly on tap, before the network call, and flip back visibly if the save fails (§28.5).

Text handling LOCKED

- Font stack declares a system fallback so text never waits on a font file.
- Card titles truncate at one line. Card summaries wrap to two lines then truncate. Utility names wrap to two lines in sheets, truncate at one in list rows. Journal notes truncate at one line in the list, full on tap. Assistant messages never truncate.
- **Numbers never truncate.** If a value doesn't fit, the container grows or the label shortens.

- Spanish runs 15–30% longer; no button width is fixed to its English label (§61).

Higher contrast (Settings) PROPOSED

The blueprint gives Settings a higher-contrast switch (§49.2) but defines no palette for it. Proposed effect: secondary text uses the body-text token, borders use Navy 700 (dark: a lighter border), and pills and focus rings get a 2px outline, implemented as a token override in both modes. Exact values DESIGN (Phase 0 adds the override token set, P6). **How the three Settings**

accessibility controls reach every page: the shell sets `data-text-scale` (100, 115, 130 or 150 percent), `data-contrast="high"` and `data-motion="reduce"` on `<html>`, and `--halo-text-scale` on `.halo-app`. All font sizes use `rem` or `calc(Npx / 16 * 1rem * var(--halo-text-scale))` (the existing CSS is `px`-based, so Phase 0 converts the foundation). CSS honors both the media query and `[data-motion="reduce"]`; reduce-motion defaults to the system setting. Settings stores these under `halo.device.*`.

Dark mode LOCKED (colors DESIGN)

There is no default mode. The app follows the phone's setting (`color-scheme: light dark`) and every color is a `light-dark()` token; Onboarding was approved in both modes. No in-app toggle. The map style, `theme-color` and charts switch with the same setting. Charts keep identical geometry with substituted colors. The health card's print output is always light. Build every color as a token from day one: the blueprint's own condition is that if colors are hardcoded early, dark mode is cut rather than retrofitted (§59, §68). Whether to instead make the app dark only is open (P10, D3).

A9. The shared components: functional contract

Every page is assembled from these (§6.6, §44). This table is what each must *do* and which states it must support. What each *looks like* is **DESIGN**, decided once in Phase 0 and frozen (Part P2), then reused unchanged. The "First page that needs it" column below is historical: with three parallel builders, Phase 0 builds every component that two or more pages use.

Component	Must do	States / variants required	First page that needs it
Button	One primary per screen max	Primary, secondary, tertiary, destructive, icon-only × default, hover, pressed, loading (label → spinner, width held), disabled (only for genuinely unavailable actions, never validation)	Onboarding
Card	Container for content	Static, tappable, expandable, severity-marked (severity visible on the card edge or equivalent)	Today
Expandable card	Collapses to a summary row; expands in place; never navigates; not a history entry	Collapsed, expanded, loading, no-data	Today
Bottom sheet	Slides over current screen; dismiss by drag, backdrop tap, Escape, or close control; pushes a history entry; traps focus; locks scroll beneath	Heights: compact (≤50%, map details), standard (~70%, Learn/Alerts/Act), tall (~90%, Assistant). Internal scroll	Today (Learn)
Score ring	Shows a 0–100 composite, higher = better, colored by severity	Populated, no-data (no arc, the words No data in the centre), partial (visibly incomplete without a caption), entrance sweep (skipped under reduced motion)	Today
Risk bar	Horizontal fill by risk, colored by severity	Populated, no-data, "shown, not scored" (no fill)	HomeGuard
Severity pill	Icon + word + color, always together	All six words	Today
Confidence pill	Quieter than severity; absent at full confidence	partial, 3+ yrs old, not evaluated	HomeGuard
Provenance pill	Same family as confidence	measured, modeled, estimate	Today
Callout	Inline note with an icon	Info, caution, notice tones	Today (incomplete-data notice)
Skeleton	Placeholder sized exactly like what replaces it	Animated, static (reduced motion)	Today
Empty state	Icon + heading + explanation + optional action	With / without action	Journal
Toast	Brief confirmation above the tab bar; one at a time	Plain, and with an Undo action. One duration for every toast: 8 seconds , long enough to reach Undo (§65.3, the blueprint's ~4s/~8s split is Tier 3)	Journal
Tab bar	Five destinations	Active, inactive, pressed	Shell
Header	Title + right controls	Today variant (bell + gear), standard (gear), Settings (back arrow), Map (over map)	Shell
Segmented control	2–3 options, switches content, no navigation	Journal mode, journal severity, letter recipient, owner/renter	Journal
Toggle chip	On/off label; flips instantly	Selected, unselected, disabled	Onboarding (household)
Switch	Settings rows	On, off, disabled	Settings
Accordion row	Settings sections; one open at a time; current value summary when closed	Collapsed, expanded	Settings
Text / numeric input, dropdown, checkbox, slider	Per A7	Default, focused, error, disabled	Onboarding
Offline banner	Under header, pushes content down	Shown / hidden	Shell
Swipe action	Reveals delete/dismiss; action is immediate with an undo toast	Journal entry delete, alert dismiss	Journal

THE BLUEPRINT'S OWN SIMPLIFICATIONS ARE THE DEFAULT (§65.4): where the table above lists more variants than a page needs, build the simpler set unless a page genuinely requires more: **buttons**, three variants (primary, secondary, tertiary); destructive is secondary in the Elevated red, icon-only is tertiary with no visible label (it keeps an accessible name). **Sheet heights**, two (standard ~70%, tall ~90%). The Assistant is tall. Learn, Alerts, Act, Journal day detail and Settings' delete sheet are standard. **Map feature details are the one exception:** sized to their content, at most 50% of the screen, so the selected feature stays visible (the auto-pan that would otherwise fix this is Tier 3). The district panel is standard. **Cards**, mark severity with the pill, not a colored card edge as well ("both is redundant emphasis"). **Skeletons**, two shapes, a card block and a text block, composed as needed. **Motion**, three durations: 150ms fades, 250ms movement, 400ms score ring.

A10. The signature visuals: what's locked vs. what's open

The blueprint's exact pixel geometry for these (§56) is visual and void. What they must *do* is not, each carries meaning, which is why they exist (§6.0).

Visual	Locked function LOCKED	Open DESIGN
Score ring (Today, HomeGuard hero)	0-100, higher is better, same direction on every screen. Fill = score/100. No-data = no arc and the words "No data in the centre. Partial = visibly incomplete, distinguishable without a caption. No "out of 100," no percent sign.	Size, stroke, how "partial" looks, entrance motion
Contribution bar (under each hero)	One stacked bar; segments sized by each factor's share of the total risk, colored by that factor's severity, each labeled (segments under 10% of the width are labeled only in a legend). Tap a segment → scroll to and expand its card. Edge cases: if the total is 0 the bar is hidden; a factor with data but zero risk gets a minimum visible sliver and a legend entry; a factor whose severity is <code>no_data</code> is a hatched "no data" segment, never zero width; if only one segment exists it fills the width.	Thickness, labels, legend treatment
Threshold bar chart (HomeGuard water, map sheets, health card)	One row per contaminant. Scale is always 0 to 2× that contaminant's federal limit, so the limit sits at the same position on every row, this is what makes different contaminants comparable. Values past 2× fill fully with a visible "continues" end and the true multiple in the label; never rescale the axis. Limit marked on every scored row. Sort worst-first by ratio. Unscored substances below a divider, de-emphasized, no limit marker, labeled "shown, not scored." Tap a row → Learn for that contaminant.	Row/bar dimensions, marker style, entrance motion
House diagram (HomeGuard)	Three zones, tap (water), foundation (radon), pipe (lead), each shown in its severity, grey when no data. Each zone is a tap target that scrolls to and expands its card. Schematic, not an illustration. If it turns into an illustration project, replace with three labeled rows carrying the same colors, the page loses character, not meaning.	Everything about how it's drawn
Calendar heat grid (Journal)	One cell per day, seven columns. Cell = that day's worst severity. No-data day visibly different from Good. Future days not drawn. Logged-symptom days marked. Today marked. Tap a day → that day's detail sheet.	Cell size, colors, markers
Seven-day trend (Today expanded cards)	Daily values, today emphasized, severity bands behind the line so a rise visibly enters a worse band. Missing days break the line, never interpolated. The band edges are display-only drawing constants mirrored from <code>lib/severity.js</code> (they position bands, they do not classify): AQI 50, 100, 150, 200; UV 3, 6, 8, 11; pollen 0, 2, 3, 4 on a 0 to 5 scale. Each point's color is the server's word from <code>history[]</code> (for a pollen point on a non-dominant day, the band that its plotted value falls in). Mold has no trend.	Size, curve, labels
Quarterly trend (HomeGuard water)	Worst contaminant's readings across sampling dates, federal limit as a reference line. Only drawn when there are 2+ dated readings.	Everything visual
Comparison bar pair (district panel, map banner)	Two bars on the same scale.	Everything visual

Every chart has a plain-language caption stating its takeaway, units on any axis, missing data as a visible gap (never zero, never interpolated), a screen-reader text equivalent, and colors from the severity scale only, no chart gets its own palette (§6.8). All drawn as inline SVG, no charting library (§35.2).

A11. The design foundation: locked

Decided once, used by every page. The blueprint's own color and type values (\$6.3, \$6.4, \$59) are replaced by these.

Severity colors LOCKED

Word	Label	Light mode	Dark mode	Style	Icon
good	Good	#1e7a3c green	#4cc27a	Solid fill	Circle with check
moderate	Moderate	#b85c00 orange	#f0913a	Solid fill	Circle with i
elevated	Elevated	#c0262d red	#f0605f	Solid fill	Triangle with exclamation
high	High	#8e2a6b red-purple	#d668ad	Solid fill	Octagon with exclamation
severe	Severe	#4f1440 deep purple	#a870f0	Solid fill	Filled octagon
no_data	No data	#111111 black	#e6e6e6 near-white	Outline only, never filled	Circle with question mark

No data is drawn as an outline so it can never be mistaken for Severe (black and deep purple are nearly the same brightness). Pill text on a filled level is white in light mode and navy (#081725) in dark mode. Every color above passes 4.5 : 1 for text in its mode (checked by calculation). Icons have clearly different shapes because severity is always color + icon + word.

Brand and surface colors LOCKED

Token	Light	Dark	Use
Navy 900	#0b1f33	#081725 (page)	Main text in light mode; the base of dark mode
Navy 700	#133b5c	#1d3a50 (border)	Headers, deep accents
Primary, deep teal	#0e5e6f	#2cc4bd	Primary buttons, active tab, links, focus ring
Teal 500	#138a8a	n/a	Secondary accents, never text (fails contrast)
Turquoise	#1fb5b0	n/a	Highlights and fills only, never text
Tint (--halo-tint)	#d9f0ef	#1d4f5c	Selected chips, soft tints. This approved value replaces the earlier light aqua #a8e6e2
Page background	#eef8f7 aqua mist	#081725	Behind cards
Card surface	#ffffff	#0f2436	Cards and sheets
Border	#d3e4e6	#1d3a50	Dividers, card outlines
Body text	#0b1f33	#e8f3f5	All primary text
Secondary text	#4a6275	#9bb4c2	Timestamps, sources, captions

No saturated green anywhere in the brand, so green always means Good. There is no default mode: the phone's setting decides (A8). The token names that carry these values are in P6.

Typefaces LOCKED

Role	Face	Notes
Headings, hero score numbers, page titles	Fraunces (variable)	A soft serif, the app's personality. Has no evenly spaced digits, so an animated score needs a fixed-width container so it doesn't wobble
Everything else: body, labels, buttons, cards, data rows	Instrument Sans	Readings use its tabular digits (font-variant-numeric: tabular-nums) so columns align, verified the font includes them
Codes only	IBM Plex Mono	Water system IDs (NC0190010), certifications (NSF/ANSI 53). Not for readings

All three are free fonts, **self-hosted** with next/font/local (public/fonts/instrument-sans-latin.woff2 as --font-instrument , public/fonts/fraunces-latin.woff2 as --font-fraunces) with a system fallback so text never waits (\$61). IBM Plex Mono is not loaded yet; Phase 0 adds it as --font-plex (P2). All include the accented characters Spanish needs. A proposed starting type scale is in A13; it's finalized in Onboarding's design pass and then used everywhere.

REPLACED, NOT TO BE USED: everything visual in the blueprint (§6.3–6.5, §56–60 pixel values), Vibhav's glassmorphic HomeGuard branch and its `docs/HALO-design-language.md` (Playfair Display + Poppins, dark gradients, glass nav, the plexus ban), the original onboarding prototype's look (aurora and plexus canvases, frosted glass card, "Proxima Soft"/"DM Sans"), and the Geist fonts and Arial body in today's `app/layout.tsx` and `globals.css`. The foundation above replaces all of them.

A12. Priorities, performance, testing, and limits

The blueprint specifies everything; it also says completeness is not priority (§65). This section is how to spend effort. Read it before building.

Tier 1: load-bearing, never cut **LOCKED** (§65.1)

Item	Why
The four universal states on every screen	A screen that silently shows nothing is indistinguishable from a broken one
Error copy distinct from empty copy	"We failed" and "the data doesn't exist yet" need completely different reactions
Severity as color + icon + word	A risk app must not fail a colorblind reader
Missing data never shown as zero or green	The single most dangerous failure, absence presented as safety
Every estimate labeled	The app's credibility rests on not overstating what it knows
The back-button contract (A1)	Without it overlays feel broken on every device
Number and date formatting (A5)	The same value formatted two ways reads as unfinished
The shared component set, built first	Otherwise every page looks like a different app
Bind to the frozen contract (Part D)	A shape change after screens exist blocks everyone
Dark mode	Ignoring the system preference reads as unfinished. Cheap only because every color is a token from day one
Health card print styling (B5)	Its stated purpose is to be handed to a doctor or landlord
Assistant's structural refusal to diagnose (server-side)	A prompt can be argued around live; a code path can't
Threshold bar's fixed 0–2× limit scale (A10)	What makes different contaminants comparable at a glance
"Felt fine today" in Journal (B4)	Without good days the pattern analysis is biased
Journal's minimum data before reporting	Four data points is the difference between credible and misleading
Stored readings shown when a source fails (A2)	An environmental app that goes blank on a weak signal fails when it's wanted

Tier 2: build in this order once Tier 1 is solid **LOCKED** (§65.2)

1. The map's contaminant selector, the most interactive element, the one most likely to hold a judge's attention
2. Household-aware wording (display the server's sentences, A4)
3. The threshold bar chart
4. The Journal calendar heat grid
5. The HomeGuard house diagram
6. The results reveal after onboarding
7. The contribution bar
8. Seven-day trend lines
9. Pull to refresh
10. Spanish (only once a person has reviewed the translation, §68)

Tier 3: cut first, in this order, without hesitation (§65.3)

Assistant button hide/show on scroll (leave it fixed) · map auto-panning above a sheet · the home pin's pulse · score number counting up · chart bars staggered · skeleton shimmer (a static block is fine) · header shadow on scroll · onboarding slide transitions (a cross-fade or nothing is fine) · air-monitor glow · hiding zoom buttons on touch · toast duration varying by content (one duration) · the map time slider (only if the data supports it, §68) · season recap notifications · landscape handling · offline journal queueing (a clear "couldn't save, try again" is acceptable).

Page cut order if time runs short (§26.3): Assistant sheet → Settings → Act → district panel. Cutting Settings means household groups can only be set in onboarding, and renting mode can't be set at all, onboarding has no renting question (B1), so every household gets owner wording. If Settings is cut, keep HomeGuard's **PROPOSED** "Showing actions for renters · Change" control above the action plan (B3) and have it save `renter_mode` through the one `PUT /api/household` save function (all seven groups). Two finished pages beat five half-finished ones; an unfinished page reads as broken rather than scoped.

Performance targets **LOCKED** (§35)

Measure	Target	Maximum
First meaningful content	under 1.5s	3s

Interactive, responding to taps	under 2s	4s
Tab change	under 100ms	300ms
Sheet open	under 250ms	n/a
Cached reading display	under 200ms	500ms
Fresh reading retrieval	under 2s	Stored readings show immediately while fresh data loads; the "couldn't reach the data source" notice appears only when the request's own deadline passes (15 seconds for readings, A2)
Map first render	under 2s	4s
Map layer switch	under 300ms	1s
Total page weight, first load	under 500KB	1MB

Measured on a mid-range phone over a mobile connection. How: charts are inline SVG (no charting library); the map library loads only when the Map tab is first opened; map data is one pre-built file per layer; stored readings render immediately while fresh data loads; images sized for display in a modern format; fonts load with a fallback shown immediately; skeletons occupy exactly the space of what replaces them (§35.2–35.3).

Security and privacy in the client LOCKED (§37)

- No analytics or tracking SDK of any kind, and the privacy statement says so (§37.4).
- No secret in browser code. The only browser-visible values are the Supabase URL and anon key, the optional Turnstile site key, the VAPID public key, and, only if the map uses Mapbox GL JS, a public `pk.` map token restricted to the app's URLs. The server's `MAPBOX_TOKEN` (geocoding) is never exposed.
- The journal note and the assistant question are the two free-text inputs: render them as text, never as HTML.
- Browser storage holds only the session (managed by the auth library), stored Today and HomeGuard readings, cached Learn content and map data, the onboarded flag, device-local accessibility settings, small interface memory (dismissed seasonal prompts, the last-used journal group), and any queued journal entries, nothing a person wouldn't already see on screen.
- Never send a household or profile id as identity; the token is the identity.

Testing: what every page must survive LOCKED (§40)

Test addresses (real, not synthetic)	Must produce
A Concord address	A utility match with PFAS above the federal limit, the primary demo case
A Union County address	A different utility with different readings
A third NC-08 address	A third utility, ideally with no published results (the no-data state)
A rural address	No utility match, the well prompt
An address entered as a well	The testing plan, never a score
An address outside North Carolina	Daily readings work; home readings degrade gracefully (radon <code>out_of_state</code> ; water says HALO's water data covers NC, B3)
A Buncombe County address (Zone 1) and an Iredell County address (Zone 2), in NC but outside NC-08	The radon-elevated paths: a radon action item, the <code>radon</code> volunteer cause, Act's radon help section, Journal's winter radon prompt, the <code>radon_season</code> alert, and a well plan's radon-in-water test

EVERY NC-08 COUNTY IS RADON ZONE 3: in the stored EPA data, every county in and around NC-08 (Cabarrus, Union, Stanly, Anson, Montgomery, Richmond, Moore, Rowan, Mecklenburg, Scotland, Robeson, Hoke) is Zone 3. So for every NC-08 demo address the radon card reads Good · Zone 3, radon never enters the action plan, and the radon prompt, alert, and help section never appear. Test those paths with the Zone 1/2 addresses above; use Zone 3 in NC-08 fixtures. "Zone 2" examples in this document are illustrative.

Cases: no household groups (general wording everywhere) · every group selected · respiratory only (wins the priority) · build years before 1950, 1950–1987, 1988 or later, and blank · renting vs owning (every action changes) · Journal with 0, 4 (below threshold), and 20 entries including illness-flagged days · every map layer and contaminant, including one with no readings anywhere · the Assistant against ten adversarial questions including direct requests for diagnosis · every empty and error state triggered on purpose · both languages on every screen.

Technical: every screen at 375px on a real phone (not a resized desktop window) · both mobile engines (Safari/WebKit and Chrome) · keyboard-only through every screen · a screen reader on Today and HomeGuard at minimum · one data source deliberately blocked · fully offline, then back online · Back with every sheet type open · deep links pasted into a fresh browser.

Live checks and safety: the local environment points at the **production** Supabase project, so destructive calls (`DELETE /api/account` , `DELETE /api/journal?all=true` , `PUT /api/household`) are never run against it from a test. Use the mock adapters, or a separate Supabase project. **Before any demo (§40.4):** run `/api/health` (it needs a session token, call it with `authedFetch` from the running app's console) · load the demo address and confirm results within target · confirm the map data was rebuilt recently · have a fallback if a government service is down.

What HALO deliberately cannot do: each is stated in the interface `LOCKED` (§69)

Limitation	Where the interface says it
It can't tell you what's in your tap water, only what the utility measured at its own sampling points	HomeGuard water card (B3)
It can't measure radon in your home, county zones predict averages	HomeGuard radon card (B3)
It can't detect mold, it estimates favorable conditions from humidity and rain	Today mold card, "estimate" pill (B2)
It can't test private wells, no agency does	Well testing plan (B3), results reveal (B1)
It can't tell you whether your plumbing contains lead, age indicates an era, not a material	HomeGuard lead card (B3)
It can't diagnose anything	Assistant disclaimer and refusal (C2)
It can't prove an exposure caused a symptom	Journal findings disclaimer (B4)
It can't promise its data is current	Every reading carries its date (A5)
It can't cover the whole country equally, water and radon detail is strongest in North Carolina	Results reveal note for non-NC addresses (B1)

What the judges actually see (§65.5)

Whether it works at all, on a real device, without errors · the map, especially the moment a contaminant selection recolors the state · whether numbers are explained in language a person understands · whether it admits what it doesn't know · whether the presenter can explain why it was built this way. They won't notice easing curves or a two-pixel shadow. "If a trade-off ever arises between looking finished and being honest, choose honest."

A13. The design process, and the proposed base visual system

Who designs, and how LOCKED

Codex (in each builder's own account) does the wireframes, the visual design and the code for the pages that builder owns; Yogi supplies reference inspiration and approves each page's look; Part P says who owns which page and which shared components. The "wireframe" is a real, final-looking page at 375px with mock scenarios, reviewed and frozen before the live build. The full procedure, and the rules Codex follows, are in **HALO_Wireframe_to_Build.md**. Per page:

1. **Read the function first.** Part P, the page's section in Part B, every shared part it uses (A1 to A13, C, D). The element table is the checklist: every element, binding, state and copy string.
2. **Study the references Yogi supplies.** Borrow what they do well: hierarchy, spacing rhythm, density, how a card or chart is treated. Don't borrow their colors or fonts (A11 governs), and don't borrow their emptiness if this page holds more data than theirs.
3. **Wireframe** the page as a preview route (`</page>/preview`) with mock scenarios for every state: populated, loading, empty, error, partial or no data, offline, in light and dark, at 375px. Use the shared components as they are; do not restyle them.
4. **Check it against the element table, line by line, independent of the look.** A visual change never drops an element, a state or a string to look cleaner. If a reference's layout can't hold a required element, the layout changes, not the requirement.
5. **Show Yogi screenshots of every state.** Iterate on visuals only. List every PROPOSED item built. When approved, freeze the screenshots.
6. **Build** with the live adapter (Part D), the same screen component, and run the definition of done (E1) plus the zero-difference check against the frozen screenshots.

The quality bar: high-end, not overboard

The blueprint's own bar (\$6.0, \$65.5): clean, deliberate, finished, not a funded consumer product. What reads as high-end is discipline: one type scale, one spacing scale, one color system applied predictably, aligned edges and even margins, text that never overflows, nothing shifting as content loads, correct contrast in both modes, and precise details (consistent icon stroke weight, optical alignment of icons with text, tabular digits in data). Personality comes from the signature elements, one per page, each carrying meaning (\$6.7): Today's time-of-day gradient behind the hero, HomeGuard's house diagram, the Map's full-bleed canvas, Journal's heat grid, Act's editorial cards, Learn's typographic sheet. Buttons, cards, and type stay identical everywhere. "If a visual decision takes more than an hour and doesn't change what a person understands, it isn't worth making."

What makes an interface look generated: avoid all of it

- Gradient washes, glassmorphism and background blur, decorative blobs or particle/plexus canvases (the only gradients are Today's time-of-day hero and Onboarding's Welcome band)
- Drop shadows on every card; colored left-border cards; everything a rounded pill
- Emoji as icons; mixed icon families (lucide-react only, one stroke weight); icons drawn for decoration
- Colors outside the A11 tokens; saturated green anywhere except the Good severity
- Center-aligned body text; title case or all caps; exclamation marks; lorem ipsum or invented numbers
- A fake phone frame or status bar, the app is the real page at the real device width (the built Onboarding already follows this: a 430 px maximum width on the real page)
- Uneven spacing, near-miss alignments, animation on everything

Proposed base visual system PROPOSED

Now built and approved through Onboarding (radius 14, gutter 20, page heading 26, tint, primary button, inputs, chips and progress are all in `app/onboarding.css`). Treat these values as final. Where this table differs from the built CSS, the built CSS wins. It is used on every page unchanged.

Type role	Face	Size / line height	Use
Hero score	Fraunces 600	56 / 60	The composite number on Today and HomeGuard, fixed-width container (Fraunces has no tabular digits)
Display (Welcome)	Handwritten "Welcome" heading (approved, animated pen strokes), then "To Halo" subtitle	56 (<code>--halo-welcome-size</code>)	Onboarding Welcome only. Shown immediately under reduced motion; never delays Get started
Step / page heading	Fraunces 600	26 / 32	Onboarding step questions
Header title	Fraunces 600	22 / 28	Page titles in the header bar
Section heading	Fraunces 600	18 / 24	Section dividers within a page

Card title	Instrument Sans 600	16 / 22	Card headers
Body large	Instrument Sans 500	17 / 24	Supporting sentences, privacy line
Body	Instrument Sans 500 (built: <code>.halo-app</code> sets weight 500)	15 / 22	All explanatory text, the floor ; nothing readable is smaller (\$25.1)
Small	Instrument Sans 500	13 / 18	Timestamps, sources, captions, field labels, error text, never explanations
Pill label	Instrument Sans 600	12 / 16	Severity, confidence, provenance pills only
Button	Instrument Sans 600	16	All button labels

Element	Proposed treatment
Spacing	4px grid: 4, 8, 12, 16, 24, 32, 48. Page gutter 20px
Radius	Cards and inputs 14px · buttons 12px · chips and pills fully rounded · sheets 20px on the top corners
Surfaces	Cards: card color + 1px border, no shadow. The primary button is the one element with a soft shadow (<code>0 2px 8px rgba(11,31,51,.12)</code>)
Primary button	Primary teal fill, white text, 52px tall, full width in forms. Pressed: 1px down. Loading: width held, label replaced by an in-place spinner
Secondary	Transparent, 1.5px primary border, primary text. Destructive = secondary in the Elevated red
Tertiary	Primary text only, no border or fill (e.g. "Skip this"). Icon-only = tertiary, 44×44 target, accessible name
Disabled	40% opacity, no shadow, only for genuinely unavailable actions, never validation (A7)
Text input / dropdown	52px tall, 1px border, 14px radius, card fill; a visible label above the field (Small, secondary text). Focus: 2px primary border plus the keyboard focus ring. Error: Elevated-red border and red error text with an icon beneath, the field fill never turns red
Toggle chip	Unselected: card fill, 1px border, body text. Selected: tint (<code>--halo-tint</code>), 1.5px primary border, and a small check icon, so the state isn't carried by color alone; <code>aria-pressed</code>
Onboarding progress	Four 3px segments with 4px gaps, filled primary / unfilled border color, no numbers shown; screen-reader text "Step 2 of 4"
Severity pill	Icon + word, fully rounded, filled in the level's color with white text (light) or <code>#081725</code> text (dark); No data outline only
Motion	150ms fades, 250ms movement (sheets; onboarding steps as a short slide or a cross-fade), 400ms score ring. Reduced motion: instant, except very short opacity fades

Part B: Pages

B1. Onboarding

Purpose LOCKED

Collect the absolute minimum needed to produce a real result, and nothing more. No account creation, no email, no password, no permission prompt, the anonymous account already exists before this screen renders (§10.1). A household that finishes onboarding lands on real, personalized results; one that abandons it sees nothing. That's why this is the one page that stays deliberately short.

Address and entry

Built and approved (on main, commit c84cbb3, refined in e3f2147). The real entry is `/`, which redirects to `/onboarding?entry=1`; the first-visit rules are in A1. `/onboarding?change=1` starts address replacement (from Settings, HomeGuard's "no county" error). `/onboarding/preview` is a development-only review route with a scenario selector (loading, errors, missing data, wells, springs, outside NC); production hides it unless `HALO_ENABLE_FRONTEND_PREVIEW=1`. Full screen: no header, no tab bar, no assistant button (§51.1).

As built REFERENCE

ONBOARDING IS THE REFERENCE FOR EVERY OTHER PAGE. Study these files before building anything: `app/OnboardingScreen.tsx` (the screen takes an `api` prop: the live adapter or a mock), `app/onboarding/page.tsx`, `app/onboarding/preview/`, `app/onboarding.css` (tokens and foundation), `lib/frontend/{api,copy,mock,onboarding,storage,types,useHaloData}.ts`, `components/ui/{Button,DataState,ErrorCallout,FieldError,SeverityPill}.tsx`, `test/ui/` (Playwright), and `docs/onboarding/` (README, preservation, integration, requirements-checklist, verification, 179 approved screenshots and a hash manifest). Verified on main: 354 of 354 node tests pass, and typecheck, lint and build succeed. Live-provider and real-phone acceptance are still to do (Vibhav's wrap-up list). The pattern each later page repeats: a screen component that takes an API object, a live adapter, a mock adapter with named scenarios, a preview route, a copy catalog, immutable approved screenshots.

WHAT ONBOARDING ALREADY DOES THAT PAGES RELY ON. Session bootstrap and its two error messages; the GPS path (sends "lng,lat", no fallback address); a fresh `request_id` (UUID) on every address submission so a timed-out write can be confirmed through the profile's `onboard_request_id`; bare five-digit ZIPs are constrained to the United States in the geocoder (without it 28025 resolves to Madrid); the identity-scoped storage and the `halo:identity-changed` event; `withDeadline` (8 seconds, 15 for onboard, daily-score and home-guard); after a location change the client clears old readings and calls `/api/daily-score?fresh=1`. The Today destination is a labeled stub (`app/today/page.tsx`) that renders `ProfileRecheck` for `?verify=1`; Today's owner replaces it and must keep that behavior.

Two follow-ups before Settings is built (Part P7): in change mode, Back on the address step should return to `/settings`, and the Household step (shown prefilled) must be described accurately in Settings.

Structure LOCKED

Four full-screen steps. A slim four-part progress indicator at the top (not shown on Welcome). A back arrow from step 2 onward returns to the previous step with every entered value intact. Each step is a history entry. Each step: the question, any supporting line, the input, then the primary button at the bottom (§10.2, §51).

Data calls, in order BIND

When	Call	Sends	Why here
Page load	<code>ensureAnonSession()</code>	n/a	Already runs app-wide; onboarding surfaces its failure
Step 2 "Continue"	<code>POST /api/onboard</code>	<code>{ address, request_id }</code> (a new UUID each submission)	So address problems surface on the address step, where the blueprint puts them (§10.4)
Step 3 "Continue" (not on Skip)	<code>PUT /api/household</code>	all seven booleans, every time	Saves composition. Skip sends nothing, leaving <code>household_set: false</code>
Step 4 "See my results"	<code>PATCH /api/profile</code>	<code>{ water_source, home_year }</code>	Saves the home details without re-sending the address, the address was used once at step 2 and never kept
Results reveal	In parallel: <code>GET /api/daily-score</code> and <code>GET /api/home-guard</code>	No parameters, both use what onboarding just stored. The onboard response's <code>state</code> decides the outside-NC note	Fills the four reveal lines and warms the cache for Today and HomeGuard

TRAP: `PUT /api/household` is a full replace: any of the seven not sent become `false`. Always send all seven, as real booleans.

Step 1: Welcome LOCKED

Element	Content	Notes
Mark	HALO logo, white, on the navy band	<code>public/halo-logo-mark.png</code> used as a mask (A5). The blue tile and <code>finallogo.png</code> are superseded
Headings	A handwritten " <i>Welcome</i> " that writes itself along the pen strokes, then a smaller, lighter " <i>To Halo</i> " subtitle that fades in beneath it. Both show immediately under reduced motion; never delay Get started	Approved; supersedes a HALO-only heading
Sentence	" <i>See what's in the air, water, and ground around your home, and what to do about it.</i> " then " <i>No account needed.</i> "	\$10.3, punctuation per A5. Built in <code>copy.welcome</code>
Primary button	<i>Get started</i>	→ step 2
Supporting line	" <i>Takes about a minute.</i> "	Below the button
Session failure	" <i>Verification could not be completed. Check your connection and try again.</i> " or " <i>Could not start a session. Please reload and try again.</i> "	From <code>ensureAnonSession</code> <code>error</code> <code>.code</code> ; Get started disabled until a session exists

Step 2: Address LOCKED

Element	Content / behavior	Binding / notes
Heading	" <i>Where do you live?</i> "	
Text field	Label "Address or ZIP code" visible; placeholder "Street address or ZIP code"; device address autofill; 16px+ text	Validation on blur: 5+ characters else "Enter an address or ZIP code."
Secondary button	<i>Use my location</i>	<code>navigator.geolocation</code> → field shows "Using your current location"; sends <code>"lng,lat"</code> as <code>address</code> . Unsupported: "This device can't share its location. Please type your address." Denied/failed: "Couldn't get your location. Please type your address instead." Typing in the field discards the coordinates
Privacy line	" <i>We use your address once to find your local data, then throw it away. We keep only your approximate coordinates, your county, and which water utility serves you. We never keep the street address itself.</i> "	Directly under the field, visible before they type (\$10.4)
Primary button	<i>Continue</i> ; while processing: loading state + supporting text "Finding your local data."	Calls <code>POST /api/onboard { address }</code>

Outcomes, by response BIND

Response	Show	Next
200	n/a	Step 3. Keep <code>service_area_status</code> , <code>county</code> for later
404	"We couldn't find that address. Try adding your city or ZIP code." Field keeps its contents	Stay
429 (either message)	"You've made a lot of requests. Please try again later." No blame, no detail	Stay
500 / network	"Something went wrong on our end, not yours. Try again in a moment." + Retry	Stay
401 / 503	Handled by the auth layer (A2)	n/a

OUTSIDE NORTH CAROLINA: a non-NC address proceeds normally (\$10.4). The onboard response carries `state`; when it isn't `NC`, the results reveal adds a note that daily air, UV, and pollen readings work everywhere while water and radon detail is strongest in North Carolina. **PROPOSED** wording: "*Daily air, UV, and pollen readings work anywhere. Water and radon detail is strongest in North Carolina.*"

NOTE: `service_area_status: "outside_known_area"` is a normal, common result in NC-08 (no utility boundary contains the point), not an error. `lookup_failed` means the boundary lookup itself failed; onboarding still continues. Either way, step 4's water-source question matters more for these households.

Step 3: Household LOCKED

Element	Content / behavior	Binding
Heading	"Who lives here?"	
Supporting text	"This lets HALO tell you what today's readings mean for the people in your home. We never ask for ages, birthdays, or names, just which groups apply."	
Seven toggle chips, wrapping, this order, all start unselected	Toddler · Child · Teen · Adult · Senior (65+) · Someone pregnant · Someone with asthma or a breathing condition	<code>has_toddler</code> , <code>has_child</code> , <code>has_teen</code> , <code>has_adult</code> , <code>has_senior</code> , <code>has_pregnant</code> , <code>has_respiratory</code>
Primary	<i>Continue</i>	<code>PUT /api/household</code> with all seven. On failure: keep selections, show the generic error, allow Continue again
Skip	<i>Skip this</i> , same row as Continue, equal visual weight so skipping never feels like failing	No call. General wording everywhere; never prompted again; editable in Settings

Step 4: Your home LOCKED

Element	Content / behavior	Binding
Heading	"A little about your home."	
Build year	Label "Year built", optional, numeric keypad, digits only. Under it: "Homes built before 1988 may have lead solder in their plumbing. If you're not sure, leave it blank. We'll tell you we don't know rather than guess."	<code>home_year</code> : number or <code>null</code> . Validation per A7 (1700-current year)
Water source	Label "Where does your water come from?"; options: City or town water · Private well · Spring · Other · Not sure. Required	<code>water_source</code> . The server normalizes these exact labels: City or town water → <code>utility</code> , Private well → <code>well</code> , Spring → <code>spring</code> , Other and Not sure → <code>other</code>
Primary	<i>See my results</i>	<code>PATCH /api/profile { water_source, home_year }</code> , set the local onboarded flag, go to the results reveal. On failure: keep both values, show the generic error, allow retry

This one choice is the largest branch in the application: well or spring sends all of HomeGuard to a testing plan instead of measurements (§10.6).

The results reveal LOCKED

A transitional screen after step 4, before Today (§33, §51.5): the logo, then four lines appearing in sequence, each starting with a small working indicator that turns into a check as its result resolves, with the result on the right. Then a primary button *See my results* → `/today` . Under reduced motion all four appear at once. Purpose: makes the work visible, introduces the four things HALO tracks, and turns a redirect into proof the app did something real.

The four line labels and the example results come from §33. Result strings not in the blueprint are marked PROPOSED .

Line	Result shown	Binding
<i>Finding your water utility...</i>	Utility name (through <code>formatUtilityName</code> , A5), or PROPOSED <i>No utility matched your address · Private well (no utility) / Spring (no utility)</i>	<code>water.pws_name</code> ; <code>water.status = no_pwsid_available / private_well (+ water.source_type)</code> . For <code>no_data_yet</code> and <code>lookup_failed</code> there's no name in the response, show PROPOSED <i>Found your water system ([pwsid])</i> from the onboard response's <code>pwsid</code>
<i>Checking federal testing results...</i>	<i>PFOS found above the limit (§33) · No published water results yet for your utility. (§33) ·</i> PROPOSED <i>: Nothing found above federal limits · Wells aren't tested by any agency · Found compounds with no federal limit · Couldn't check right now</i>	Measured: the worst entry in <code>water.scored_contaminants</code> with <code>exceeds_limit: true</code> (the server decides, the frontend never compares a value to a limit); none → "Nothing found above federal limits." Other states by <code>water.status: no_data_yet , private_well , no_pwsid_available</code> ("No utility to check"), <code>detected_unregulated</code> , <code>lookup_failed</code>
<i>Looking up radon for [County]...</i>	Zone 2 (§33's example, every NC-08 county is actually Zone 3, A12) · or PROPOSED <i>Not available for this county · for out-of-state addresses</i> PROPOSED <i>Covers North Carolina only</i>	<code>radon.zone</code> ; <code>radon.error</code> present → not available; <code>radon.out_of_state</code> → the out-of-state line
<i>Getting today's air quality...</i>	Severity label, e.g. <i>Good</i>	<code>air.severity</code> → label

If a lookup produced nothing, the line says so plainly rather than disappearing (§33). For an address outside North Carolina

(`state` isn't `NC`), a `no_data_yet` water result reads **PROPOSED** *HALO's water testing data covers North Carolina utilities* instead of the "not published yet" line, which would be false there. If a call fails outright, its line reads *Couldn't check right now* and the button still appears, Today and HomeGuard have their own error states.

VOICE CHECK: "Nothing found above federal limits" is a statement about a measurement, which the rules allow. Never "Your water is safe" (§30.3).

ROUTING: the local `halo.onboarded` flag is set only after step 4 succeeds. The server's `onboarding_complete` turns true at the same moment (coordinates and water source both stored); `onboarded` turns true earlier, at step 2, which is why routing never uses it (A1).

Proposed content **PROPOSED**

None beyond the outside-NC note. This page is deliberately minimal (see the note in section 0).

Visual decisions: made **DECIDED**

Onboarding's design pass is done and approved in light and dark. Everything below is now the foundation for every page.

Decision	Outcome
Palette, typefaces, tokens	A11 and P6. Appearance follows the phone; every color is a <code>light-dark()</code> token
Overall tone	"Something in between," quiet and immersive: the page background everywhere, a brand moment on Welcome only. Steps 2 to 4 and the reveal stay light and calm
Welcome band	A navy band (<code>-halo-band</code> , 220px) behind the white logo mark, on Welcome only. This band and Today's time-of-day hero are the only gradients or bands in the app
Base components	Built: buttons (primary, secondary, tertiary), inputs and dropdown, toggle chips, progress segments, severity pill, error callout, field error, type scale, spacing (gutter 20, radius 14), motion. These are the values in A13, and the built CSS wins if they ever differ
Step transition and reveal	As built and approved. Reduced motion: instant, all four reveal lines at once

REMOVED, AND MUST NOT COME BACK. The fixed 390×844 phone shell, the aurora and plexus canvases, the frosted glass card, the "Proxima Soft" and "DM Sans" fonts, the old tagline "Know what surrounds your family.", the non-link "Terms of Service" footer, and the single-screen form.

B2. Today

Purpose LOCKED

Answer, in under two seconds of looking: *is it a good day to be outside, and is there anything I should do differently today?* This is the page a household opens habitually, legible at a glance, and it never requires scrolling to deliver its headline (§11.1).

Address and entry

`/today` : default landing for onboarded households, Tab 1. `/today?learn=air|uv|pollen|mold` opens that Learn sheet over it (any other value is ignored; on a deep link Today rebuilds the reading and context from its loaded data). **Today replaces the stub `app/today/page.tsx` and must keep its behavior:** render `<ProfileRecheck />` when `?verify=1`, which keeps its redirect to `/onboarding?entry=1` for an incomplete profile (A1). The shell renders the header, bell and assistant button; Today renders content only (Part P2).

Data calls BIND

Call	Params	Feeds
GET <code>/api/profile</code> through the shared <code>useProfile()</code> (Part P3)	None	<code>profile.county</code> , <code>household</code> , <code>profile.locale</code>
GET <code>/api/daily-score</code>	none, the server uses the household's stored location. <code>?fresh=1</code> on pull to refresh	Hero, four cards
GET <code>/api/history?days=7</code>	None	Trend lines (today plus the six days before) and yesterday's row for the proposed change line. Use <code>history.to</code> as "today" (A5)

Today makes no call to `/api/alerts` : the shell owns the alerts fetch and the badge (Part P3).

DON'T SEND COORDINATES: call `/api/daily-score` without `lat / lng` so it uses the household's stored location. Readings for any other coordinates (a GPS fix, a rounded value) are fetched but never saved to the household's history or cache.

Layout, top to bottom LOCKED

- Header (rendered by the shell): *Today* · alerts bell with unread badge · gear
- Hero block (roughly the top third, no scroll needed): score ring → headline → notices → location/freshness line → contribution bar, on the time-of-day background
- Four reading cards, fixed order: Air quality, UV index, Pollen, Mold risk. Order never changes, habitual users navigate by position (§11.4)
- Floating assistant button (rendered by the shell)

Hero LOCKED

Element	Content	Binding	States
Score ring	The composite, 0–100, higher is better	<code>score.display_score</code> (never <code>score.score</code> , which is a two-decimal float; never <code>score.risk</code> , which runs the other way)	Partial when <code>score.is_partial</code> . No-data when <code>display_score</code> is null, and see the guard below
Ring color / headline level	n/a	<code>score.severity</code>	No data when null
Headline	By level: Good → "Good day to be outside" · Moderate → "A mostly good day, with one thing to watch" · Elevated → "Worth taking some care outside today" · High or Severe → "A rough day out there. See what's driving it." · No data → "We couldn't get readings for your area today"	From the level above	
Incomplete notice	"Based on 3 of 4 readings. Pollen data isn't available right now." Two or three missing: "Based on 2 of 4 readings. Pollen and UV data aren't available right now." (join with commas and "and"; singular "isn't", plural "aren't"; the count	Count = <code>score.included_inputs.length</code> ; names from <code>score.missing_inputs</code> (air → "air quality", uv → "UV", pollen → "pollen", mold → "mold")	Only when <code>is_partial</code> and <code>display_score</code> is not null (with nothing available it would read "0 of

	is a data string, so it keeps its numeral). The score still shows; it's never presented as complete when it isn't		4")
Estimate notice	"Includes an estimated value, not only measurements."	<code>score.includes_proxy === true</code>	Nearly always true (mold is always an estimate)
Location line	"Union County · updated 14 minutes ago"	<code>profile.county + retrieved_at</code> as relative time (A5)	Manual refresh that returns cached data still updates this line
Contribution bar	What's driving today's score, without opening a card	Segment size = each factor's share of the total of <code>score.inputs.air/uv/pollen/mold</code> (these are risks, higher = worse); segment color = that factor's <code>.severity</code> . Factors with <code>null</code> input are absent. Labels: Air quality, UV, Pollen, Mold. Edge cases per A10 (total 0 hides the bar; a zero-risk factor with data gets a sliver and a legend entry; segments under 10% are labeled only in the legend; mold's risk is capped at 30 by the server, which is correct)	Tap segment → scroll to + expand that card
Time-of-day background	Changes with the device clock. Fixed local-device-clock ranges (PROPOSED): 05:00 to 09:59 dawn, 10:00 to 16:59 midday, 17:00 to 19:59 dusk, 20:00 to 04:59 night. Light and dark mode change the gradient stops. Atmospheric only; carries no information; text contrast verified in all four looks in both modes (Phase 0 adds the stop tokens, P6)	Device clock (the one place the device clock is right: it is decoration)	Concept LOCKED (§6.7, §11.3, Today's signature element); the color stops are DESIGN , drawn from the A11 tokens, with text contrast verified in every state

WHEN EVERY SOURCE FAILS: the server returns `display_score: null` with every input in `missing_inputs` and `reason` set. The hero shows its No data state and the headline "We couldn't get readings for your area today", never a number built from nothing, and the incomplete notice is hidden. Today's `isEmpty(data)` is `display_score === null` and every card's severity is `no_data`.

The four reading cards **LOCKED**

Collapsed (every card): severity icon, card name, measured value (tabular digits, A11) with unit, severity pill, any provenance/confidence pills, the one-sentence household-aware explanation beneath, and a chevron. **Expanded:** the detail below, and a "What does this mean?" link that opens Learn for that topic.

Card	Collapsed value	Pills	Sentence	Expanded contents
Air quality	<code>air.aqi</code> , whole number, unit "AQI". Null → No data	<code>air.severity</code> ; <i>measured</i> if <code>is_measured</code> true, <i>modeled</i> if false, none if null	<code>air.sentence</code>	7-day trend (history <code>air.words + values.aqi</code>) · provenance line: true → "Measured at a monitoring station near you"; false → "Modeled estimate. The nearest monitoring station is more than 25 miles away."; null → PROPOSED "The source of this reading wasn't recorded." · static note PROPOSED : "The air quality index reflects the single worst pollutant, not an average." · the dominant pollutant with a one-line explanation, held by the frontend and hidden when null (<code>air.dominant_pollutant</code>), PROPOSED strings: <code>"PM2.5"</code> → "PM2.5: fine particles from smoke, traffic and industry that can reach deep into the lungs."; <code>"PM10"</code> → "PM10: coarser dust and particles."; <code>"O3"</code> → "Ozone: a gas formed when sunlight reacts with vehicle and industrial emissions. It is highest on hot afternoons." · Learn: <code>topic=air</code> with its context (C1)
UV index	<code>uv.index</code> , one decimal. Null → No data	<code>uv.severity</code>	<code>uv.sentence</code>	7-day trend · the hours UV is elevated, e.g. "UV is 3 or higher from 11am to 3pm. Highest today: 7.4." The window is the first hour at or above UV 3 to the end of the last such hour, and <code>max</code> is the day's highest value (<code>uv.peak_window.start</code> , <code>.end</code> , <code>.max</code>). Format <code>"HH:MM"</code> as "11am" (drop ":00"); an end of <code>"00:00"</code> reads "midnight". When null, hide the row (it means UV never reached 3 or data is missing; do not claim "UV stays low") · static note PROPOSED : "UV passes through cloud cover, so a cloudy day can still burn." · Learn: <code>topic=uv</code> with its context (C1)
Pollen	The worst category named:	<code>pollen.severity</code> (severity of the	<code>pollen.sentence</code>	All three categories: <code>pollen.tree</code> , <code>.grass</code> , <code>.weed</code> each with its own value and severity pill

	pollen.dominant + its value, e.g. <i>Grass</i> 3. All null → No data	dominant category)		(pollen.categories.tree / grass / weed → value , severity) · 7-day trend of the dominant category · seasonal note by month, static, NC (PROPOSED , content owner to confirm): February to April " <i>Tree pollen is usually highest now.</i> "; May to July " <i>Grass pollen is usually highest now.</i> "; August to October " <i>Ragweed pollen is usually highest now.</i> "; November to January " <i>Pollen is usually low this time of year.</i> " (device month; hidden when profile.state is a two-letter code other than NC) · Learn: topic=pollen with its context (C1)
Mold risk	No numeric value exists and the value slot stays empty: the estimate pill and the severity pill carry the card. Do not display mold.risk ("low", "moderate", "high"): a "high" label beside an Elevated pill would contradict itself	mold.severity + estimate always	mold.sentence	Always, without exception: " <i>This is an estimate based on humidity and rainfall forecasts, not a sensor in your home.</i> " · the humidity and chance of rain the estimate is built from (mold.basis.humidity_pct , .precip_pct) · static PROPOSED : " <i>Humid, rainy stretches raise the estimate. Dry, breezy days lower it.</i> " · Learn: topic=mold with its context (C1)

NEVER BIND status: each reading object also carries a status field (unhealthy , very_high , low , none ...) from older code with different thresholds. It disagrees with severity for many real values. Only severity exists as far as the UI is concerned.

MOLD NEVER READS HIGHER THAN ELEVATED: the server caps it on purpose because it's an estimate. When the weather forecast couldn't be retrieved, mold is no_data with risk: null , not a quiet "low."

Page-level states

State	Shows
First load, nothing stored	Skeleton: ring-shaped placeholder, headline bar, location bar, four card-shaped blocks, same sizes as the real content (§52)
Stored readings exist	Show them immediately with their original time; fetch fresh in the background
Fresh fetch fails, stored exists	Keep stored + " <i>Showing your last readings from [time]. We couldn't reach the data source.</i> "
Request fails, nothing stored	Error state (A2): " <i>Something went wrong on our end, not yours. Try again in a moment.</i> " + Retry
Request succeeds but every reading is missing (display_score null)	The page-level empty state: " <i>We couldn't get readings for your area right now. This is usually temporary. Pull down to try again.</i> " plus each card's own No data sentence. Error and empty copy stay different (§24)
One reading missing	That card shows No data and its server sentence (e.g. " <i>We couldn't get a UV reading for your area right now.</i> "); the other three render normally; the hero shows the incomplete notice
Trend with under 2 days of history	" <i>Your trend will appear here after a few days of readings.</i> "
Offline	Banner + stored readings

Interactions **LOCKED**

- Pull down to refresh (disabled mid-request) → refresh({ fresh: true }) , which sends /api/daily-score?fresh=1 , bypassing the server's one-hour cache. The server allows one provider refresh per household every 5 minutes (a plain call that misses the cache is limited too); a repeat inside that window returns 429 { retry_after_seconds: 300 } : keep the current readings on screen and show the short toast from A2, never the error state. Show stored readings immediately; the "couldn't reach the data source" notice appears only after the 15 second deadline (A2).
- A 400 "*No location yet...*" means there's no stored location, route to onboarding (A1).
- Auto-refresh on returning to the app after 15+ minutes away.
- Tap a card → expand/collapse in place (not a history entry). Tap a contribution-bar segment → scroll to and expand that card.
- Bell → Alerts sheet, gear → Settings, assistant button → Assistant with page=today : all three belong to the shell (Part P2). The badge shows the unread count and clears as rows are read, not when the sheet opens (decided in P3).

Proposed content **PROPOSED**

Addition	From	Why
Change since yesterday under the ring, e.g. " <i>6 lower than yesterday</i> "	history : yesterday's score vs today's display_score . Show it only when yesterday's row has no no_data factor and today's score.is_partial is false (a new household's backfilled "yesterday" is a partial, modeled score, and comparing it with a	Answers "is today worse than usual" without opening anything

	full one would be wrong). Otherwise hidden. Ask 4 in P9 makes this exact	
A "worded for your household" line, e.g. <i>"Readings explained for: a toddler, someone with asthma · Edit"</i>	<code>household</code> from profile; hidden when no household boolean is true (<code>household_set</code> alone is not enough, B6)	Makes the personalization visible, a judge sees immediately that the sentences are tailored, and the household can fix it if wrong
A week summary inside each expanded card, e.g. <i>"Air quality was Elevated or worse on 2 of the last 7 days."</i> (the same sentence is the chart's caption; the screen-reader text lists each day and its value)	<code>history</code> severity words, counted	Turns the trend line into a sentence (every chart needs a takeaway caption anyway, \$6.8)
Provenance explained once in the air card: measured vs modeled, and that roughly half the district has no monitor within 25 miles	Static, from \$41.2	This disclosure is unusual for a consumer app and worth making visible (\$11.4)

Visual decisions to make DESIGN

The shared pieces (score ring, contribution bar, pills, expandable card, callouts, skeletons, the shell) are decided in Phase 0 and frozen (Part P2). Today's own: the seven-day trend line (`TrendLine` , also used by Journal if needed), pull to refresh, the time-of-day background's color stops, and the hero's composition. Static strings marked PROPOSED above are built and listed for Yogi to keep or cut.

B3. HomeGuard

Purpose LOCKED

Answer *what is true about this specific building, and what should we do about it?* These facts change over years, not hours, so the page is read carefully rather than glanced at, it can carry more depth and more text than Today (\$12.1). It should read as a building inspection, not a weather report.

Address and entry

`/home`, Tab 2. `/home?risk=water|radon|lead` opens with that card expanded; `/home?learn=pfas|radon|lead` opens that Learn sheet. One card is open at a time; expanding uses `router.replace` to keep `?risk=` in step (never a history entry); an unknown value is ignored. The shell renders the header and assistant button.

Data calls BIND

Call	Params	Feeds
GET <code>/api/profile</code> through <code>useProfile()</code>	None	<code>pwsid</code> (shown with the utility name), <code>home_year</code> , <code>renter_mode</code> , <code>state</code> , <code>water_source</code> , <code>locale</code>
GET <code>/api/home-guard</code>	none, the server uses the stored county, utility, water source, and build year	Everything on the page

DON'T PASS THE HOME'S DETAILS: call `/api/home-guard` with no parameters so it always reflects what's stored. (The endpoint also accepts `county` / `pwsid` / `water_source` / `home_year` overrides; no page uses them.) Store the last response locally so it shows instantly on the next open (A2).

Response shape at a glance BIND

```
{ success, assembled_at, radon, water, score, lead, action_plan, breakdown }
water → five shapes, discriminate on water.status (table below)
score → four shapes: private well (no score) · neither input available · normal ·
    PARTIAL (one input only: is_partial true, missing_inputs ['water'] or ['radon'],
    display_score NOT null; happens when water is no_data_yet, lookup_failed,
    no_pwsid_available, or radon is out_of_state or errored)
lead → object for public water; null for well/spring
action_plan → [] for well/spring
breakdown → [water box, radon box, lead box (public only), one box per excluded substance (lithium)]
```

Layout, top to bottom LOCKED

- Header (rendered by the shell): *Your home* · gear
- Hero: score ring → headline naming the dominant contributor → basis line → contribution bar (water vs radon)
- House diagram
- Three risk cards, fixed order: Drinking water, Radon, Lead
- Section heading *Your Action Plan* → numbered items (or the testing plan's summary for wells)
- Floating assistant button (rendered by the shell)

Hero LOCKED

Element	Content	Binding / rule
Score ring	Home score 0-100, higher is better	<code>score.display_score</code> . Ring color: <code>score.severity</code> , water and radon only, never lead (A3). Null in two cases: private well/spring (<code>score.reason</code> : "No score can be calculated for a private well or spring...") and no inputs at all. Show the no-data ring with the reason text beneath, a missing score here is an expected state, not an error. Partial when <code>score.is_partial</code>
Floor	n/a	The score never goes below 6 (<code>score.floored</code>). A 6 means "as bad as it gets," not "6% good." No special copy needed; don't treat it as near-zero
Cap notice	When the water result couldn't be fully evaluated, the score is capped (70 or 85) and says why	Show <code>score.reason</code> when non-null for partial or unscoreable coverage (through <code>displayText</code>). Gate any wording that says "capped" on <code>score.capped === true</code> (the server writes "capped" even when no cap applied; ask 8 in P9). <code>score.ceiling</code> and <code>score.capped</code> are otherwise unused
Headline	Names the dominant contributor, e.g. "Water is driving most of this" /	Name a driver only if that factor's <code>breakdown.severity</code> is Elevated or worse , choosing the larger of <code>score.water_risk</code> and <code>score.radon_risk</code> (two server numbers). Otherwise: <code>action_plan</code> non-empty → PROPOSED "A few things here are worth doing."; <code>action_plan</code> empty and the score is not null → "Nothing here needs urgent attention."; well or spring (score null) → PROPOSED "Your well water hasn't been tested."; no score and not a well →

	"Radon is driving most of this"; or "Nothing here needs urgent attention."	PROPOSED "We couldn't check your home right now." (Naming the larger raw risk alone would call a Good radon zone "the driver" when lead is the only real item.)
Basis line	"Based on EPA testing of your water utility and county radon data."	Well/spring variant PROPOSED : "Based on county radon data. No agency has tested your well water." Partial variant (<code>score.is_partial</code>) PROPOSED , replaces the basis line: "Based on 1 of 2 readings. Water data isn't available right now." (or "Radon data")
Contribution bar	Water vs radon share of the home risk	Sizes from <code>score.water_risk</code> , <code>score.radon_risk</code> , only for a box whose severity is not <code>no_data</code> (unscorable water has <code>water_risk</code> 0, not null). A <code>no_data</code> box is a hatched "no data" segment, never zero width. Colors from each box's <code>severity</code> . Lead is never in this bar; it isn't scored. Tap → scroll to that card. Edge cases per A10. Bind cards by <code>breakdown[].key</code> , never by index; the lithium box (its <code>key</code> is the lowercase name) belongs to the water card's Excluded group, not a fourth card

House diagram **LOCKED**

Three zones, tap (water), foundation (radon), pipe (lead), each in its severity, grey with no data. Each zone taps to scroll to and expand its card. Purpose: shows three unrelated readings as facts about one building, and visually separates this page from Today (§12.2b). Zone severity: water and radon from their `breakdown[].severity` ; lead from `lead.level` . Well households: lead zone grey with the lead card explaining why (below).

Card 1: Drinking water LOCKED

Collapsed: name, risk bar, severity pill, confidence pill, and one line naming the driving fact, by status:

<code>water.status</code>	One line (PROPOSED where marked)	Confidence pill
measured	"[worst scored contaminant] [value] ppt · [ratio]× the limit" from <code>scored_contaminants</code> sorted by ratio; when there are none: the <code>no_detections</code> sentence below	Shown, from the water box <code>confidence</code>
<code>detected_unregulated</code>	"Compounds detected, no federal limit"	Shown
<code>private_well</code>	"Not tested by any agency. See your testing plan."	Suppressed
<code>no_data_yet</code>	"No published results yet"	Suppressed
<code>lookup_failed</code>	"Couldn't check right now"	Suppressed
<code>no_pwsid_available</code>	"No utility matched"	Suppressed

Risk bar state keys on the box severity, not on `risk`: no-data when the box severity is `no_data` (its `risk` is 0, not null, and a zero-fill bar would look empty-good); "shown, not scored" (no fill) when `contributes_to_score` is false; otherwise fill = `risk` (0 to 100).

Expanded, the deepest single view in the app (§12.3). Contents depend entirely on `water.status`:

<code>water.status</code>	What it means	What the card shows
anything else (e.g. <code>measured</code>)	Real federal testing results exist	The full measured view, next table
<code>detected_unregulated</code>	Compounds were detected but none has a federal limit or benchmark	The measured view, but the threshold chart has no scored rows. Lead with <code>water.message</code> (ends "Detected is not the same as safe.") and the detected names from <code>water.detected_unregulated</code> . Score is capped, show <code>score.reason</code>
<code>private_well</code>	No agency has ever tested this water	The testing plan instead of any score (below)
<code>no_data_yet</code>	Valid utility, no published results	Neutral, not a warning: " <i>Testing results for your utility haven't been published yet. This is normal. Federal testing is still rolling out nationally.</i> " (blueprint copy, not the server's message, which names the dataset). Show the utility's ID (<code>profile.pwsid</code>) so the household can check it's theirs; there's no name in this response. Outside North Carolina means <code>profile.state</code> is a two-letter code other than <code>NC</code> (use <code>isOutsideNC()</code> ; a null or missing state counts as NC, as the server does). HALO's water data covers NC utilities only, so this copy would be false there. Show PROPOSED " <i>HALO's water testing data covers North Carolina utilities.</i> "
<code>lookup_failed</code>	Something broke on our side	" <i>This is a problem on our end, not a finding about your water.</i> " + Retry. Radon and lead still render. Never the same wording as <code>no_data_yet</code> : one means wait, the other means try again (§12.4)
<code>no_pwsid_available</code>	Address outside every mapped utility boundary	" <i>We couldn't match your address to a water utility. Are you on a private well?</i> " + choices (PROPOSED labels) <i>Private well · Spring · No, city or town water.</i> Well/Spring → save it with <code>PATCH /api/profile { water_source }</code> , then reload HomeGuard, which now shows the testing plan. "No" → no change; the card keeps the prompt's explanation

NEVER RENDER THESE SERVER FIELDS. `breakdown[].note` (for radon it can be a raw error such as "County 'X' not found in North Carolina radon dataset."), `water.message` for `no_data_yet` , `lookup_failed` and `no_pwsid_available` (they name datasets), `water.hint` , `water.next_step` , and any `radon.error` text. Only `water.message` for `detected_unregulated` and `private_well` , and `score.reason` , are displayable (through `displayText`). A `water_source` of `other` is treated as public water (utility data if a `pwsid` exists); PROPOSED note under the utility name: "*You told HALO you're not sure where your water comes from. Change this in Settings.*"

Measured sub-states BIND

The measured view depends on `water.coverage` : `complete` , `partial` , `unscoreable` , `no_detections` , `excluded_only` . Many utilities have no stored detections (the loaded EPA data kept only detections), so `contaminants` is empty.

	Card shows
complete , partial	The full measured view (next table)
no_detections	PROPOSED "No compounds were reported above detection levels in your utility's federal testing." Never "safe" (non-detects aren't stored). No chart, no trend. Dates show "No data"
excluded_only	The excluded group (lithium) plus the same sentence
unscorable	The <code>detected_unregulated</code> treatment above

The chart and the quarterly trend are hidden whenever `scored_contaminants` is empty. Freshness line under the hero (**PROPOSED**): "Checked 14 minutes ago" from `assembled_at` . Water data age comes from the sample dates, not from this line.

Measured view, in order **BIND**

Element	Content	Binding
Utility	Name and ID, so the household can check it's the right one	<code>water.pws_name</code> (through <code>formatUtilityName</code> , A5) + <code>profile.pwsid</code> (Plex Mono)
Threshold bar chart	One row per scored contaminant, worst first; unscored below a divider (A10)	Scored rows: <code>water.scored_contaminants[]</code> → <code>contaminant</code> (through <code>contaminantLabel</code>), <code>value_ppt</code> , <code>limit_ppt</code> ; <code>ratio = value/limit</code> ; sort by ratio. Row color per A3 (interim rule until ask 1 in P9). Below the divider: <code>water.excluded_from_score[]</code> (lithium, shown in µg/L, A5) and <code>water.detected_unregulated_latest[]</code> → <code>contaminant</code> , <code>value_ppt</code> , <code>date_iso</code>
Detail rows	Per scored contaminant: name, value with units, the limit, the ratio, date sampled. Worst first	<code>value_ppt</code> , <code>limit_ppt</code> , <code>ratio</code> , <code>date_iso</code> (→ "sampled July 2025")
Limit type	Per row: enforceable federal limit vs. a health benchmark that isn't enforceable	<code>is_enforceable</code> , <code>basis</code> , <code>source</code> . Rows scored against a benchmark: label "Health benchmark, not an enforceable limit". The words "above the federal limit" appear only when <code>exceeds_limit</code> and <code>is_enforceable</code> are both true. Print <code>limit_ppt</code> as the limit (not <code>mcl_ppt</code>)
Rescission marker	Per row whose limit is proposed for repeal	<code>proposed_for_rescission</code>
Quarterly trend	Worst contaminant across sampling dates with the limit as a reference line, "is it getting better?"	<code>water.contaminants[<worst name>]</code> = every reading as <code>{ date_iso, value_ppt }</code> . Draw only with 2+ distinct dated readings (one point per <code>date_iso</code> , the highest value that date); plot by <code>date_iso</code> . Caption PROPOSED : "Only detections are shown."
Unregulated group	Detected, no enforceable limit, shown, with the statement that these are not enforceable limits	<code>water.detected_unregulated</code> ; <code>water.message</code> when present
Guidance-scored group	Scored against a health benchmark rather than a federal limit	<code>water.guidance_scored[]</code> → <code>contaminant</code> , <code>limit_ppt</code> , <code>basis</code> , <code>source</code> ; <code>water.includes_guidance</code>
Excluded from score	Lithium: measured, shown, not scored, with the reason written out. Units are µg/L: "EPA reference level 10 µg/L (non-regulatory)" from <code>health_reference_level_ppt / 1000</code> ; never ppt	<code>water.excluded_from_score[].reason</code> (through <code>displayText</code>), <code>value_ppt / 1000</code> , <code>date_iso</code> , <code>health_reference_source</code> ; the lithium box adds <code>health_reference_level_ug_l</code> and <code>detail[].value_ug_l</code>
Regulatory notice	Callout when the score rests on a limit proposed for rescission	<code>water.regulatory_notice</code> (present only when it applies; through <code>displayText</code>). The server's wording names the specific compound ("The federal limit for PFHxS is currently the subject of a proposed EPA rescission..."); the blueprint's example is general ("Three of the five federal PFAS limits have been proposed for rescission. This score uses one of them.", §12.3). Recommendation: use the server's, it's more specific and stays correct as the rules change
Sample age	Caution callout when the latest sample is roughly 3+ years old: "This sample is about 3 years old."	Water box <code>confidence === "stale"</code> (server decides), <code>data_age_days</code> , <code>latest_sample_iso</code>
Learn link	"What does this mean?" → PFAS	<code>openLearn</code> with <code>topic: 'pfas'</code> , <code>reading</code> from the worst scored row, <code>context</code> <code>contaminant</code> (the raw key, e.g. HFPO-DA , not the display label), <code>value</code> , <code>limit</code>

Private well / spring: the testing plan **LOCKED** **BIND**

No score, ever, inventing a reading for untested water would be dishonest (§12.5).

Element	Binding
Framing	<code>water.message</code> : the well version ("Private wells are not covered by the Safe Drinking Water Act...") or the stronger spring version, chosen by <code>water.source_type</code>
Why these tests	<code>water.test_plan.reasons[]</code> , already tailored to the county's radon zone, the build year, and well vs spring
Tests, in priority order	<code>test_plan.tests[]</code> → <code>rank</code> , <code>name</code> , <code>priority</code> (words: <i>Critical</i> , <i>Recommended</i> , <i>Optional</i>) , <code>cadence</code> , <code>why</code> , <code>where</code> , <code>estimated_cost.low</code> - <code>.high</code> . When <code>needs_local_context</code> is true (metals panel, PFAS): PROPOSED caveat " <i>Local conditions matter for this test.</i> "
Totals	<code>test_plan.start_here_cost</code> (all critical tests) and <code>test_plan.full_panel_cost</code> , as ranges

NOTE: test costs are estimates (`estimated_cost.is_estimate: true` , same on both totals), label them as such.

Card 2: Radon **LOCKED**

Element	Content	Binding
Collapsed	Name, risk bar, severity pill, <i>estimate</i> pill, one line e.g. " <i>Zone 3 · a county average, not your home</i> "	Radon box <code>risk</code> ; severity per A3; <code>radon.zone</code>
County and zone	<i>Union County · Zone 3</i> (every NC-08 county is Zone 3, A12)	<code>radon.county</code> , <code>radon.zone</code>
What a zone is and isn't	<i>"Zones predict the average indoor level across a whole county. They are not a measurement of your home. Homes in low-zone counties still test high, and neighboring houses can differ."</i>	Static, \$12.3
Action threshold, testing, cost	The federal action level (4 pCi/L), what a test involves, rough cost	Static; cost matches the radon action item (<i>\$15-40 for a test kit</i>)
Seasonal note	October–February only: winter readings are the most accurate (sealed, heated homes draw in more soil gas)	Device month
County not found	PROPOSED " <i>Radon zone data isn't available for your county.</i> "	<code>radon.error</code> present , no <code>out_of_state</code>
Outside North Carolina	PROPOSED " <i>HALO's radon zone data covers North Carolina addresses. A home test is the only way to know your level anywhere.</i> " Card in its no-data state	<code>radon.out_of_state === true</code> (also carries <code>radon.state</code> , e.g. "SC"). Never show an NC zone for an out-of-state county with a matching name
Learn	" <i>What does this mean?</i> "	<code>topic=radon&county=&zone=</code>

Card 3: Lead **LOCKED**

Informational only. It has a severity for display and ordering and can produce an action item, but it never contributes to any score, settled, final (§12.3). The card header carries a permanent *shown*, *not scored* label, matching lithium.

Element	Content	Binding
Collapsed	Name, <i>shown</i> , <i>not scored</i> , severity pill, <i>estimate</i> pill when estimated, one line	<code>lead.level</code> , <code>lead.is_estimate</code> , <code>lead.contributes_to_score</code> (always false). Risk bar: no fill
Household sentence	e.g. " <i>...Young children absorb lead far more readily than adults, so this is worth testing sooner rather than later.</i> "	<code>lead.sentence</code>
Basis statement	Every age-based result says it's an estimate from the home's age, not a measurement of the plumbing	<code>lead.text</code> (includes that sentence)
By build year	Before 1950 → High · 1950-1987 → Elevated · 1988 or later → Good · unknown → No data	<code>lead.basis</code> : <code>built_before_1950</code> , <code>built_1950_1987</code> , <code>built_1988_or_later</code> , <code>unknown_year</code>
Unknown year	<i>"We don't know when this home was built, so we can't estimate lead risk. Testing is the only way to know."</i> + link to add the build year in Settings. For an unknown year, <code>lead.sentence</code> , <code>lead.text</code> and this copy are the same sentence, show it once . In general, render <code>lead.text</code> only when it differs from <code>lead.sentence</code>	<code>lead.basis === "unknown_year"</code> . The link goes to <code>/settings?open=home</code> (Part P5), copy " <i>Add the year your home was built in Settings</i> ". Saved via <code>PATCH /api/profile { home_year }</code> (B6)
Private well	PROPOSED " <i>For a private well, lead is part of your testing plan above.</i> " Show it only when <code>test_plan.tests</code> contains an entry with <code>id</code> <code>lead_copper</code> (the plan adds it only for homes built before 1988 or of unknown age). Otherwise show nothing for lead and keep the zone grey	<code>lead === null</code>
Learn	" <i>What does this mean?</i> "	<code>topic=lead&home_year=</code>

DECIDED (backend, recorded): THE SEVERE LEAD LEVEL IS KNOWINGLY UNREACHABLE: the Severe level (`basis: "utility_lead_service_line"`) would need to know that this specific address has a lead service line. No public address-level inventory exists for NC-08 utilities, so the backend never produces it. Lead is always age-based (High, Elevated, Good, or No data), with the estimate pill whenever `lead.is_estimate` is true (A3). The mapping still handles `severe` in case that changes.

Element	Content	Binding
Your utility's service lines	PROPOSED Context under the lead estimate, only when the utility appears in the state's published inventory: <i>"[Utility] reported [needs_replacement] service lines that need replacing (lead or galvanized) and [unknown] of unknown material. This is for the whole utility, not your address."</i> + source line <i>"NC DEQ lead service line inventory, July 2025"</i> . It never changes the lead level	<code>lead.utility_inventory: { utility, reported, lead, galvanized_requiring_replacement, non_lead, unknown, needs_replacement, address_level: false, source: { name, url, retrieved } } or null</code> (hide the row). Format counts with commas (A5)

Your Action Plan **LOCKED** **BIND**

Numbered, worst problem first, only items at Elevated or above. **Collapsed:** rank number, severity pill, title. **Expanded:** reason, cost, certification, action, and the tag. Each item expands to its full guidance (§12.6, §54).

Element	Binding
Order and number	<code>action_plan[].rank</code> (the server already moves water and lead up when a toddler or pregnancy is present)
Severity	<code>action_plan[].severity</code>
Title	<code>title</code> , e.g. "Filter your drinking water for PFOS"
Reason naming the finding	<code>reason</code> , e.g. "PFOS measured at 8.3 ppt, about 2.1× the federal limit of 4 ppt." Show as sent (through <code>displayText</code>); the value in it is already rounded to one decimal
Cost range	<code>cost</code> (already formatted, e.g. "\$30-90"; lead and radon costs include words such as "\$20-50 for a lead test"). Label it <i>"Estimated cost: [cost]"</i> , since every cost is an estimate
Certification to look for	<code>certification</code> (null for radon), shown inline, so it can be read standing in a store
What to do	<code>action</code> , already written for renter or owner by the server
Renter / owner tag	From <code>profile.renter_mode</code> : the text is already tailored; the tag says for whom. PROPOSED strings: <i>"For renters"</i> / <i>"For owners"</i> . <code>renter_can_self_serve</code> is unused
Empty	<i>"Nothing here needs action right now. We'll tell you if that changes."</i> , but never for a well: a well's action plan is always empty because the testing plan replaces it. Show a pointer to the testing plan instead

VERIFY BEFORE LAUNCH: every certification number (NSF/ANSI 53, P473, NSF/ANSI 58, NSF/ANSI 42) against the certifying body's own documentation (§12.6, §66). A wrong filtration claim is materially harmful to a household acting on it.

Page-level states

State	Shows
Loading, nothing stored	Skeleton: ring, headline, house-diagram block, three card blocks
Request fails (500)	Stored response if any + notice; else <i>"Something went wrong on our end, not yours. Try again in a moment."</i> + Retry
Radon unavailable	<code>radon.error</code> present (county not matched, or <code>out_of_state</code>), the radon card shows its no-data state; water and lead render normally

Proposed content **PROPOSED**

Addition	From
A "what this is based on" line in the water card: <i>"Based on 6 readings from March 2023 to July 2025"</i>	<code>water.reading_count</code> , <code>earliest_sample_iso</code> , <code>latest_sample_iso</code>
Detected count: <i>"4 compounds detected · 3 scored · 1 with no federal limit"</i>	<code>water.detected_count</code> , <code>scored_count</code> , <code>detected_unregulated.length</code>
Renter/owner line above the action plan: <i>"Showing actions for renters · Change"</i>	<code>profile.renter_mode</code> . Change opens <code>/settings?</code> <code>open=renting</code> or, if Settings is cut, saves through <code>saveHousehold</code> (Part P3)
A small "what HomeGuard can't tell you" footer: what's at your tap depends on your own plumbing; zones aren't home measurements; build year isn't a pipe inspection	Static, from §69, the blueprint calls stating limits plainly an asset in judging

Visual decisions to make DESIGN

HomeGuard's own: the house diagram (the page's signature), risk bar, threshold bar chart, quarterly trend, testing-plan list, action-plan item anatomy, and how this page reads as "inspection" rather than "weather." The confidence pill, pills, cards, sheet and callouts are shared and frozen in Phase 0. Learn is this owner's sheet (C1).

B4. Journal

Purpose LOCKED

Connect environmental readings to how the people in the household actually feel. The only feature where the household puts information in and gets something specific back, what separates a tool from a display (§14.1).

THREE BINDING RULES FROM THE RETENTION RESEARCH (§14.2): (1) The app must be fully useful to someone who never logs anything, Journal is optional depth, never a prerequisite. (2) No streaks, badges, points, or celebration animations. (3) The first insight must arrive immediately, not after two weeks.

Address and entry

`/journal` (Log mode), Tab 4. `/journal?mode=trends` opens Trends. Switching the segmented control uses `router.replace` (no history entry) and keeps the address in step. The page may already be open in Trends when an alert navigates to `/journal?mode=trends`; handle the parameter changing while mounted.

Data calls BIND

Call	Params	Feeds
GET <code>/api/profile</code> through <code>useProfile()</code>	None	household → group selector; state
GET <code>/api/journal</code>	default = last 90 days. For the "no entries at all" test use <code>from</code> = 365 days back, so a household whose entries are all older than 90 days does not see the first-visit prompt again	Entries: <code>entries[]</code> → <code>id</code> , <code>entry_date</code> , <code>band</code> , <code>symptoms[]</code> , <code>severity</code> , <code>note</code> , <code>retrospective</code> , <code>possibly_illness</code>
GET <code>/api/history?days=90</code>	None	Heat grid, day detail, each entry's day condition, line chart. (The retrospective result is computed by the server, never here)
POST <code>/api/journal/retrospective</code>	<code>{ dates }</code> (at most 62; future dates are dropped silently)	The first-visit result: <code>{ ready, factor, statement, flagged_days_with_readings, flagged_avg, month_avg, unflagged_avg }</code>
GET <code>/api/journal/findings</code>	None	Trends findings or not-ready progress
GET <code>/api/journal/summary</code>	None (defaults to the season that most recently ended)	Season summary callout
GET <code>/api/home-guard</code>	Once, only in October to February and only when no stored home reading exists	The radon zone for the winter radon prompt
POST <code>/api/journal</code>	see the form table	Save / update / "Felt fine today" / retrospective days

Shape LOCKED

Header *Journal*. Directly beneath, a two-option segmented control: *Log* · *Trends*. Switching changes the content below without navigating (§14.3).

Log mode: first visit (no entries at all) LOCKED

Element	Content / behavior	Binding
Prompt	"Think back over the last few weeks. Which days were rough? Think headaches, congestion, trouble sleeping. Tap them."	Shown when <code>entries</code> is empty
Month calendar	Weekday initials header (Sunday first); past days selectable, future days dimmed and not selectable; month arrows, back no more than three months; at most 62 days selectable (the route's limit). A day with no readings in <code>/api/history</code> is still selectable but marked "no readings" (only about the last 28 days are backfilled, and pollen is never backfilled). Keyboard and screen-reader model per A8	n/a
Count	"3 days selected." (singular "1 day selected.", zero "0 days selected.")	n/a
Primary	See what was in the air: active at three or more days; below that it is disabled with the helper PROPOSED "Pick at least 3 days to compare." (an explicit exception to A7's "never disabled for validation", because the comparison has nothing to say below three)	Saves each selected day with <code>POST /api/journal</code> and <code>retrospective: true</code> , see the rule below
Result	The calendar collapses and a callout appears, e.g. "On the 3 days you flagged, air quality averaged Elevated. The month's average was Moderate. Days you didn't flag averaged Moderate." Then a tertiary <i>Start logging daily</i> . Show also PROPOSED	<code>POST /api/journal/retrospective { dates } → statement</code> ,

	"HALO has readings for 2 of your 3 days." from <code>flagged_days_with_readings</code> (the statement's count is the days that have readings, not the number selected). A multi-month selection reads "Those months averaged"; <code>factor: null</code> comes with a statement that none of the tracked readings were higher on flagged days: show it as returned	see below
--	--	-----------

HOW THE RETROSPECTIVE WORKS: after saving the flagged days, call `POST /api/journal/retrospective` with `{ dates: [...] }`. The server compares the flagged days' readings with the rest of that month and returns `{ ready, factor, statement }`, the frontend shows `statement` as-is and claims nothing more (§14.4). `ready: false` means too few of those days have readings; show its `statement` the same way. The flagged days are saved with `retrospective: true`, `band: "household"`, and `symptoms: ["unspecified"]`: the findings analysis counts a day as a symptom day only if `symptoms` is non-empty, and the person didn't say which symptoms. **Order and failure:** POST every selected day first (a POST is an upsert, so a retry is safe), then call the retrospective. If any POST fails, keep the selection, show the save-failure copy and retry only the failed dates. If the retrospective returns `flagged_days_with_readings` of 0 right after onboarding (backfill runs after the onboarding response), show its statement and a Retry.

WHY THIS WORKS ON A FIRST VISIT: the server records a day's readings for every household every day, and fills in recent weeks at onboarding where a historical source exists, so the flagged dates usually have readings before the person ever logs anything (§14.4). Dates without readings (older than the backfill, or a day a source had nothing) are left out of the comparison.

Log mode: ongoing LOCKED

Opens straight to today's form once any entry exists. The whole interaction must fit in under ten seconds (§14.5).

Element	Content / behavior	Binding (POST body)
Date header	"Today, March 14" with left/right arrows to log an earlier day (no future days; no earlier than 90 days back, the history window; arrows disabled at the limits). Last seven days use the weekday ("Tuesday"), older days "March 4" (A5)	<code>entry_date</code> , YYYY-MM-DD, the household's local calendar date (joined to history by the same string, see the date note in A5)
Who	Toggle chips, single-select , naming each group present; hidden entirely for a single-group or skipped household; last-used group preselected (stored at <code>halo.frontend.v1.<identity>.journal.lastBand</code> ; if it no longer exists, the first group). Tag labels: Toddler, Child, Teen, Adult, Senior, Pregnant, Breathing, Household (<code>band: "household"</code> , also what retrospective entries show)	<code>band</code> , the group key (e.g. <code>has_toddler</code>), or <code>"household"</code> when there's one group or none. The server accepts only these values
Symptoms	Wrapping grid of seven chips, multi-select. Labels: Headache · Congestion · Cough · Itchy or watery eyes · Fatigue · Trouble sleeping · Wheezing or shortness of breath	<code>symptoms[]</code> , send stable keys, not display text: <code>headache</code> , <code>congestion</code> , <code>cough</code> , <code>itchy_eyes</code> , <code>fatigue</code> , <code>trouble_sleeping</code> , <code>wheezing</code> (so Spanish labels don't split the data)
Severity	Three-option control: Mild · Moderate · Bad. Appears only after a symptom is chosen; no default (null until chosen)	<code>severity: mild / moderate / bad</code> , or <code>null</code> . Not the same scale as environmental severity
Illness flag	Checkbox: "I think this might be a cold or flu, not the environment."	<code>possibly_illness</code> . The analysis drops these days entirely, which is the point: cold and flu season overlaps heating season (§14.5)
Note	Single line, expands; placeholder "Anything else? (optional)"; counter "412/500" past 400; max 500	<code>note</code> (the server truncates at 500); send "" when empty
Save	Primary, full width: <i>Save</i> , or <i>Update</i> if an entry exists for this day and group. On save: brief toast, form collapses into a header row <i>Log another day</i> (opens the form for the header date) and the list shows the new entry on top. Tapping a list row opens the form pre-filled for that date and group with Update. A first visit also offers a tertiary <i>Log today instead</i> that opens today's form. Save with no symptom, no severity and no "Felt fine" shows PROPOSED "Choose a symptom, or tap Felt fine today."	Also <code>retrospective: false</code> . While GET /api/journal is loading or has failed, Save is disabled ("Loading your entries..." or Retry): a POST without the loaded entry would overwrite it with blanks
Felt fine today	Tertiary beneath: <i>Felt fine today</i> . Saves an entry with no symptoms, toasts. On a day that already has symptoms it replaces them (the tap is the intent)	<code>symptoms: []</code> , <code>severity: null</code> , and the form's <code>note</code> and <code>possibly_illness</code> . This matters: without good days recorded, the analysis only ever sees bad ones (Tier 1, §65)

Save failure	"Couldn't save that entry. It's still here. Try again."	Form keeps everything
--------------	---	-----------------------

TRAP, EVERY SAVE REWRITES EVERY FIELD. Re-saving a day/group with a partial body resets whatever was left out (note → empty, symptoms → none, flags → false). Always send the whole entry, pre-filled from the existing one.

Recent entries list LOCKED

Below the form, up to seven recent entries. Each row: date, group as a small tag, symptoms as comma-separated display labels ("Felt fine" when empty; the retrospective key `unspecified` displays as **PROPOSED** "Rough day"), and that day's dominant condition as a severity pill. Note truncates to one line, full on tap (§47.4, §61).

BIND Day condition = worst of `history[date].air/uv/pollen/mold` (no `data` if the date has no history). Offline entries show a pending marker until sent, only if the offline queue is built (Tier 3, A2); otherwise a failed save keeps the form filled and says so.

Delete: swipe left reveals delete, and every row also has a visible 44px Delete control (`SwipeRow` , A8). Deletion is immediate (`DELETE /api/journal?id=` ; the id must be a UUID, and `{ deleted: 0 }` means it was already gone) with an undo toast. Undo re-saves the **full deleted entry** (keep it in memory: band, symptoms, severity, note, retrospective, possibly_illness) with `POST` , which gives it a new id (§47.4, §31.5).

Seasonal prompts: only on this page LOCKED

Seasonality appears in Journal and nowhere else (§14.7). As dismissible callouts at the top of Log mode, above the date header, never a modal, never interrupting the form. Dismissal lasts for that season (stored locally).

- **Winter radon prompt:** October to February, household in radon zone 1 or 2 (no NC-08 household qualifies, A12): *"Radon usually peaks in winter in homes like yours. This is the most accurate time of year to test, because a summer test can read low and miss the real number."* Zone from the stored HomeGuard response (`radon.zone`); if none is stored, fetch `GET /api/home-guard` once when the Journal opens in those months and store it; skip if still unknown.
- **Pollen season note:** NC only (`profile.state` is NC) and only when today's pollen severity is Elevated or worse. Months (**PROPOSED** , content owner to confirm): February to April tree, May to July grass, August to October ragweed. Copy: *"It's tree pollen season in your area. Elevated readings this month are normal."* (with "grass" or "ragweed").
- **Season summary:** *"This spring you logged 22 days. On the 9 high-pollen days, you flagged symptoms 7 times."* (the server's wording: "high-pollen", "poor-air", and so on). A milestone, not a nag. **BIND** `GET /api/journal/summary` → `statement` , `logged_days` , `season` , `year` . Request it on load; show it only when `logged_days` is above 0 and the season ended within the last 30 days. Shown at the top of **both** Log and Trends (D6), so the `season_summary` alert, which opens Trends, lands on it.

Order when several show: radon, pollen, summary; at most two visible. Dismissal is per season and stored at `halo.frontend.v1.<identity>.dismiss.<prompt>.<season>.<year>` (winter belongs to the year it starts).

Trends mode LOCKED

Element	Content / behavior	Binding
Calendar heat grid	The page's signature visual: one cell per day, colored by that day's worst severity, dot on days symptoms were logged. Answers Journal's whole question visually before any statistics, and shows gaps in logging without nagging about them. Twelve weeks, Sunday first (seven columns), rows aligned to weekday, oldest at the top ; dates before the first history row and after today are not drawn	Cell = worst of the available factor words in <code>history[date]</code> (ignore <code>no_data</code>). All four <code>no_data</code> , or no history row → the no-data outline cell. Some factors missing (backfilled days always lack pollen) → the cell keeps its color plus a small hollow corner mark, and the day sheet lists the missing ones as No data (a missing factor is never read as green). Dot = any entry that date with non-empty <code>symptoms</code> . Today marked. Each cell's label: <i>"March 14, worst reading High, symptoms logged"</i>
Grid legend	Severity colors with labels + the dot labeled <i>you logged</i>	n/a
Day detail (tap a cell)	Standard sheet (Part A9): date as title; every reading that day with its severity pill; what was logged (every group's entry for that date), if anything; a button <i>Add or edit entry</i> that closes the sheet and switches to Log mode with that date	<code>history[date]</code> → <code>air</code> , <code>uv</code> , <code>pollen</code> , <code>mold</code> + <code>values</code> . Rows: Air quality "37" (whole number) + pill from <code>history.air</code> ; UV "5.2" + pill; Pollen "Grass 3" for the highest of <code>values.pollen</code> + pill from <code>history.pollen</code> (the worst category); Mold "Low", "Moderate" or "High" with the estimate pill and the mold severity pill (the value here is the only place <code>mold_risk</code> is shown)
Findings	One card per finding: the statement, the underlying counts, a small pair of bars comparing the two rates. Factor labels: air = "Air quality", pollen = "Pollen", uv = "UV", mold = "Mold". Bar labels <i>"On elevated days"</i> and <i>"On other days"</i> (the rates are already percentages). Counts line <i>"12 elevated days, 40 other days"</i> . Caption: a descriptive one (the statement itself); screen-reader text is a small table	<code>findings[]</code> → <code>statement</code> , <code>elevated_rate</code> , <code>other_rate</code> , <code>counts.elevated_days</code> , <code>counts.other_days</code> , <code>factor</code>
Disclaimer	Permanent, never dismissible, under the findings	<code>disclaimer</code> , <i>"This is a pattern in your own logs, not a medical finding. Many things affect how you feel on a given day."</i>
Not ready	Replaces findings with progress: <i>"Keep logging. There isn't enough here yet to say anything reliable. You've logged 3 days; patterns need about 5."</i>	<code>ready === false</code> . The n is counted from <code>entries</code> : distinct dates with non-empty <code>symptoms</code> and possibly <code>illness</code> false. Not <code>flaggedDays</code> (see below). When n already meets <code>needed.flagged</code> but <code>flaggedDays</code> is still below it (the logged days lack readings), the message is PROPOSED <i>"HALO needs readings for more of the days you logged. It has [flaggedDays] so far."</i>
Line chart	One line of <code>history[].score</code> (0 to 100, higher is better; a gap where the score is missing) with the severity bands behind it (A10 edges do not apply; the bands are the score's own severity words). A marker on each logged date; the marker's shape per group PROPOSED (design confirms in the wireframe): toddler circle, child square, teen triangle, adult diamond, senior star, pregnant cross, breathing hexagon, household ring. More than one entry on a date draws one marker per group, offset. Caption <i>"Each mark is a day you logged symptoms."</i> (neutral, no finding). Screen-reader text: the logged dates with that day's worst severity. Horizontally scrollable	<code>history</code> + <code>entries[].band</code>

DON'T SHOW `flaggedDays` AS "DAYS LOGGED": the server's `flaggedDays` counts only symptom days that also have a day of readings and aren't flagged as illness, it's "comparable days," not "days logged." Count logged days from `entries` for the message. Findings need two things: 5 comparable symptom days (`flaggedDays` vs `needed.flagged`) **and** 14 days of readings (`totalDays` vs `needed.total`). When the household has logged enough but there aren't 14 days of readings yet (a new household before its history fills in), use **PROPOSED** *"HALO needs about two weeks of readings for your area to compare against. It has [totalDays] so far."*

MISSING COPY, READY BUT NO FINDINGS. When there's enough data but no factor shows a big enough difference, the server returns `ready: true, findings: []`. The blueprint has no copy for this. **PROPOSED** : *"No clear pattern yet. Your symptom days don't line up with any one condition so far."*, the disclaimer still shows.

WHY THE RESTRAINT IS THE FEATURE: findings are rates of co-occurrence, never a fitted model; illness-flagged days are excluded; factors with under three days on either side are skipped; differences under about 25 points aren't reported (§14.6). The frontend shows exactly what's returned and adds nothing. This is one of the strongest things to explain to a judge.

Page-level states

State	Shows
No entries at all	Log: the retrospective first visit. Trends: <i>"Nothing logged yet. Start by marking a few days that felt rough, and we'll show you what was in the air."</i>
Loading	Trends: grid-shaped placeholder cells + two finding-card placeholders
Entries fail to load	Generic error + retry; the form still works for today
Offline	Default: the form stays filled and a failed save shows <i>"Couldn't save that entry. It's still here. Try again."</i> (PROPOSED). With the Tier 3 queue built: entries queue with a pending marker (A2)

Connection to the Assistant LOCKED

Journal is the one feature that gives the Assistant information about the household itself: a question like "why does my son cough more on some days?" is answered from that household's own entries and environmental history, alongside cited guidance (§14.8). The server supplies this context to the Assistant; nothing for this page to do beyond saving entries, see C2.

Proposed content PROPOSED

Addition	From
Progress toward findings shown as two small counters in Trends: comparable symptom days 3 of 5, days of readings 9 of 14, so it's clear which one is holding things up. Note the readings counter excludes illness-flagged days, so it can drop when a day is flagged	<code>flaggedDays</code> , <code>totalDays</code> , <code>needed</code> (labeled as comparable days, not days logged)
In day detail, the household sentence for that day's worst reading isn't available from history, instead show the values with units	<code>history[date].values</code>
Per-group filter on the heat grid dots when more than one group logs	<code>entries[] .band</code>

Visual decisions to make DESIGN

The Journal's own: the heat grid (signature), the month calendar for the retrospective step, the findings card with its bar pair, the scrollable line chart with its marker shapes. Segmented control, toggle chips, toast, empty state and swipe row are shared and frozen in Phase 0.

B5. Act

Purpose LOCKED

Everything a household can do beyond its own four walls, contributing, advocating, sharing. Distinct from Learn: Learn is personal protection (filters, sunscreen, closing windows); Act is acting on the problem itself. This is the layer that makes HALO civic rather than merely personal, and the clearest answer to why it belongs in a competition run by a member of Congress (§15.1, §17.2).

Address and entry

`/act` , Tab 5. Each of the four features opens a bottom sheet (history entries). **Each sheet fetches what it needs independently, on open, with its own skeleton; a failed part shows "No data" with a Retry for that part and never blanks the rest** (for example a failed `/api/sources` must not blank the health card). Act's sheets always fetch fresh rather than reading the stored copy, since Settings changes can make the stored copy stale (Part P3).

Shape LOCKED

Header *Act*. A framing line: *"Environmental problems aren't solved one household at a time. Here's what else you can do."* Then four full-width cards, each with one strong icon, a title, a one-line description, and a chevron. Styled editorially, taller cards, larger type, more whitespace, no charts, because these are invitations, not readings. Same card component; only proportions and type scale differ (§15.2, §48.1).

Data calls BIND

Call	Feeds
GET <code>/api/profile</code> through <code>useProfile()</code>	County, water source, renter mode, coordinates (for the facility distance)
GET <code>/api/home-guard</code> (fresh on sheet open)	Which risks are elevated (volunteer causes, help sections), letter values, health card
GET <code>/api/daily-score</code> (fresh on sheet open, never fresh=1)	Health card daily block
GET <code>/api/volunteer?causes=...</code>	Volunteer sheet
GET <code>/api/district</code>	PROPOSED district context card

Card 1: Volunteer near you LOCKED

Description: *"Groups working on the problems HALO found in your area."* The list is filtered by this household's own findings, a household with elevated water sees the riverkeeper working on exactly that; one with clean water and high radon doesn't (§15.3).

Element	Content	Binding
Request	None	GET <code>/api/volunteer?causes=</code> . The frontend derives causes; the server doesn't. From HomeGuard: water item in <code>action_plan</code> → <code>water,pfas</code> ; radon item → <code>radon</code> ; lead item → <code>water</code> . For a well or spring, use <code>wells</code> from <code>profile.water_source</code> (a well's <code>action_plan</code> is always empty, so the plan cannot supply it). Nothing else, adding a broad tag like <code>advocacy</code> would match every organization and undo the filtering. Nothing elevated → send no causes (every county-serving org is returned, unranked). County defaults to the profile's
Sheet title + basis line	<i>Volunteer near you</i> , then <i>"Groups working on [water quality / PFAS / radon / private wells] in [County]."</i> (join several with "and"); with no causes <i>"Groups serving [County]."</i>	<code>county</code> , <code>causes</code> from the response. Cause tag labels: water "Water quality", pfas "PFAS", radon "Radon", air "Air quality", wells "Private wells", advocacy "Advocacy"
Each organization	Name; description (2 lines max); cause tags; counties served (<i>Statewide</i> when <code>counties</code> includes <code>"statewide"</code> ; else the household's county if listed, else the first two and <i>"and N more"</i>); <i>Visit site</i> opening externally; when it was last verified by a person	<code>orgs[]</code> → <code>name</code> , <code>description</code> , <code>causes[]</code> , <code>counties[]</code> , <code>url</code> , <code>last_verified</code>
Verification	<code>last_verified</code> set → <i>"Last verified March 2026"</i> ; null → <i>"Not yet verified"</i>	Never claim verification that didn't happen. Today all 13 seeded organizations have <code>last_verified</code> empty , so every row says "Not yet verified" and the footer below never appears until a person verifies them (P10, D9)

Footer	"Every group here was checked by hand. Last verified [month year]."	Show only if every listed org has <code>last_verified</code> , otherwise the claim would be false
Empty	"We don't have groups listed for your county yet." (the list includes unverified groups, so the word "verified" would be wrong)	<code>orgs</code> empty

NOTE: don't render `match_count` when no causes were sent (it's always 0 then). A household with nothing elevated sees every organization serving its county, unranked.

Card 2: How you can help LOCKED

Description: "Ways to act on what's affecting your area." Guidance tied to this household's actual readings, never generic (\$15.4). Sections appear only for risks actually elevated here; each has a heading naming the risk and two to four numbered actions, each one line with an optional supporting sentence. A closing *Always useful* section applies to everyone (\$48.3).

Section appears when	Covers	Trigger binding
Radon zone is elevated	How to test, and why it matters beyond one home, radon is geological, so a high result is information the neighborhood should have too	<code>radon.zone</code> 1 or 2
Water contamination is elevated	Finding the utility's public comment periods; who to contact; what a Consumer Confidence Report is and how to read your own	<code>action_plan</code> has a <code>water</code> item
Nearby facility violations	How enforcement works, looking up a facility's record, filing a complaint	A facility with open violations (<code>open_violations</code> > 0) within PROPOSED 5 miles of the home, from <code>/api/map?layer=facilities</code> (distance from the profile's <code>lat</code> / <code>lng</code>)
Always	How to read a water quality report; what to ask at a town or utility meeting	None

FACILITY SECTION HONESTY. The facilities layer holds only active major facilities and repeat violators in NC-08, so the section appears only when the layer returns a facility with `open_violations` above 0 within 5 miles (haversine from the profile's rounded coordinates), and it is naturally absent outside NC-08. `open_violations` counts programs currently in violation (0 to 4), not violations. Tests need a fixture with such a facility.

CONTENT TO WRITE. The blueprint names each section's topic but not its actual numbered actions. Draft actions are supplied below as **PROPOSED** so builders are not blocked; the content owner (Aadit, P10 D5) must verify every factual claim, and a claim that cannot be verified is cut, not softened. This is design/content-owner work (\$26.3b), not engineering.

Section	PROPOSED draft actions (numbered in the sheet)
Radon	1. Test your home for radon. A short-term kit costs about \$15 to \$40 and takes a few days. 2. If the result is 4 pCi/L or higher, run a second test to confirm, then talk to a certified radon mitigation contractor. 3. Tell your neighbors. Radon comes from the ground, so a high result on one street is useful to the people around you. 4. Ask your state health department whether reduced-price test kits are offered.
Water	1. Find your utility's public comment periods. Utilities and state agencies post them on their websites. 2. Read your utility's Consumer Confidence Report. Utilities send one every year, by July 1; find the column that compares each result with its legal limit. 3. Ask your utility what its plan is to meet the federal limit, and when. 4. Contact your city council, county commission or the utility's board, which set water budgets.
Nearby facility	1. Look up the facility's record on EPA's ECHO site (Enforcement and Compliance History Online). 2. Report a concern to your state environmental agency (in North Carolina, the Department of Environmental Quality). 3. Keep notes: date, time, what you saw or smelled, and photos.
Always useful	1. Learn to read a water quality report. 2. At a town or utility meeting, ask: What are our most recent test results? What is the plan and timeline if a result is over a limit? How will households be told?

Card 3: Write a letter LOCKED

Description: "A ready-to-send message about your water, filled in with your numbers." The single strongest fit for a congressional competition: a tool that helps a constituent engage an institution, built from their own measured data (\$15.5).

Element	Content / behavior	Binding
Recipient	Three-option control at the top: <i>Utility</i> · <i>Landlord</i> · <i>Representative</i> . Default: Landlord for a renter, Utility otherwise. Changing it shows that recipient's draft; each recipient keeps its own edited draft, and Start over resets only the current one	None

Data notice	"Filled in with your utility's most recent test results."	None
Draft	Editable multi-line text area filling most of a tall sheet	Utility template (verbatim below)
Actions	Primary <i>Copy</i> (toast confirms) · secondary <i>Share</i> (the device share sheet) · tertiary <i>Start over</i> restores the generated text after edits	<code>navigator.clipboard</code> , <code>navigator.share</code> (fall back to Copy where Share isn't supported; a cancelled share is not an error; if the clipboard is refused, select the text and say so)

I am a customer of [utility name], public water system [identifier]. Your most recent federal sampling, dated [date], reported [contaminant] at [value] parts per trillion. The federal maximum contaminant level is [limit] parts per trillion. This system serves approximately [population] people. I am writing to ask what your compliance timeline is, and what interim steps are available to households in the meantime.

Placeholder	Binding
utility name / identifier	<code>water.pws_name / profile.pwsid</code>
contaminant, value, limit, date	The cited entry: from <code>water.scored_contaminants</code> keep entries with <code>is_enforceable === true</code> and <code>exceeds_limit === true</code> , take the highest <code>risk</code> ; if none, show the explanation state. → <code>contaminant</code> (through <code>contaminantLabel</code>), <code>value_ppt</code> (1 decimal), <code>limit_ppt</code> (the limit; never say "federal maximum contaminant level" for a non-enforceable entry), <code>date_iso</code> as "July 2025"
population	<code>water.population_served</code> (formatted per A5, "serves approximately 120,000 people", no plus sign); drop the sentence if null. Utility name through <code>formatUtilityName</code>

CONTENT TO WRITE AND EDGE CASES: (1) The blueprint gives the utility letter only; the landlord version ("asks about testing and tenant remedies rather than compliance schedules") and the representative version need writing (content owner). (2) HALO doesn't store who the household's representative is; the letter opens "***Dear Representative,***" for the person to complete. (3) When there's no measured water result to cite, show a short explanation instead of an empty template; for wells, hide the utility option.

PROPOSED draft	Text (the person completes bracketed parts; HALO never stores the address or a name)
Utility, greeting and closing	Opens "To whom it may concern," then the body above, then "Thank you," and "[Your name]".
Landlord	"Dear [landlord or property manager], I rent at [your address]. My water comes from [utility name], public water system [identifier]. That utility's most recent federal sampling, dated [date], reported [contaminant] at [value] parts per trillion. The federal maximum contaminant level is [limit] parts per trillion. I am writing to ask whether you have tested the water at this property, whether you plan to, and what steps you can take, such as providing a filter certified to reduce [contaminant]. Please reply in writing. Thank you, [Your name]"
Representative	"Dear Representative, I am a constituent in [county]. My water comes from [utility name], public water system [identifier]. Its most recent federal sampling, dated [date], reported [contaminant] at [value] parts per trillion; the federal maximum contaminant level is [limit] parts per trillion. The system serves approximately [population] people. I am writing to ask what is being done to help utilities like this one meet federal limits, and what help is available to households in the meantime. Thank you for your time, [Your name]"
No result: well	"Private wells aren't covered by the federal drinking water rules, so there is no utility result to cite. See your testing plan on the Home page."
No result: other cases	No data yet: "Your utility's federal results haven't been published yet, so there is nothing to cite." Lookup failed: "We couldn't check your utility's results right now. Try again in a moment." Nothing over a limit: "No enforceable federal limit was exceeded in your utility's most recent results, so this letter has no measurement to cite." Outside North Carolina: "HALO's water data covers North Carolina utilities, so it can't fill in this letter."

Card 4: Your home's health card LOCKED

Description: "*A one-page summary you can save or share.*" Its practical purpose is to be handed to someone: a doctor asking about exposure, a landlord being asked to test, a relative being persuaded to test their own home. Formats data every other page already produces, plus the source list (\$15.6, \$48.5).

Section	Content	Binding
Header	HALO mark, county, generation date with the year (" <i>March 14, 2026</i> ", because it is printed)	<code>profile.county</code> , <code>now</code>
Two scores	Daily and home scores side by side with their rings and labels	<code>score.display_score</code> from each stored response
Category table	Every category with its reading, severity, and the date behind it: air, UV, pollen, mold, water, radon, lead	Today readings + time retrieved; water worst contaminant + <code>latest_sample_iso</code> ; radon zone ("county estimate"); lead level + build year
Household notes	The household-aware sentences, if any	<code>*.sentence</code> , <code>lead.sentence</code>

Top actions	Top three action items. For a well or spring the action plan is always empty, so show the top three tests instead, never "Nothing here needs action"	<code>action_plan[0..2]</code> (title, cost, certification); well: <code>water.test_plan.tests[0..2]</code> (name, priority, estimated cost)
Sources	Every source with the date it was retrieved	<code>GET /api/sources</code> → <code>name</code> , <code>provides</code> , <code>last_retrieved</code> , rendered as returned (no hard-coded list); <code>last_retrieved</code> null → "Retrieval date not recorded"
Buttons	Primary <i>Save as image</i> · secondary <i>Print</i> (the card's description says "save or share": there is no separate Share button, so use "save or print" unless the image share below is built)	Client-side image render, always light, with a library approved by the team before use (the first-load budget is 500 KB; load it only when the button is first pressed); optionally <code>navigator.share</code> of the image. Print stylesheet per the box below

PRINT IS TIER 1. Printing must produce a clean black-on-white page with no interface chrome, no backgrounds, severity conveyed by label (and pattern) rather than color, and it's always light even in dark mode (\$48.5, \$59.3, \$65.1). Implementation: `@media print { body > *:not(.halo-health-card) { display: none } .halo-app { color-scheme: light } }`; severity shown as icon plus word; rings replaced by text such as "Score: 78, High". Missing values print "No data".

Page-level states

State	Shows
HomeGuard not yet loaded this session	Cards still render; each sheet fetches what it needs on open with its own skeleton
Loading (page)	Four card-shaped placeholders (the page itself needs almost no data)
A sheet's request fails	The sheet shows the parts it has, "No data" for the failed part, and a Retry for that part

Proposed content **PROPOSED**

Addition	From	Why
A district context card above the four features: e.g. "[<i>affected_population</i>] people in NC-08 are served by a water system over a federal PFAS limit." ("at least" when <code>affected_population_complete</code> is false). Show only when <code>affected_population</code> is a number and the household's county appears in <code>counties[]</code> (in the district, wholly or partly); hide it otherwise (a household in Buncombe, Iredell or out of state is not in NC-08). Tap → <code>/map?panel=district</code>	<code>GET /api/district</code> → <code>affected_population</code> , <code>affected_population_complete</code> , <code>assembled_at</code>	Puts the household's result in civic context, "is this just my problem?", on the civic page. Strongest single line for a judge
In the letter sheet, a line showing which reading the letter cites, with a link to it in HomeGuard	The chosen <code>scored_contaminants</code> entry	Transparency: the person sees exactly what they're sending

Visual decisions to make **DESIGN**

Act's own: the editorial card treatment (taller, larger type, one strong icon each), organization list rows, the letter editor sheet, the health card as a formatted document (on screen and in print), the district context card. Bottom sheet, segmented control, toast and callouts are shared (Phase 0).

B6. Settings

Purpose LOCKED

Everything entered during onboarding must be changeable, and everything HALO knows must be inspectable. A household that can't correct a wrong address or remove a group will stop trusting the results (§16.1).

Owner: Yogi (Lane 1, Part P). Settings also owns `saveHousehold()`, which HomeGuard and Onboarding import, so it is built early in Lane 1.

Address and entry

`/settings`, from the gear in the shell header on every tabbed page. Header *Settings* with a back arrow (not a gear). No tab highlighted. Eight accordion sections (seven until reviewed Spanish exists: the Language section is hidden until then, see the Language box). Only one open at a time; each row shows its current value when closed. Every control saves immediately and toasts only on failure (§54). `/settings?open=household|home|renting|language|accessibility|notifications|privacy|about` opens that section on arrival (used by HomeGuard's "unknown year" link, `/settings?open=home`). Opening or closing a section does not change the URL after arrival.

Back arrow rule. Back goes to the page the person came from (`router.back()`). If the app was opened directly on `/settings` (no in-app history), back goes to `/today`.

Loading. Every control is disabled, with a skeleton for its summary, until `useProfile()` has resolved. A control that saves the household is never enabled with guessed values, because the save replaces all seven groups.

Data calls BIND

Call	Feeds
GET <code>/api/profile</code> (through <code>useProfile()</code>)	<code>profile.county</code> , <code>pwsid</code> , <code>water_source</code> , <code>home_year</code> , <code>renter_mode</code> , <code>locale</code> , <code>household</code> , <code>household_set</code> , <code>zip</code> (null after onboarding; never bind it), <code>onboard_request_id</code> (never displayed)
PUT <code>/api/household</code> (through <code>saveHousehold()</code>)	Household groups, renting or owning, language
PATCH <code>/api/profile</code>	Build year, water source (send only the changed field)
Onboarding address step (<code>/onboarding?change=1</code>)	Change address (<code>POST /api/onboard</code> , made by the onboarding screen, not by Settings)
GET / PUT <code>/api/notifications</code> , POST / DELETE <code>/api/push/subscribe</code>	Notification switches and device notifications
GET <code>/api/journal?from=2000-01-01</code>	Entry count for the Clear journal confirmation (read <code>count</code>)
DELETE <code>/api/journal?all=true</code>	Clear journal (returns <code>{ deleted: n }</code>)
DELETE <code>/api/account</code>	Delete everything
GET <code>/api/sources</code>	About: data sources and dates
GET <code>/api/assistant</code>	Only <code>sends_household_context</code> , for the privacy statement (Part P9 ask 2; assume false until it exists)

ONE SAVE FUNCTION FOR `/api/household`: `PUT /api/household` replaces all seven groups on every call. Send only `renter_mode` and the household's groups are wiped. Every Settings save that touches this endpoint goes through `saveHousehold({ household, renter_mode?, locale? })` (Part P3), which always sends all seven current booleans plus whatever changed. Its response only echoes `renter_mode` and `locale` if you sent them, so keep local state from the profile, never from the PUT response. Two quick taps on different chips must not race: queue saves so each one sends the latest full state, and after the last one settles refresh the profile once.

AFTER ANY SAVE THAT CHANGES WORDING: household, renting or owning, water source, build year and language all change what Today, HomeGuard and Learn say. After a successful save Settings calls `clearReadings()` and dispatches `halo:settings-changed` (Part P3). The other pages refetch when they next show. Accessibility, notification and About changes do not dispatch it.

The eight sections LOCKED

Section	Collapsed summary	Expanded	Binding
Household	"Toddler, adult" or "Not set". "Not set" whenever no group is true. Don't rely on	The same seven chips as onboarding (the shared <code>ToggleChip</code>), same order; each tap saves; takes effect on the next reading refresh. Failure-only	Summary from <code>household</code> + <code>household_set</code> . Save: <code>saveHousehold</code> . Chip flips

	<code>household_set</code> alone: saving renting or language through <code>PUT /api/household</code> also sets it to true	toasts everywhere in Settings (the blueprint conflicts with itself: \$49.2 toasts on save, \$54 only on failure; failure-only is quieter and is the rule)	instantly, reverts on failure with the toast <i>Couldn't save that. Try again.</i>
Your home	" <i>Union County · City water</i> "	Build year field (A7 rules), water source dropdown (same five options as onboarding; "Not sure" is saved as <code>other</code> , so a stored <code>other</code> preselects "Other"), and secondary <i>Change address</i>	Build year and water source: <code>PATCH /api/profile</code> , saved on change (build year on blur). Change address: see the flow below
Renting or owning	<i>Owner / Renter</i>	Two-option control + one sentence that it changes which actions are recommended everywhere	<code>renter_mode</code> via <code>saveHousehold</code> (with all seven groups). Changes every action recommendation (\$12.7)
Language	<i>English</i>	English · Español, applying immediately. Hidden unless <code>NEXT_PUBLIC_SPANISH_REVIEWED=1</code> (see the Language box)	<code>locale</code> via <code>saveHousehold</code> . Interface strings switch from the translation layer; Learn content is requested with the new <code>locale</code>
Accessibility	<i>Default</i> , or <i>Custom</i> if any of the three is changed	Text size slider with live preview sentence; higher-contrast switch; reduce-motion switch. Details in the Accessibility contract below	Device-local (stored in the browser under <code>halo.device.*</code>): these are about this device, not the household. Reduce motion defaults to the system setting
Notifications	"4 of 5 on"	One switch per alert type with its plain-language name and one-line description, all default on; plus one switch for device notifications on this browser (hidden until D8, see below)	Types: <code>GET / PUT /api/notifications</code> → <code>preferences.air_quality_change</code> , <code>weather_advisory</code> , <code>new_water_results</code> , <code>radon_season</code> , <code>season_summary</code> . Device notifications: see the push block below
Data and privacy	<i>Your data</i> (no value, so the row reads as a label)	"What HALO knows about you" (the stored values), the full privacy statement, then <i>Clear journal</i> and destructive <i>Delete everything</i>	<code>useProfile()</code> , <code>GET /api/journal</code> , <code>DELETE /api/journal?all=true</code> , <code>DELETE /api/account</code>
About	" <i>Version 1.0</i> "	Every data source with the date it was last retrieved; how scores are calculated; who built HALO; link to the source repository	<code>GET /api/sources</code> + static text (below)

Change address flow BIND

1. Tapping *Change address* opens a compact confirm sheet: "*This will replace your current results. Continue?*" with *Continue* and *Cancel*.
2. Continue goes to `/onboarding?change=1`, which starts at the address step (B1). The onboarding screen does the geocoding and the `POST /api/onboard` (with a fresh `request_id`), and discards the typed address afterward, exactly as in first-time onboarding.
3. Onboarding, not Settings, clears the stored Today and HomeGuard readings and requests the fresh reading. When the (rounded) location changes, the server clears that day's stored score and resets the daily-score limiter, so a one-hour-old cache can no longer serve the old address's readings. Re-entering the same address changes nothing and clears nothing. (If a `fresh=1` request still returns 429, keep the stored-readings notice and retry after `retry_after_seconds`.)
4. In change mode the flow continues to step 3 (Household), step 4 (Your home), then the results reveal, then Today. **Both steps are shown, prefilled** with the profile's current groups, build year and water source, because a new home may have a different water source and different people living in it. **DECISION D2** (Part P10): default keep, as built. Yogi's call if he wants Household skipped.
5. Back on the address step in change mode returns to `/settings` (Part P7).

Accessibility contract LOCKED

Control	Stored key	What it does
Text size	<code>halo.device.textScale</code>	Slider with 5 steps: 100%, 115%, 130%, 150%, 175%. Sets <code>--halo-text-scale</code> on <code>.halo-app</code> and <code>data-text-scale</code> on it. All font sizes are rem-based and multiply by that variable (Part P2), so nothing is a fixed pixel size
Higher contrast	<code>halo.device.contrast</code>	Sets <code>data-contrast="high"</code> . Tokens swap to the high-contrast values (text and border colors only; status colors keep their hues, and the six severity words stay printed next to color)
Reduce motion	<code>halo.device.motion</code> (<code>reduce</code> <code>full</code>	Sets <code>data-motion="reduce"</code> . Absent means follow the system <code>prefers-reduced-motion</code> . When reduced, the ScoreRing draws its final state with no sweep, sheets appear without sliding, and

	absent)	skeleton shimmer stops
--	---------	------------------------

Live preview sentence under the slider, in the current size: *"Your air today is Moderate. Take it easy outside if you're sensitive."* (Fixed sample, not real data.) The three values are read once at app start by the shell, before first paint where possible, so the page never flashes at the wrong size. These keys are device settings, never sent to the server, and never cleared by `clearReadings()`. They are cleared by Delete everything.

Notification switch copy **PROPOSED**

Preference	Name	Description	When it really fires (from the server)
<code>air_quality_change</code>	Air quality changes	When the air gets worse than yesterday. Earlier for homes with someone sensitive	Today is worse than yesterday (or yesterday has no reading), and today is Elevated or worse (High or worse for homes with no sensitive group). Message: "Air quality dropped to [label] today."
<code>weather_advisory</code>	Weather advisories	Flood and heat advisories for your county	Any National Weather Service alert with "flood" or "heat" in its name for the home's point. That includes watches and warnings, which the message still calls an advisory. Boil-water advisories are not sent: no national source exists, so the description must not promise them (the blueprint's "boil-water" wording is removed)
<code>new_water_results</code>	New water results	When new federal testing results are published for your utility	New testing results were published for the utility
<code>radon_season</code>	Radon testing season	A winter reminder when testing is most accurate	October to February, radon Zone 1 or 2, and no radon test recorded (the server has no record of radon tests yet and always treats it as "no test", so the reminder fires every winter in Zone 1 or 2). Zone 3 never gets it (every NC-08 county is Zone 3)
<code>season_summary</code>	Season summaries	A look back at your journal when a season ends	A season just ended and there are journal entries

The switch names and descriptions are PROPOSED for content-owner review (D5). The "when it fires" column is verified against the server and is for builders and the Alerts owner, not shown in the app. Turning a type off stops both inbox alerts and device notifications of that type. Alert titles and messages as the server sends them are in C3.

Device notifications (push) **BIND** **DECISION D8**

Device notifications are built **after the core pages** (Part P10, D8). Until then the switch is not rendered, and the Notifications section shows only the five type switches. The backend is ready: `POST /api/push/subscribe` and `DELETE`, and the daily job already sends pushes to every subscription once the four VAPID variables are set on Vercel.

Piece	What it is
Files	<code>public/sw.js</code> (service worker), <code>lib/frontend/push.ts</code> (subscribe, unsubscribe, permission state), the switch in <code>components/settings/NotificationsSection.tsx</code>
When the switch shows	Only when <code>'serviceWorker'</code> in navigator, <code>'PushManager'</code> in window and <code>NEXT_PUBLIC_VAPID_PUBLIC_KEY</code> is set. On iPhone this needs the app added to the home screen, so when push isn't available show nothing rather than a broken switch
Turning on	Ask browser permission only on this tap, and only after the person has used the app at least once (\$19.5). Register <code>/sw.js</code> , convert <code>NEXT_PUBLIC_VAPID_PUBLIC_KEY</code> from base64url to a <code>Uint8Array</code> for <code>applicationServerKey</code> , call <code>pushManager.subscribe({ userVisibleOnly: true, applicationServerKey })</code> , then <code>POST /api/push/subscribe</code> with the subscription JSON (<code>endpoint</code> , <code>keys.p256dh</code> , <code>keys.auth</code>)
Permission denied	Switch returns to off, with the line <i>Notifications are blocked in your browser settings.</i> under it
Turning off	<code>DELETE /api/push/subscribe</code> with <code>{ endpoint }</code> , then <code>subscription.unsubscribe()</code>
Delete everything	Unsubscribe the browser subscription before calling <code>DELETE /api/account</code> (the server removes its own rows with the account)
Service worker handlers	<code>push</code> : parse the JSON <code>{ id, type, title, message, url }</code> and show a notification. <code>notificationclick</code> : focus an open HALO tab or open <code>url</code> . Landing pages by type: air and weather to <code>/today</code> ; new water results to <code>/home?risk=water</code> ; radon to <code>/home?risk=radon</code> ; season summary to <code>/journal?mode=trends</code>
Rate limit	429 "Too many changes. Try again later." on subscribe or unsubscribe: switch returns to its previous state with the failure toast

Privacy statement LOCKED

What HALO stores: your approximate coordinates, your county, which water utility serves you, your home's build year, and which household groups you told us about.

What HALO never stores: your street address, anyone's name, anyone's age or birthday, and any medical record. Your journal notes stay on your account and are never shared with anyone.

The third paragraph depends on which Assistant mode is live (the server reports it as `sends_household_context` on `GET /api/assistant`; assume false until that exists). **DECISION D4** default is off.

When false (default): When you ask HALO's assistant a question, your question is sent to an AI service to write the answer. Your readings and symptom log are not sent. Don't include names or addresses in your question.

When true: When you ask HALO's assistant a question, the readings and logged symptoms that help answer it are sent to its AI service to write the answer. Your notes are never sent.

Also stated here, per §37.4: HALO collects no behavioral analytics.

OPEN: THE "STORES" LINE IS INCOMPLETE. §16.2 asks for exactly what is and isn't stored. The locked line omits water source, renting or owning, language, state, daily reading history, journal entries, alerts, device notification subscriptions, notification settings and the rounded coordinates' precision. **PROPOSED** addition for Yogi to approve (don't change the locked text without it): *"It also stores your water source, whether you rent or own, your language, your daily readings, the journal entries you write, the alerts it sends you, your notification settings, and, if you turn on device notifications, a code that lets your browser receive them."* Coordinates are rounded to about 100 meters before saving, and street addresses are never stored.

DECIDED: THE ASSISTANT AND JOURNAL DATA. The Assistant can send readings and journal symptom names and dates to its AI provider (never the free-text notes) to answer questions like "why does my son cough more on some days?" (§14.8, §18.2). The team approved that for this build, but the free model may log prompts, so it ships **off** until Yogi turns it on. The server variable `ASSISTANT_SEND_HOUSEHOLD_CONTEXT` must be exactly the string `false` for that: any other value, including unset, currently turns household data on (Part P9 ask 2 fixes the default). The statement above changes with the switch, so it can never claim more privacy than the server provides.

What HALO knows about you, and Clear journal BIND

- **"What HALO knows about you"** at the top of Data and privacy shows the stored values back: county, water utility ID, water source, build year, renting or owning, language, household groups. From `useProfile()`. A missing value reads *No data*. Makes "everything must be inspectable" (§16.1) literal. This is a required part of the page, not optional.
- **Clear journal** sits above Delete everything. Tapping it first calls `GET /api/journal?from=2000-01-01` and reads `count`. Confirm sheet: *"This deletes all [n] of your entries permanently."* with a destructive confirm. At `count = 0` the row is disabled with *No entries to clear*. Confirm calls `DELETE /api/journal?all=true`; on success clear any stored journal data and toast *Journal cleared* (the one success toast in Settings, because the effect isn't visible on this page). On failure, stay on the sheet with the generic error.

Delete everything LOCKED BIND

Tapping *Delete everything* opens a compact sheet stating exactly what gets deleted: *"This permanently deletes everything HALO knows about your household. This cannot be undone."* With a field that requires typing *DELETE* (exactly, capitals), and a destructive confirm button that stays disabled until it matches (§49.3). Steps on confirm:

1. Unsubscribe this browser from push if it is subscribed (see above). A failure here does not stop the deletion.
2. `DELETE /api/account` with body `{ "confirm": "DELETE" }`. On failure: stay on the sheet with the generic error *Couldn't delete your data. Nothing was deleted. Try again*. Do not clear anything locally.
3. On `{ deleted: true }`: remove every `halo.frontend.v1.*` key, every `halo.device.*` key and `halo.onboarded`; call `supabase.auth.signOut()` and discard the session; then `router.replace('/onboarding')`.
4. Onboarding opens on Welcome with the toast *Your data has been deleted*. (8 seconds, the shell's toast).

Tier 1 per the blueprint's own triage. Never test this against production data (Part P4).

About LOCKED

- **Data sources.** Render `GET /api/sources` exactly as returned: `sources: [{ name, provides, last_retrieved }]` (currently 10 entries). Show `name` as the row title, `provides` as the description, and `last_retrieved` as *Retrieved September 27, 2026* (date only, in the person's time zone). A `null` date reads *Retrieval date not recorded*, never a guess and never "No data" (this is a fact about the record, not a missing reading). Don't hard-code the list of sources in the app, so a new source appears without a release.
- **How scores are produced**, in plain language (static copy in `copy.settings`): daily conditions combine as independent risks; home conditions combine multiplicatively; missing data is excluded rather than counted as zero; one severe problem can't be averaged away by several mild ones.
- **Version.** The collapsed summary and the top of the expanded section read *Version 1.0*, a constant in `copy.settings` (changed by hand at release).
- **Who built HALO and the repository link.** **PROPOSED** text, content owner (Aadit, D5) to confirm names before submission: *"HALO was built by four students in North Carolina's 8th Congressional District for the 2026 Congressional App*

Challenge." and a link *View the source code* to the repository URL (a placeholder constant until the team publishes it; hide the link while the constant is empty). No names or personal contact details in the app itself.

- **Volunteer organizations.** No mention here. (All 13 organizations are still unverified, D9; verify before submission.)

Language and Spanish

DECISION: the blueprint's rule is that machine-translated health guidance isn't acceptable; ship English-only rather than unreviewed Spanish (§68). The Language section appears only when a person has reviewed the Spanish interface strings and Learn content and the environment flag `NEXT_PUBLIC_SPANISH_REVIEWED=1` is set. Until then the profile's `locale` stays `en` and there are seven sections. Either way, every string goes through a translation layer from day one, no button's width is fixed to its English label, and Learn trusts the `locale` it gets back (it reports the language actually served).

Visual decisions DESIGN

Accordion rows with value summaries (`AccordionRow` , Part P2), switches (`Switch`), the text-size slider with live preview, the destructive button and delete sheet (built on `BottomSheet`), the formatted privacy statement. The wireframe for Settings shows: every section closed; each section open (eight screenshots); Household loading and save-failed states; Notifications with push available and unavailable; Clear journal with 0 and with several entries; the Delete sheet empty, matching and failed; About with a null date. Font sizes and colors come from the existing tokens (A11, Part P6); nothing new is invented here.

B7. Map and the District panel

Purpose LOCKED

Answer *is this just my problem, or is it everyone's?* Every other page is about one household; this one places it in its region, which is what turns a personal utility into something civic. The most visually striking screen, so it matters disproportionately in a demo (§13.1).

Owner: Vibhav (Lane 3, Part P), after Phase 0. The Map is the largest single page (four data layers, a slider, a district panel, a map library), so it starts early in Lane 3. It has no dependency on other pages except `openLearn` (Part P5) and the shell.

Address and entry

`/map` (Water layer, overall severity), Tab 3. The map library loads only when this tab is first opened, never on app start (§35.2). The header sits over the map. On wide screens the map column widens to 600px, the one exception to the 430px cap (§53.2); the shell does this by setting `data-wide` on `.halo-app` for `/map` (Part P2).

Parameter	Accepted values	History
<code>layer</code>	<code>water</code> <code>radon</code> <code>facilities</code> <code>air</code>	replace
<code>contaminant</code> (water only)	<code>pfoa</code> <code>pfos</code> <code>pfhxs</code> <code>pfna</code> <code>genx</code> <code>hazard</code> (means the combined hazard index; the default overall view has no parameter)	replace
<code>quarter</code> (water only)	Must match <code>^\d{4}Q[1-4]\$</code> and be one of the response's <code>quarters[]</code> ; otherwise ignored. Applies only when the slider exists	replace
<code>system</code>	A water system ID (<code>pwsid</code>), or a facility <code>id</code> , air station <code>id</code> or county name for the active layer. Opens that feature's sheet once its layer has loaded. A value not found in the loaded layer is ignored (no error)	push (a sheet, Part P3)
<code>panel</code>	<code>district</code> : opens the district panel	push (a sheet, Part P3)

Precedence when the page opens: address parameters, then the last state this session (kept in memory only: viewport, layer, contaminant, quarter, selection, per §46.3; the tab bar link is a plain `/map`, so returning to the tab restores it), then the defaults. Unrecognized or invalid values are ignored and the default is used, never an error page (A1). Changing layer, contaminant or quarter uses `router.replace` so the Back button never steps through every chip tap.

Map library, tiles and county shapes DECISION D7

Default: Mapbox GL JS on the team's existing Mapbox account, token `NEXT_PUBLIC_MAPBOX_TOKEN` (already in the environment table). Package `mapbox-gl`, loaded with a dynamic import inside the Map page only. A muted, low-saturation base style, and a dark variant of the same style when the phone is in dark mode (§59.3). The map's required attribution control (Mapbox and OpenStreetMap) stays visible in every state; it is never hidden to save space. Data layers are drawn as WebGL layers (GeoJSON sources with circle and fill layers), not DOM markers, because the water layer has hundreds of systems.

County shapes. Radon needs county polygons. The repository already has them: `lib/data/ncCounties.json` (U.S. Census Bureau TIGERweb, simplified to about 200 m, 100 counties, about 160 KB, each with `name` such as `"Union County"` and `geoId`). Phase 0 or the Map owner converts it once to `public/data/nc-counties.geojson` and loads it only when the Radon layer or the district panel needs it. The radon API's county names carry the same "County" suffix, so join by exact name; trim and compare case-insensitively as a guard. A county in the response with no shape is skipped, never guessed. Do not import the JSON under `lib/` into the browser bundle (it is server code and would add 160 KB to every page).

Cached map data. Layer responses are cached for the session and offline (§36) in an IndexedDB database named `halo-map`, keyed by identity + layer (+ quarter), holding the response and its `assembled_at`. It contains only public data, but it is keyed and cleared like household data (`clearReadings()`, Delete everything), per Part P2. Never use `storeReading` for map data: it strips coordinates. Offline, show the cached layer with *Showing saved map data from [date]*. and disable the quarter slider if that quarter isn't cached.

Data calls BIND

Call	Feeds
<code>GET /api/profile</code> (<code>useProfile()</code>)	Home pin (<code>lat</code> , <code>lng</code> , rounded by the server to about 100 m, so the pin is approximate), own system (<code>pwsid</code>), <code>county</code>
<code>GET /api/map?layer=water</code>	<code>features[]</code> , <code>counts</code> (the live count line), <code>count</code> and <code>located</code> , <code>quarters[]</code> (oldest first), <code>assembled_at</code> , <code>population_source</code> , <code>prebuilt</code>
<code>GET /api/map?</code> <code>layer=water&quarter=2024Q3</code>	The same shape (except no <code>prebuilt</code>), only systems that have a reading in that quarter, <code>quarter</code> set. <code>assembled_at</code> here is the request time, not the data date (see Freshness)

GET /api/map?layer=radon	features[] → county, zone, severity; joined to the county shapes by name. One feature per county (100)
GET /api/map?layer=facilities	features[] → facility points (fields in the sheet table), plus scope ("NC-08") and selection (a server sentence, see the honesty line)
GET /api/map?layer=air	features[] → monitoring stations
GET /api/district	Comparison banner (household_county), district panel
Static asset public/data/nc-counties.geojson	County shapes for the radon fill

Each layer is fetched the first time it is selected and cached for the session (and offline). The daily job builds every layer once a day, so a normal request is fast; a cold rebuild can take up to about 20 seconds and up to 60 on the server, hence the 20-second client deadline (Part P3). The water layer is the default layer: request it at once; request /api/district only when the panel opens or the water banner needs it (after the water layer has drawn).

Layout LOCKED

The map fills everything between header and tab bar, edge to edge. Floating on top: a row of layer chips; beneath them (Water only) the contaminant selector, a live count line, and a conditional rescission notice; a locate button (crosshair icon; tapping it animates to the home pin at the default zoom, §54) and zoom buttons on the right edge (the blueprint hides zoom buttons on touch devices; that's Tier 3: **default: visible everywhere**, §65.3); a legend lower left; the time slider (Water only, when the data has quarters); a comparison banner; and a handle at the very bottom, *See the full NC-08 picture*, opening the district panel (§13.2, §46.2).

OVERLAP BUDGET PROPOSED : on a small phone (375 x 667, so about 545 px of map after the header and tab bar) the floating controls must never cover more than 60% of the map height, or the map stops being a map. Suggested numbers: top stack (chips 44, selector 44, count 24, notice 24, gaps) about 160 px; bottom stack (handle 48, banner 56, slider 64, legend 28, gaps) about 210 px. That is already 68%, so the design owner must use these hiding rules in the wireframe: (1) below 600 px of map height the comparison banner hides while the slider is showing; (2) the legend collapses to a single *Legend* button that opens a small popover; (3) the rescission notice moves into the contaminant selector's popover as an info line. Design confirms the actual sizes; the 60% ceiling is the rule.

Drawing order (bottom to top): base map, radon fill, water circles (largest population first, so small systems on top of big ones stay tappable), facilities, air stations, the outline of the household's own system, the selected feature's ring, the home pin. **Tap targets**: a feature's drawn size can be smaller than 44 px, so a tap selects the nearest feature within a 22 px radius of the tap point (draw an invisible larger hit layer). Water circle radius is a square-root scale of population, from a 6 px minimum to a 22 px maximum **PROPOSED** .

Initial state and memory LOCKED

Centered on the household at a zoom showing roughly their county and its neighbors: close enough to identify their own system, wide enough to show others exist. Home pin visible; Water layer, overall severity. If the profile has no lat / lng (should not happen after onboarding) or isOutsideNC(profile, county) is true, center on NC-08 as a whole, hide the locate button, and hide the pin unless it falls inside the view. This is an NC-08 app, so an outside-NC household still sees the district; the comparison banner is hidden for them because household_county is null or unranked. On return within a session, restore the last viewport, layer, and selection (§46.3).

Layer chips LOCKED

Water · Radon · Facilities · Air. One active; active chip visibly filled. Switching re-renders data without moving the viewport, so the household's position stays anchored (§13.3). The row is a radiogroup : Left and Right arrow keys move between chips and select (roving tabindex), Tab moves on.

What each layer draws LOCKED BIND

Layer	Drawn as	Binding
Water	One circle per public water system at its service-area center. Size by population served (square-root scale so big systems don't swamp small ones); fill by severity. No reading for the selected compound: small grey marker, legend PROPOSED <i>No detection reported</i> , not "Not tested for this". The loaded EPA data kept only detections (non-detects were dropped), so a system with no reading may have been tested and found nothing. The same limit affects the district's tested counts until Vibhav reloads the data with non-detects or adds a per-compound tested flag (F4, still open). In a quarter view, systems with no reading in that quarter are absent from the response and are not drawn at all. Water and air layers never cluster (§46.5): the pattern is the point. Systems without a location are not drawn; they are listed in the district panel (see below)	lat, lng, population (may be null: draw at the minimum size), overall_severity or the selected compound's severity . located vs count tells you how many can be drawn
Radon	Filled county shapes by the county's radon-zone severity, low enough fill opacity that county names stay legible. No clustering, no points	features[].severity, zone . Zone 1 is severity high , Zone 2

		elevated , Zone 3 good
Facilities	Points colored by compliance; clusters that separate as you zoom in; cluster badge shows a count; tapping a cluster zooms to fit it rather than opening a sheet (§46.5). Use the library's built-in clustering (cluster radius about 40 px, stop clustering at about zoom 12)	lat , lng , severity , compliance_status (no_violation good · violation elevated · significant_violation high · unknown no_data)
Air	One point per monitoring station by current air level, visually distinct from water circles (a square or diamond marker, not a circle, so the shape carries the difference and color isn't the only cue)	lat , lng , severity , aqi
Home pin	Always on top of every layer, never recolored by severity: the household never loses itself	Profile lat / lng

HONESTY LINE (FACILITIES). The facilities layer is not every regulated site. The server selects **active major facilities and repeat violators (more than 4 of the last 12 quarters in noncompliance), inside NC-08 only**. Show, under the legend when Facilities is active: *Shows major facilities and repeat violators in NC-08, not every regulated site*. (static copy; do not display the server's `selection` sentence). A facility outside NC-08 will never appear, even if it is near an out-of-district household.

Legends LOCKED

Layer	Legend entries, with the same colors, icons and words as everywhere else (A10)
Water, overall	Good · Moderate · Elevated · High · Severe, then grey <i>No detection reported</i> . When a system has <code>coverage: "unscorable"</code> : PROPOSED <i>Detected, no federal limit to compare</i>
Water, one compound	The same, plus the limit for that compound: <i>Limit: 4 ppt</i> (from the count line's response, <code>contaminants[X].limit_ppt</code>)
Radon	<i>Zone 1, highest potential</i> (High color) · <i>Zone 2, moderate</i> (Elevated) · <i>Zone 3, lowest</i> (Good). EPA zone definitions: Zone 1 predicted county average above 4 pCi/L, Zone 2 between 2 and 4, Zone 3 below 2. The legend never uses the word "safe"
Facilities	<i>No violation</i> · <i>Violation</i> · <i>Significant violation</i> · <i>Unknown</i> (No data color)
Air	Good · Moderate · Elevated · High · Severe (by AQI: 50, 100, 150, 200 boundaries). Caption <i>Latest hourly reading</i>

The contaminant selector (Water) LOCKED

The single most interactive element in the app: it lets someone answer their own questions rather than read a fixed answer (§13.4). Choosing a compound recolors every system by that compound alone, against its own limit, animated over about 300ms (motion off: instant).

Option	Colors by	Binding
Overall water health (default)	Each system's combined severity across everything measured	<code>overall_severity</code>
PFOA · PFOS	That compound vs its 4 ppt limit	<code>contaminants.PFOA.severity , contaminants.PFOS.severity</code>
PFHxS · PFNA · GenX	vs its 10 ppt limit, with the rescission notice	<code>contaminants.PFHxS , .PFNA , ["HFPO-DA"]</code> (short label <i>GenX</i> on the selector chip; <code>contaminantLabel()</code> , which reads <i>GenX (HFPO-DA)</i> , in sheets and tables); notice when <code>proposed_for_rescission</code>
Combined hazard index	The federal combined measure: 1.0 or above is a violation even when no single compound exceeds its own limit. Applies only to systems where two or more of four PFAS were detected, so systems without a hazard index (<code>hazard_index: null</code>) get the No detection reported marker	<code>hazard_index.severity</code> (<code>value</code> and <code>exceeds</code> shown in the sheet), with the rescission notice when <code>hazard_index.proposed_for_rescission</code>

- An untested entry is `{ severity: "no_data", limit_ppt, proposed_for_rescission }` with no value, ratio, or date; check before reading them. Absence of a measurement is never shown as safety.
- `overall_severity: "no_data"` with `coverage: "unscorable"` means compounds were detected but none has a federal limit; see the water legend.
- **Count line** beneath the selector, updating with the selection: the most quotable line on the page. Bind the map response's own `counts` ; they describe exactly the systems drawn, statewide. The district panel's figures are NC-08 only. Strings (**PROPOSED** , numbers with thousands separators, singular when 1):
 - Overall: *[counts.over_limit] of [counts.total] systems exceed a federal limit.*
 - One compound: *[by_contaminant[X].over] of [by_contaminant[X].tested] systems tested for [PFOS] exceed the limit.*

(denominator is systems tested for that compound, matching the district panel's wording)

- Hazard index: *[counts.hazard_index_over] systems exceed the combined PFAS hazard index.* (no denominator exists)
- Zero: *No systems exceed the limit for [PFOS].*
- In a quarter view append *in Q3 2024*. The counts then describe only the systems that reported that quarter.
- Rescission-flagged selection (a compound's `proposed_for_rescission`, or `hazard_index.proposed_for_rescission` for the hazard index): one-line notice beneath the count that the standard itself is contested. **PROPOSED** *EPA has proposed removing this limit. It still applies today.* (Content owner verifies against current federal status before launch.)
- The household's own system (`pwsid` matches the profile) stays outlined and labeled regardless of selection.
- Any other layer active: the selector and count line are replaced by that layer's legend and the count line *[count] [stations | counties | facilities] shown*.

Tapping a feature LOCKED

Every feature opens a compact bottom sheet over the map (never navigates away); the feature shows as selected while its sheet is open and deselects when it closes. Height is the one exception to standard sheets: sized to its content, at most 50% of the screen, so the selected feature stays visible (A9). The blueprint pans the map just enough to keep it visible above the sheet (\$13.6, \$46.6 and \$46.7); that is Tier 3: **default is no auto-pan**; the sheet may cover part of the map and people drag (\$65.3). Each sheet: title row, severity pill, body, footer with the data date. Opening a sheet adds one history entry (address `?system=`); Learn opened from inside it replaces that entry (Part P3).

Feature	Sheet contents, in order	Binding
Water system	System name (title); ID beneath; severity pill; <i>Serves [n] people</i> (or <i>No data</i> when <code>population</code> is null); a bar row for each of PFOA, PFOS, PFHxS, PFNA, GenX using the threshold-row component (value, limit, ratio, severity; a row with no reading says <i>No detection reported</i>) and each row's own sample date <i>Sampled March 4, 2025</i> (from <code>date_iso</code>); the hazard index row, <i>Hazard index 1.4. 1.0 or above is a violation.</i> with <i>Based on the latest sample.</i> (hidden when <code>hazard_index</code> is null; show the rescission notice when flagged); footer <i>Latest sample: March 4, 2025</i> (from <code>latest_sample_iso</code> , or <i>No data</i>); <i>Learn about PFAS</i>	<code>name, pwsid, overall_severity, population, contaminants[X] (value_ppt, limit_ppt, ratio, severity, date_iso), hazard_index (value, exceeds, basis, components), latest_sample_iso.openLearn({ topic: 'pfas', reading, context })</code> uses the selected compound if it has a value, else the compound with the highest ratio; context is that compound, value and limit (C1)
County (radon)	County name; zone as a severity pill; two sentences on what the zone means; the sentence that zones predict county averages rather than measuring a home; <i>Learn about radon</i> (<code>openLearn({ topic: 'radon', context: { county, zone } })</code>)	<code>county, zone, severity.</code> PROPOSED copy: Zone 1 <i>Homes in this county average above 4 pCi/L, the level where EPA recommends fixing a home.</i> Zone 2 <i>Homes in this county average between 2 and 4 pCi/L.</i> Zone 3 <i>Homes in this county average below 2 pCi/L.</i> then always <i>Any home can have high radon. Only a test tells you about yours.</i> (content owner verifies against EPA)
Facility	Facility name; compliance status pill; which programs regulate it (small tags, the program names as the server sends them); <i>Quarters with a violation in the last 3 years: 5 (of 12); Programs currently in violation: 2</i> (only when <code>open_violations</code> is more than 0); <i>Penalties in the last 5 years: \$12,500</i> (only when <code>penalties</code> is more than 0); <i>Last inspection: June 3, 2026</i> (or <i>No data</i>); county	<code>name, compliance_status, severity, programs[], recent_violations</code> (this is quarters in noncompliance out of the last 12, not a count of violations), <code>open_violations, penalties</code> (dollars), <code>last_inspection, county</code>
Air monitor	Station name; <i>AQI 87</i> and severity pill; dominant pollutant (through <code>displayText</code>); <i>[d] miles from your home</i> ; last reading time (format below)	<code>name, aqi, severity, dominant_pollutant, last_reading_at.</code> Distance is the haversine distance from the profile <code>lat / lng</code> to the station, in miles to one decimal, minimum <i>0.1</i> ; hidden when the profile has no coordinates
Home pin	<i>Your home</i> ; county; today's score (only when a stored Today reading exists; never fetched here); primary <i>Go to Today</i> (closes the sheet by replace, then pushes <code>/today</code>)	<code>profile.county, stored daily display_score</code>

Reading time format (air stations): reading today *Today at 3:00 PM*, yesterday *Yesterday at 3:00 PM*, older *September 26 at 3:00 PM*, in the device's time zone. **Freshness format** (below) is the same rule, prefixed *Updated* .

Time slider (Water) LOCKED BIND

Federal water testing is quarterly, so the water layer can move through time: a horizontal slider with quarter labels above the comparison banner; dragging recolors every system to its severity at that quarter, showing whether contamination is spreading, receding, or steady (\$13.7). **Shown only when** `quarters` **has two or more entries**; per the blueprint, if the data holds only the latest reading, the slider is cut and the layer is static (\$68). Yogi confirmed the data has the history: about 12 quarters, 2023Q1 to 2025Q4 (F4).

- **Positions.** The response's `quarters[]` (oldest first) followed by one final position, *Latest*, which is the default and means "each system's most recent reading", i.e. the normal prebuilt water layer. Quarter labels read *Q3 2024* (from `2024Q3`). Show first, last and every third label if space is tight.
- **Loading a quarter.** Each quarter is a live request (`?layer=water&quarter=2024Q3`), sent when the person releases the thumb, or 250 ms after the last keyboard step. Abort any request still in flight when a new position is chosen. Keep the previous colors on screen with a small *Loading Q3 2024* label until the response lands; on failure, snap back to the previous position and show the layer error callout. Cache each quarter's response for the session (and in `halo-map`).
- **What the quarter shows.** Only systems that reported in that quarter; the rest are not drawn (not grey). The count line says *in Q3 2024* and counts only those systems. The contaminant selection persists across quarters.
- **Freshness while scrubbing.** Use the `assembled_at` of the default (Latest) payload. A quarter response's `assembled_at` is the moment it was requested and would say "Updated today" over years-old data.
- **Keyboard.** The slider is a native range input (or `role="slider"` with `aria-valuetext="Q3 2024"`): Left and Right step one quarter, Home and End jump to first and Latest.

Comparison banner LOCKED

A small floating card near the bottom with one sharp, specific comparison, updating with the layer; tap opens the district panel (§13.8). Deliberately a single number rather than a chart: specific rankings are what people repeat to each other.

Layer	Banner	Binding
Water	<i>Cabarrus County ranks 34th of 100 North Carolina counties for water quality. with a second, smaller line 1st means the fewest people on water over a federal limit. (§13.8, and the rank's meaning stated, because "34th" alone doesn't say which end is good)</i>	<code>district.household_county → county, water_rank, of</code> . Bind <code>of</code> : it is the number of counties actually ranked (<code>ranked_counties</code>), which can be below 100. Hide the banner when <code>household_county</code> is null, or <code>water_rank</code> or <code>of</code> is null. Ordinals come from one shared <code>ordinal()</code> in <code>format.ts</code> . An unranked county comes back as an object with <code>water_rank: null</code> (not a null <code>household_county</code>), so test the rank, not the object. <code>share_over</code> is a 0 to 1 fraction of the <i>people served</i> on over-limit systems; it is not comparable with the district-versus-state pair below, which is a share of <i>systems tested</i>
Radon	PROPOSED <i>Union County is Zone 3. 8 of 100 North Carolina counties are Zone 1. (counts from the radon features, never hard-coded)</i>	Profile county + radon features (<code>count</code> is the county total)
Facilities / Air	PROPOSED <i>[n] facilities within 10 miles have open violations. (<code>open_violations</code> more than 0, haversine from the profile point) · The nearest air monitor is [d] miles away. Zero case: No facilities in this layer are within 10 miles of you. Hidden when the profile has no coordinates</i>	Layer features + profile location

What the ranking is (verified in the server code): counties are ranked by the **share of the people served by mapped water systems whose most recent sample exceeds a federal limit**; 1 is the lowest share; ties share a rank; counties with no mapped system and population figure are not ranked. The district panel repeats this in one plain line, *Ranked by the share of people served by water systems over a federal limit*. (do not display the server's `ranking_rule` string).

Freshness LOCKED

Every layer shows when its data was assembled, e.g. *Updated today at 7:00 AM*; undated environmental data invites the reasonable suspicion that it's stale (§13.9). **BIND** the active layer's `assembled_at` (for Water always the default payload's, see the slider). The line sits with the legend. The daily job builds layers at 11:00 UTC, so "today at 7:00 AM" (6:00 AM in winter, Eastern) is normal; a layer older than 36 hours is rebuilt by the server on request, so the client needs no staleness rule of its own.

The district panel LOCKED

An expandable panel inside Map, not a separate page. Opened by dragging the handle up or tapping the comparison banner or handle; address `/map?panel=district` (push, so Back closes it). It is a standard-height sheet (about 70%). Shows no household data: it's the view a community group or legislative office could open (§20). Not tied to the active layer.

Content, in order	Binding and rules
Headline figure: people affected across the district, at hero size, with a one-line caption PROPOSED <i>people are served by water systems over a federal</i>	Data guard first: if <code>total_systems</code> is 0, or <code>unlocated_systems</code> is 90% or more of <code>total_systems</code> , the server's geography lookup failed and every figure below is meaningless (the server would otherwise report 0 systems over the limit): show the district error state, never an all-clear (Part P9 ask 9 fixes it at the source). Then <code>affected_population</code> . Prefix <i>At least</i> when <code>affected_population_complete</code> is false. If <code>affected_population</code> is null (population data couldn't be joined), show <code>systems_over_limit</code> as the hero instead, with <i>water systems over a federal limit</i> . If it is 0 and <code>systems_over_limit</code> is 0: <i>No water system in NC-08 is over a federal limit</i> .

<i>limit</i>	
Comparison bar pair: district vs state, share of tested systems over a limit	<code>state_comparison.district_share_over</code> and <code>.state_share_over</code> are fractions from 0 to 1 : multiply by 100 for the percent (0.214 is 21%). Either can be null: hide the pair. Caption <i>Share of tested water systems over a federal limit</i> . Denominator is systems tested for anything regulated (<code>systems_tested_any</code>)
County table: one row per NC-08 county, always nine rows: county, worst water severity, radon zone, systems over limit	<code>counties[] : county, in_district ("whole" or "partial"), systems, systems_over_limit, worst_water_severity, radon_zone, radon_severity</code> . Partial counties (Cabarrus, Mecklenburg, Robeson) carry a small <i>Partly in NC-08</i> tag and a footnote <i>Only water systems inside the district are counted for counties that are partly in it</i> . A county with 0 systems shows <i>No data</i> for water (the server sends <code>no_data</code>)
Per-compound detail: how many systems tested, how many over, how many people they serve	<code>by_contaminant[X]</code> for PFOA, PFOS, PFHxS, PFNA, HFPO-DA (label <code>contaminantLabel()</code> : GenX (HFPO-DA)): <code>tested, over, limit, population_over, over_population_unknown</code> . Compose the sentence in the app copy: <i>[over] of [tested] systems tested for [PFOS] exceed the limit</i> . (the same wording as the server's <code>exceedance_lines</code> , with GenX in place of HFPO-DA). When <code>over_population_unknown</code> is more than 0 prefix the people figure with <i>At least</i> . Same caveat as the map: "tested" counts systems with a detection on record
Named local specifics: the Concord treatment plant serving 120,000+ customers above the federal limit; Union County's reading against the federal maximum; compliance timelines extending to 2031	Static content in <code>lib/frontend/copy/district-facts.ts</code> : an array of <code>{ text, source, dateVerified }</code> , empty at first. Each item is added only after a person verifies the figure against its source (\$66), and the section is omitted entirely while the array is empty . Nothing here is typed from memory. Content owner: Aadit (D5)
Systems not shown on the map	<code>unlocated_systems</code> is a number , not a list. Show <i>[n] water systems have no mapped service area and aren't shown on the map</i> . (hidden at 0). The number counts every located-less system statewide, the same as the water layer's <code>count</code> minus <code>located</code> . The names and severities come from the water layer's features whose <code>lat</code> or <code>lng</code> is null (statewide, filtered to unlocated); an expandable list of those, if the water layer is loaded, is PROPOSED
Footer: sources and date	<i>Data assembled September 28, 2026</i> (<code>assembled_at</code>). <i>District and county lines: [boundaries.district]; [boundaries.counties]</i> . (server strings via <code>displayText</code>). Never show the response's <code>note</code> or <code>ranking_rule</code> text: <code>note</code> still claims there is no per-county breakdown, which is no longer true

OPEN (VIBHAV AND YOGI): WHICH DISTRICT MAP. The district lines in the app are the 119th Congress NC-08 (the 2023 map). A newer 120th Congress map exists for the 2026 elections (`lib/data/nc08District.json` holds the one in use). The panel should say which map it uses. **PROPOSED** footer line: *District lines: 2023 map*. Decide before submission whether to switch maps; if switched, the counties list and the "partial county" tags change.

Loading and failure **LOCKED**

- Loading: neutral canvas with a centered spinner and *Loading map data*. Controls present but disabled (the one place a spinner is right, \$46.9).
- **Layer error** (the map request returns 503 because a source is down, or 500, or times out at 20 seconds): base map + home pin + a callout *This layer's data couldn't load. with Try again*. Never show the server's message (a 503 carries one, but it names internal sources). The other layers stay usable. A 400 means the app sent a bad request; treat it as a bug: same callout, and log it in development.
- **District error** (`/api/district` 500 or timeout): the map keeps working; the comparison banner is hidden; opening the panel shows the panel chrome with an inline error and *Try again*.
- Map library fails entirely (no WebGL, missing or rejected token, or the import fails): show the district panel's content as a full page, so the tab is never empty, with a one-line notice *The map couldn't load on this device*.
- No data in view: *No data for this layer in the area you're viewing. Try zooming out*. (shown when the layer loaded but no feature is inside the viewport).
- Offline: last loaded datasets from `halo-map`, with the saved-data line. A layer never loaded offline shows the layer error callout without Try again until the app is back online.

Keyboard and screen reader model **LOCKED**

Focus order: header, layer chips (one tab stop, arrows inside), contaminant selector, count line (read-only, `aria-live="polite"` so a change is announced), locate, zoom in, zoom out, the map itself, time slider, comparison banner, district handle. The map canvas is one tab stop labeled *Map of North Carolina. Use the arrow keys to move and plus and minus to zoom. Open the district panel for a text summary*. Arrow keys pan and + and - zoom (the library's keyboard handler). Individual features on the canvas are not tab stops (hundreds of them would trap keyboard users). The equivalent content is available as text: the district panel (county table, per-compound counts, headline) for the region, HomeGuard for the household's own system, and an opened sheet through `?system=`. Severity is always a word or a shape beside the color (A10). Sheets follow the standard focus rules (A8): focus moves into the sheet on open and returns to the map on close.

Proposed content **PROPOSED**

Addition	Status
"Systems not shown on the map": a list with names and severities	In the district panel table above (the count is real; the list comes from water features with null coordinates)
In the household's own water-system sheet, a line comparing it to the district: "Your system is one of [n] in NC-08 over a federal limit"	Deferred, not in the first build: it needs to know whether the household's system is inside NC-08, which the current API does not say

Decisions to make

Technical: D7 above (Mapbox GL JS by default; MapLibre with free tiles only if the Mapbox account is a problem, because it changes the token, attribution and style files); the county shapes come from the existing `ncCounties.json` (first-load budget under 500 KB, §35.1: the map library and the 160 KB shapes load only when the Map tab opens). Performance targets: first render under 2s, layer switch under 300ms (§35.1). **DESIGN** : a muted, low-saturation base map where the data layer is the only saturated thing on screen (§57.1's principle), switching to a dark variant with the same muted treatment in dark mode (§59.3); floating control styling; chip row; legend; home pin; cluster badges; the overlap budget; district panel. The Map wireframe shows every layer; the Water layer with each contaminant option; the slider at two positions with the loading label; each feature sheet; the district panel full, with a null population, and with the error state; the layer error callout; the no-WebGL full-page fallback; offline with saved data.

Part C: Overlays

All three are bottom sheets over the current page: they add one history entry, trap focus, lock the page's scroll, dismiss by drag / backdrop / Escape / close control, and only one is ever open (A1). The opening functions, history rules and the sheet host are Phase 0 (Part P3): `openLearn` , `openAlerts` , `openAssistant` in `lib/frontend/sheets.ts` . Each overlay has one owner who builds the sheet's content; the entry points in other pages call the function and never build sheet content themselves.

C1. Learn

Owner: Naggi (Lane 2, with HomeGuard). Other pages (Today, HomeGuard, Map) call `openLearn` ; until Learn merges, the sheet host shows a labeled placeholder in development only (Part P5).

Purpose and the rule that governs it LOCKED

Explain what a number means, at the moment someone is looking at it. Never reachable from the tab bar, no index, no browsable encyclopedia: content that can only be reached from a household's own reading, and references that reading in its second sentence, can't be generic by construction (§17.1).

The opening contract BIND

```
openLearn({
  topic: 'air' | 'uv' | 'pollen' | 'mold' | 'pfas' | 'radon' | 'lead',
  reading?: { severity: Severity; valueText: string }, // the pill under the title
  context?: Record<string, string | number>           // sent to the server as query parameters
}): void
```

Standard height. One history entry: `router.push` with `?learn=<topic>` (Part P3 gives the close rules, including deep links). A second sheet opened from inside a Learn sheet (none exist) would replace it. `valueText` per topic is in Part P3 (for example *AQI 37, 8.3 ppt · 2.1×, Zone 3, Built 1975*).

Opened from, and the context to send BIND

The server builds the "why yours" sentence only from these exact parameter names. Send every one that the page has; a missing one is left out (never sent empty or as "undefined"). Always add `locale=en|es` (the profile's language) and `severity` as the canonical word when the reading has one.

Where	Topic	Context params BIND
Today, Air card	air	<code>source</code> (<code>airnow</code> or <code>open-meteo</code> , from <code>air.source</code>), <code>value</code> (<code>air.aqi</code>), <code>severity</code> , <code>pollutant</code> (the dominant pollutant as Today shows it)
Today, UV card	uv	<code>value</code> , <code>severity</code> , <code>peak_start</code> and <code>peak_end</code> (the same clock strings Today shows for the burn window; omit both when Today shows no window)
Today, Pollen card	pollen	<code>value</code> and <code>category</code> of the dominant pollen type (for example <code>Grass</code>), <code>severity</code>
Today, Mold card	mold	<code>risk</code> , <code>humidity</code> (percent), <code>precip</code> (percent chance of rain), <code>severity</code> . There is no numeric <code>value</code> . Omit humidity or rain when not shown
HomeGuard Water card or threshold row; Map water sheet	pfas	<code>contaminant</code> (send the label the person sees, from <code>contaminantLabel</code>), <code>value</code> (ppt), <code>limit</code> (ppt), <code>severity</code> . The tapped row, or the worst row when opened from the card
HomeGuard Radon card; Map county sheet	radon	<code>county</code> , <code>zone</code>
HomeGuard Lead card	lead	<code>home_year</code> . Omit it when unknown: the server then writes its "we don't know when this home was built" text

Call: `GET /api/learn?topic=<topic>&locale=<locale>&...context` . Deadline 8 seconds (Part P3). Topics are exactly those seven; anything else is a 400.

CACHING: cache only the parts that don't depend on the reading or the household: `title` , `what_it_is` , `protect` , `sources` , `locale` , keyed by topic + locale, so the sheet can open offline (§36). Always request fresh `why_yours` and `household_note` while online (they depend on the reading passed and on the household's groups). Offline, show the cached parts and omit sections 2 and 3 without a notice. Clear this cache with `clearReadings()` and on `halo:settings-changed` .

Anatomy, top to bottom LOCKED BIND

Element	Content	Binding
Title	Topic name. While loading, the static topic name from <code>copy.learn</code> ; then the server's title	<code>title</code>

Your reading	The household's own reading as a severity pill with its value, right under the title: what makes this contextual rather than generic. Omitted when the sheet was opened by a deep link the page couldn't rebuild a reading for (P3)	Passed in from the page that opened it, not from this endpoint
1. What it is	Two sentences, plain language	what_it_is
2. Why yours reads this way	References the household's actual value and situation. Section omitted (heading too) when why_yours is null . The server returns null when the context wasn't passed, except for Lead (always a sentence, using the "we don't know when this home was built" wording when home_year is missing) and Air (a sentence from source alone). Never filler	why_yours
3. What it means for your household	Branches on household composition	household_note : section omitted when null
4. How to protect yourself	A short checklist, most effective first; every item a concrete, checkable step; certifications named inline so they can be read standing in a store aisle: set text that matches /NSF\ ANSI \d+ P473/ in the mono font (IBM Plex Mono, loaded in Phase 0). The section that makes Learn worth opening twice	protect[]
5. Sources	Every claim's origin, each with the date retrieved, each a working link, visually distinct. Format: <i>EPA, PFAS drinking water regulation</i> then <i>Retrieved September 27, 2026</i> on the next line. Links open in a new tab (rel="noopener noreferrer") and have at least a 44 px target	sources[] → label , url , retrieved (an ISO date)

Every string from the response passes through `displayText()` : the content and the source labels still contain em dashes (Part P9 asks Vibhav to remove them at the source), and `displayText` turns them into commas. Render as plain text (list items as list items), never as HTML.

LANGUAGE: the response's `locale` is the language actually served. If Spanish was requested but a topic hasn't been translated and reviewed yet, it comes back in English with `locale: "en"` : show it, don't pretend otherwise. (Until Spanish is reviewed the Language setting is hidden and only `en` is ever requested, B6.)

THE LEARN / ACT BOUNDARY: Learn is personal protection (sunscreen, filters, flushing taps, closing windows, humidity). Act is collective action (volunteering, advocacy, sharing). Keeping that line clean keeps both coherent (§17.2).

States LOCKED

- **Loading:** skeleton (title bar, a pill shape, five paragraph blocks).
- **Error** (400, 404, 500, timeout, network): generic error *Couldn't load this explanation.* with *Try again*. Never the server's message (the 400 lists topic names). Reading pill and title still show.
- **Offline with a cached topic:** shown per the caching rule. Offline with no cache: the error state without Try again.

DESIGN : typographic treatment: strong headings, key facts given weight, sources as visible footnotes rather than hidden links (§6.7). The wireframe shows each of the seven topics with a reading, PFAS with a household note, a topic with `why_yours` null (section 2 absent), a deep link with no reading pill, loading and error.

C2. Assistant

Owner: Vibhav (Lane 3, after Map; the last build and first cut, \$26.3, \$68). The shell renders the entry button and hides it when `NEXT_PUBLIC_HALO_ASSISTANT=off` (Part P2), so it can be switched off without leaving a dead button.

Purpose LOCKED

Answer questions about the household's own readings and about environmental health generally, grounded entirely in cited sources, without ever practicing medicine (§18.1). It draws on curated source documents and, only when the server's household-context switch is on, the household's current readings and symptom entries; never the web, never general knowledge (§18.2).

HOW IT KNOWS THE HOUSEHOLD: the server assembles any household context itself, from the verified session, and weighs it by the page the question came from. The frontend sends only `{ question, page }`; it never sends household data in the request. **Whether household data is used at all is a server switch** (`ASSISTANT_SEND_HOUSEHOLD_CONTEXT`), decided by Yogi (D4). The team default is **off**: free models may log prompts, so only the question and public agency sources are sent, and responses say `uses_household_data: false`. When on, the context is today's readings, the home assessment, and journal symptom names and dates, **never the free-text notes**. The Settings privacy statement changes with the switch (B6). When an answer uses household context, the response says so (`uses_household_data: true`).

Entry LOCKED

The floating button on all five tabbed pages. Opens as the tall sheet (about 90%; the blueprint says two-thirds, a Tier 3 difference, tall wins). Carries the current page as context. The conversation isn't kept in the URL.

Calls BIND

Call	Sends	Returns
<code>GET /api/assistant? page=...</code> (on open, once per app session per page)	<code>page</code> : Today → <code>today</code> , HomeGuard → <code>homeguard</code> , Journal → <code>journal</code> , Map → <code>map</code> , Act → <code>act</code>	<code>suggestions</code> (exactly 3), <code>disclaimer</code> . Pages with their own set: today, homeguard, journal, map. Any other page (Act) gets the default three (<i>What is PFAS?</i> , <i>How is my score calculated?</i> , <i>What does NSF/ANSI 53 mean?</i>)
<code>POST /api/assistant</code>	<code>{ question, page }</code> (question trimmed, 1 to 1,000 characters; <code>page</code> as for GET; the server accepts today, home, homeguard, map, journal, act, settings, learn and ignores anything else)	One of the branches below
<code>GET /api/assistant field sends_household_context</code>	Not built yet (Part P9 ask 2)	Read by Settings only. The assistant sheet does not depend on it

The chips and disclaimer are server text: pass them through `displayText`. Because chips are server-supplied, every chip's answer must be tested against the source corpus before a demo (each should return a grounded answer, not a decline). Known problem: the Map chip *What does 'not tested' mean on the map?* disagrees with the map's own wording, *No detection reported* (Part P9 ask 7).

Anatomy LOCKED

Element	Content / behavior
Title row	Ask HALO + close control
Suggested questions	On first open with an empty conversation only: three tappable chips chosen by the page it came from (§18.6). Tap sends it. Hidden after the first question
Conversation	The person's messages on one side, answers on the other. Answers never truncate. Rendered as plain text , never as HTML or Markdown; blank lines make paragraphs. In memory for the app session (survives closing the sheet, not a reload), never sent to the server as history (each question stands alone), cleared on Delete everything and on an identity change
Thinking state	An in-progress indicator in the answer position until the response arrives. Answers arrive complete, not streamed; the indicator carries the wait
Markers in the answer text	The model writes <code>[1]</code> style markers and <code>[H]</code> (household readings). Strip every <code>[H]</code> . Show each <code>[n]</code> as a small superscript number matching the citation with that <code>n</code> (use <code>n</code> as sent; don't renumber); a marker with no matching citation is removed
Citations	Under each answer: each cited source as a tappable name that opens it externally, with its retrieval date. The server's rule: a grounded answer cites at least one source or the household's own readings (<code>[H]</code>). So <code>citations</code> can be an empty array on an answer that used household data; then show no sources list, only the marker below. An answer with neither is never sent
"Uses your readings"	A small icon and the label <i>Uses your readings</i> above answers that used household data (§50.2):

marker	<code>uses_household_data === true</code> . Never shown when the switch is off
Declined answer	Shown in a caution callout, visually distinct from a normal answer
Disclaimer	Fixed above the input, always: <i>Not medical advice. Talk to a clinician about health decisions.</i> A small tertiary link under it, PROPOSED <i>How your data is used</i> , opens <code>/settings?open=privacy</code>
Input	Expanding text field + send button. The blueprint conflicts: §50.2 says send is disabled while empty; §34.1 says submit is never disabled for validation. Follow §34.1: send is always enabled; an empty or whitespace-only question (trimmed client-side) shows <i>Ask a question to get started.</i> ; over 1,000 characters shows <i>That question is too long.</i> (the server's own two messages, so the client never causes a 400). Send is disabled only while a question is in flight (one at a time, no queue)

Response branches, how to tell them apart **BIND**

Check in this order:

Condition	Meaning	Show
HTTP 429	Rate limited: 20 questions per hour per household, or the app-wide daily cap of 450 (default; the cap protects the free model's quota and fails closed, meaning it also pauses the assistant if the rate-limit service is down). The body has <code>rateLimited: true</code> and a <code>message</code>	<code>message</code> : <i>You've reached the question limit for now. Please try again in a little while.</i> Same text for both causes. No Retry button
<code>declined === true</code>	Diagnostic refusal: a check that runs before any answer is composed. Has <code>message</code> and <code>citations: []</code>	<code>message</code> (through <code>displayText</code> , since it contains an em dash) in the caution callout
<code>configured === false</code>	The assistant isn't configured on this server (no model key, e.g. a development environment). The server's <code>message</code> reads as "no source for that", which would mislead	PROPOSED <i>The assistant isn't available right now.</i> Not the server's <code>message</code> , no Retry
<code>typeof answer === "string"</code>	Grounded answer (<code>grounded: true</code> , <code>reason: null</code>)	<code>answer</code> text (markers handled as above) + <code>citations[]</code> → { <code>n</code> , <code>label</code> , <code>url</code> , <code>retrieved</code> } + <code>uses_household_data</code> . The <code>[n]</code> markers match each citation's <code>n</code> ; they aren't guaranteed to be in order or consecutive
<code>answer === null</code> and <code>reason: "no_source"</code>	No source covers the question	<code>message</code> : <i>I don't have a reliable source for that. I can only answer from the health and environmental agencies I've been given.</i> (§31.3). No Retry
<code>answer === null</code> and <code>reason: "unavailable"</code>	The model or the source lookup failed. The only case with Retry	<code>message</code> : <i>I couldn't reach my sources just now. Please try again in a moment.</i> + <i>Try again</i>
Timeout, network error, HTTP 5xx (including a platform 504), or a body that isn't JSON	Infrastructure failure	Treated like <code>reason: "unavailable"</code> : the same message with Try again. The server has no explicit limit on this route yet (Part P9 ask 3 sets one at 30 seconds)

Retry re-sends the same question; it does not add a second copy of the person's message. HTTP 400 should never happen because of the client checks above; if it does, show the generic unavailable message.

Required behavior **LOCKED**

- Household data and general guidance always distinguished in wording: "your reading today is 68" and "federal guidance suggests..." are separate statements (§18.4). With the household switch off, answers use public sources only, and that is the correct behavior, not a bug.
- Diagnosis questions declined before any answer exists, reliably: a demo audience may try (§18.4). The refusal is code, not an instruction. It also declines a question that mixes an environmental word with anything about a person's body or care ("Should I take medication since my radon is high?").
- No source: says so rather than improvising.
- Answers can take up to about 20 seconds: the thinking state holds that long, and the client timeout for `POST /api/assistant` is 25s, not the 8s default. A client timeout is treated as unavailable.
- Keyboard and focus.** Focus moves to the input on open. Enter sends; Shift+Enter makes a new line; Escape closes. The conversation list is `aria-live="polite"` so a new answer is announced once, and the list scrolls to the start of the new answer (not the end of a long one). Chips and citation links are 44 px targets.
- Offline:** the input and chips are disabled with **PROPOSED** *You're offline. Reconnect to ask a question.*

WHAT THE REFUSAL COVERS: questions asking for a diagnosis, whether a symptom means an illness, or what treatment or medication to take. Questions about how serious a *reading* is ("Is an AQI of 160 serious?") or how to fix an environmental problem ("What are the treatment options for radon?") are environmental questions and get answered.

BEFORE LAUNCH: federal guidance on wellness and decision-support software was revised in January 2026; how the assistant is described publicly should be checked against the current text (§18.5, §66). Test it against ten deliberately adversarial questions before any demo (§66, Vibhav's checklist item).

DESIGN : message bubbles, citation chips, the thinking indicator, the declined-answer callout, the input row. The wireframe shows: empty with chips; a grounded answer with citations; a household-data answer with the marker and no source list; declined; no source; unavailable with Try again; rate limited; not configured; offline.

C3. Alerts

Owner: Naggi (Lane 2, after HomeGuard and Learn; with Act). Alerts also exports `useAlerts()` and `<BellBadge/>` (Part P3), which the shell header renders; until Alerts merges the bell slot renders nothing.

Purpose LOCKED

Tell a household when something meaningfully changes, without them opening the app to find out (\$19.1). Opened only from the bell on Today's header.

Calls BIND

Call	Use
<code>GET /api/alerts</code>	<code>count</code> , <code>unread (badge)</code> , <code>alerts[]</code> newest first (max 100, dismissed ones excluded). Each alert: <code>id</code> , <code>type</code> , <code>title</code> , <code>message</code> , <code>severity (air only)</code> , <code>read</code> , <code>fired_at</code>
<code>POST /api/alerts { id }</code>	Mark one read (it's POST, not PATCH)
<code>POST /api/alerts { all: true }</code>	Mark all read
<code>POST /api/alerts { id, dismissed: true false }</code>	Dismiss, and undo. Success returns <code>{ ok: true, updated: n }</code> ; <code>updated: 0</code> on an id or dismiss call means the alert no longer exists, so refetch (Mark all read returns 0 when nothing was unread). A 501 means migration 0010 isn't applied: treat as a failure

The badge BIND DECIDED

`<BellBadge/>` shows the `unread` count as a small number on the bell (9+ above nine), hidden at zero, with the accessible name *Alerts, 3 unread* (or *Alerts* at zero). The count comes down by **reading rows** (or Mark all read), not by opening the sheet, so a person who opens it and leaves still sees that something is unread. `useAlerts()` fetches once when the shell mounts, again when the sheet closes, and when the app returns to the foreground after more than five minutes. A failed fetch keeps the last known count and shows no error on the bell.

The inbox LOCKED

Element	Content / behavior	Binding
Title row	<i>Alerts</i> + tertiary <i>Mark all read</i> , hidden when nothing is unread	<code>{ all: true }</code>
Row	Reverse chronological. Unread: small colored marker + bolder title. Title, one-line message (cut with an ellipsis), relative time (<i>Just now</i> , <i>2 hours ago</i> , <i>Yesterday</i> , then <i>September 26</i> ; one shared <code>relativeTime()</code> in <code>format.ts</code>)	<code>title</code> , <code>message</code> , <code>fired_at</code> , <code>read</code> ; marker color from <code>severity</code> when set (air alerts only), neutral otherwise
Tap	Expands to the full message (season summaries can be several sentences) and marks it read	<code>POST { id }</code> . Optimistic; on failure the marker returns and the toast <i>Couldn't update that alert. Try again.</i> shows
Go to the related page	The blueprint conflicts: \$19.4 says tapping an expanded row again navigates; \$50.3 puts a tertiary button inside the expanded row instead. Use the button (\$50.3): a second tap on the same row is easy to trigger by accident. Closing the sheet first (by replace), then pushing the route	By <code>type</code> (labels PROPOSED): <code>air_quality_change</code> → Today, <i>See today's air</i> · <code>weather_advisory</code> → Today, <i>See today</i> · <code>new_water_results</code> → /home?risk=water, <i>See water results</i> · <code>radon_season</code> → /home?risk=radon, <i>See radon</i> · <code>season_summary</code> → /journal?mode=trends, <i>See your summary</i> · unknown type → no button
Swipe to dismiss	Dismiss, immediate, with an undo toast (8 seconds, <i>Alert dismissed</i> + <i>Undo</i>). A visible tertiary <i>Dismiss</i> button in the expanded row does the same (swipe is never the only way: <code>SwipeRow</code> , Part P2). On failure the row returns with the toast <i>Couldn't dismiss that alert. Try again.</i>	<code>POST { id, dismissed: true }</code> ; <code>undo</code> → <code>POST { id, dismissed: false }</code> , then refetch
Empty	<i>No alerts yet. We'll let you know when something changes.</i>	<code>alerts</code> empty
Loading / error	Three skeleton rows; an inline error <i>Couldn't load your alerts.</i> with <i>Try again</i> (no toast)	Request state

History is kept rather than cleared, so the alert system's behavior can be seen without waiting for a live trigger, which is useful for a household back from a week away and for anyone evaluating the app (\$19.4). Dismissed alerts don't return.

What generates alerts **LOCKED** (verified against the server rules)

type	Fires when	Title and message as the server sends them	Opens
air_quality_change	Today's air is worse than yesterday's (or yesterday has no reading) and today is at least Elevated for a household with a sensitive group (a respiratory condition, a toddler, or a senior), or at least High for everyone else (§19.3). At most one per day	<i>Air quality dropped / Air quality dropped to Elevated today. (the label is one of the six severity words; the blueprint's "Unhealthy for Sensitive Groups" is not what is sent)</i>	Today
weather_advisory	The National Weather Service has an alert with "flood" or "heat" in its name for the home's point (watches and warnings too; the message still calls it an advisory). Boil-water advisories are never sent: no national source exists (the rule text for them is in the code but nothing feeds it). One alert per advisory, not one per day	<i>Weather advisory / Flood advisory for Union County. Flooding can affect well water, retest after it recedes. or Heat advisory for Union County. Limit strenuous outdoor activity, especially for older adults and young children.</i>	Today
new_water_results	New federal testing results were published for the household's utility	<i>New water results / New water testing results were published for your utility.</i>	HomeGuard, water card
radon_season	October through February, household in radon Zone 1 or 2, and no radon test on record (the server has no record of radon tests yet, so it always counts as none). Once per winter. (Never sent in Zone 3, which is every NC-08 county)	<i>Time to test for radon / Winter is the most accurate time to test for radon.</i>	HomeGuard, radon card
season_summary	A season ends and the household has journal entries. The <code>message</code> is the full summary statement (a sentence or more), not a short teaser	<i>Your spring summary is ready / the summary statement</i>	Journal, Trends mode (<code>/journal?mode=trends</code>). Journal shows the seasonal-summary callout in both Log and Trends (B4, D6)

Types the household switched off in Settings are never generated, for both this inbox and device notifications. Show any unrecognized type with its title and message and no button. Every title and message passes through `displayText`, which turns the server's remaining em dash (in the flood text) into a comma. Words like "Elevated" in a message are the same six severity words used everywhere.

Device notifications **LOCKED**

Built after the core pages (D8, and specified in B6, "Device notifications"). Use the browser's notification capability where permission was granted; otherwise alerts simply appear in the inbox on next open. Ask permission only after the household has used the app at least once, never on first load (§19.5): in practice from the device-notifications switch in Settings. A notification tap opens the related page from the `url` in its payload (same destinations as the table above). Needs a service worker (which can also serve the offline behavior, §36).

DESIGN: row anatomy, unread marker, the bell badge, the expanded row with both buttons. The wireframe shows: empty; several rows mixed read and unread; one expanded per type (five); a long season summary expanded; the dismissed toast; the failure toast; loading; error.

Part D: API Reference (the finished backend)

Every endpoint and field this document binds to, in one place. This is the contract between the frontend and the finished backend: the backend was built to exactly these names. Field names are verbatim, bind to them as written.

D0. Conventions that apply to every endpoint

Rule	Detail
Identity	Every route except the daily job requires <code>Authorization: Bearer <session token></code> . No route accepts identity from a body or query parameter.
401 vs 503	<code>401</code> = the session is definitively invalid (the account no longer exists) → the client discards the session and starts a new one, silently. <code>503</code> = identity couldn't be verified right now → the client retries and never resets the session. This distinction is what protects a household's data during an auth-service outage.
Errors	<code>{ "error": string }</code> . Never shown to people verbatim; mapped to the copy in A2. This is a hard rule, not a preference: many routes still return raw database text in this field on a 500 (Part P9 ask 6), so the frontend never displays a server error string. The only server text ever shown is the Assistant's fixed <code>message</code> sentences (C2), and those pass through <code>displayText()</code> too.
Absent fields	Any field can legitimately be <code>null</code> or missing, and the interface handles that explicitly, the words No data, an explicit state, a hidden element, never a crash and never a zero (§62.4). This is also what lets a page be built before every field exists.
Severity	A closed set: <code>good</code> , <code>moderate</code> , <code>elevated</code> , <code>high</code> , <code>severe</code> , <code>no_data</code> . Anything else renders as No data. Always read the field named <code>severity</code> (or <code>level</code> for lead). Some objects also carry an older <code>status</code> field kept for compatibility; never bind it. Any server string shown to people passes through <code>displayText()</code> first (it still contains em dashes in places).
Dates	Daily readings and journal entries: <code>YYYY-MM-DD</code> in the household's local date. Timestamps: ISO 8601. Water sample dates: bind the <code>*_iso</code> fields (<code>YYYY-MM-DD</code>); the older <code>M/D/YYYY</code> strings remain for compatibility only.
Freshness	Every cached or pre-built response states when its data was retrieved or assembled (<code>retrieved_at</code> , <code>assembled_at</code>).
Change discipline	Additive only once frontend work begins: fields may be added, never renamed, removed, or re-typed (§62.4).

D1. Household and account

Endpoint	Request	Response
<code>GET /api/profile</code>	None	<code>{ onboarded, onboarding_complete, profile: { county, state, zip, pwsid, lat, lng, home_year, water_source, renter_mode, locale, onboard_request_id } null, household: { has_toddler, has_child, has_teen, has_adult, has_senior, has_pregnant, has_respiratory }, household_set }</code> . <code>onboarding_complete</code> is true once coordinates <i>and</i> water source are stored, route on this, not <code>onboarded</code> . <code>profile</code> is null when no row exists yet. <code>zip</code> is null once a household has onboarded again (older rows may still hold one; never bind it). <code>onboard_request_id</code> is the id of the address request that stored this location (only onboarding uses it, to confirm a timed-out save; never displayed). <code>lat</code> / <code>lng</code> are rounded to about 100 m for locations saved after migration 0009. A database failure returns 500 with a fixed message, never an empty profile.
<code>PATCH /api/profile</code>	<code>{ water_source?, home_year? }</code>	Same shape as GET. Only the keys sent change, and at least one is required (400 otherwise). <code>home_year</code> : integer 1700-current year or <code>null</code> (null clears it); an invalid year is a 400 whose message the app never shows (validate first, A7). <code>water_source</code> : <code>utility</code> · <code>well</code> · <code>spring</code> · <code>other</code> (the onboarding labels are also accepted and normalized)
<code>POST /api/onboard</code>	<code>{ address, request_id?, water_source?, home_year? }</code> . <code>address</code> may be "lng,lat" from device location. <code>request_id</code> is a UUID the client makes for each address submit and reuses on a retry of the same submit (if a POST times	<code>{ profile_id, lat, lng, county, state, pwsid, service_area_status, water_source, home_year, onboard_request_id }</code> . <code>state</code> is the two-letter code, e.g. "NC". <code>lat</code> / <code>lng</code> are rounded to about 100 m before storage. The response also echoes <code>zip</code> , which is no longer stored, don't bind it. <code>service_area_status</code> : <code>measured</code> (a utility boundary contains the point) · <code>outside_known_area</code> (none does) · <code>lookup_failed</code> (the boundary lookup failed; a previously stored utility is kept). When the (rounded) location changes, the server also

	out, the client re-checks <code>GET /api/profile</code> for a matching <code>onboard_request_id</code> , B1). Water source and build year are normally saved with <code>PATCH /api/profile</code> ; onboard also accepts them	clears today's stored reading and resets the daily-score limiter, so the results reveal's fresh reading works for the new home. Errors: 400 no address (or an invalid <code>home_year</code>) · 404 not found · 429 (a per-device daily limit, or the app's daily geocoding budget; both say to try tomorrow) · 500. Onboarding maps these to its own copy (B1)
<code>GET /api/household</code>	None	<code>{ household: { 7 booleans }, household_set, renter_mode, locale }</code>
<code>PUT /api/household</code>	<code>{ has_toddler ... has_respiratory, renter_mode?, locale? }</code>	Full replace of the seven booleans, always send all seven . Echoes <code>renter_mode / locale</code> only when sent
<code>DELETE /api/account</code>	<code>{ confirm: "DELETE" }</code>	<code>{ deleted: true }</code> ; 400 if the word is missing or not exactly <code>DELETE</code> . Deletes the account and everything tied to it. The client then unsubscribes push, clears its own storage and signs out (B6)

D2. Today

`GET /api/daily-score`, no parameters: uses the household's stored location. `?fresh=1` bypasses the one-hour cache (pull to refresh). Any request that has to call the providers (a fresh request, or no reading for today from the last hour) is limited to one per household per five minutes, and gets a 429 otherwise, even without `fresh`. Response:

```
{
  air: { aqi, severity, sentence, source: "airnow"|"open-meteo"|null, is_measured: true|false|null,
         dominant_pollutant }, // "PM2.5" | "PM10" | "O3" | null
  uv: { index, severity, sentence, peak_window: { start, end, max } | null }, // local "HH:MM"
  pollen: { tree, grass, weed, dominant: "tree"|"grass"|"weed"|null, severity, sentence,
            categories: { tree: { value, severity }, grass: { value, severity }, weed: { value, severity } } },
  mold: { risk: "low"|"moderate"|"high"|null, is_proxy: true, severity, sentence,
          basis: { humidity_pct, precip_pct } | null }, // no_data + null risk when the forecast failed
  score: { display_score, score, risk, severity, is_partial, missing_inputs[], included_inputs[],
           includes_proxy, inputs: { air, uv, pollen, mold }, reason },
  cached, retrieved_at
}
```

Errors specific to this route: 400 { error: "No location yet. Finish onboarding, or pass lat and lng." } when the household has no stored location (route to onboarding) · 429 { error: "Readings were just refreshed. Try again in a few minutes.", `retry_after_seconds: 300` } when a provider call is repeated within 5 minutes (keep the readings on screen; A2). `retry_after_seconds` is always 300, not the time remaining. Onboarding a changed location clears today's stored reading and resets this limit. `lat / lng` are still accepted as optional overrides; the frontend doesn't send them.

`GET /api/history?days=N` (or `from / to`; `days` is clamped to 1 to 365, default 90; 0 or a non-number gives 90) → `{ from, to, count, history: [ { date, score, air, uv, pollen, mold, values: { aqi, uv_index, pollen: { tree, grass, weed }, mold_risk } } ] }`. One record per local date. Recorded every day for every household by the daily job (not only on days the app is opened), with recent weeks backfilled at onboarding where a historical source exists; a day with no data is simply absent.

D3. HomeGuard

`GET /api/home-guard`, no parameters: uses the stored county, utility, water source, and build year. Optional `county`, `pwsid`, `water_source`, `home_year` override them for this request only. Computed on every request: no server cache, no `fresh` parameter. The response also carries `success: true` and `assembled_at` (when it was put together; not a measurement time). A `state` override parameter works like the others (radon is looked up only when the state is `NC`). Error: 400 { error: "No county on file..." } when no county is stored (A2). Water results exist for North Carolina systems only, an out-of-state utility comes back `no_data_yet` (B3 handles the wording).

Field	Shape
radon	<code>{ county, zone: 1 2 3, risk_level, severity }</code> , or <code>{ error }</code> when the county isn't matched, or <code>{ error, out_of_state: true, state }</code> for addresses outside North Carolina (radon zones are looked up only when the stored <code>state</code> is <code>NC</code>)
water	Discriminate on <code>water.status: private_well · no_data_yet · lookup_failed · no_pwsid_available · detected_unregulated · anything else = measured</code> . Measured fields: <code>pws_name, population_served, contaminants (every reading: { name: [{ date, date_iso, value_ppt }] })</code> , <code>detected_unregulated_latest[] ({ contaminant, value_ppt, date_iso })</code> , <code>coverage, latest_sample_iso, earliest_sample_iso, data_age_days, reading_count, scored_count, detected_count, detected_unregulated[], excluded_from_score[], scored_contaminants[], includes_guidance, guidance_scored[], regulatory_notice (only when it applies), message (only when relevant)</code> . Private well: <code>source_type, message, test_plan</code> . Never display these fields raw: <code>message</code> and <code>regulatory_notice</code> are

	shown only where B3 says so; internal keys (<code>risk</code> , <code>ceiling</code> , <code>basis</code> codes) never appear in the interface
<code>scored_contaminants[]</code>	<code>{ contaminant, value_ppt, limit_ppt, mcl_ppt, basis, source, is_enforceable, proposed_for_rescission, exceeds_limit, date_iso, risk }</code>
<code>excluded_from_score[]</code>	<code>{ contaminant, value_ppt, date_iso, reason, health_reference_level_ppt, health_reference_basis, health_reference_source, needs_local_context }</code>
<code>test_plan</code>	<code>{ tests: [{ rank, id, name, priority: critical recommended optional, cadence, why, where, estimated_cost: { low, high, is_estimate }, needs_local_context? }], reasons[], start_here_cost: { low, high, is_estimate }, full_panel_cost: { low, high, is_estimate } }</code>
<code>score</code>	<code>{ display_score, score, risk, severity, is_partial, floored, capped, ceiling, missing_inputs[], included_inputs[], water_risk, radon_risk, reason }</code> (<code>capped</code> , <code>ceiling</code> and <code>floored</code> are internal: never display them or the word "capped" in <code>reason</code> ; Part P9 ask 8). <code>display_score</code> and <code>severity</code> are both <code>null</code> whenever there's no score, a private well or spring, or nothing available (with <code>reason</code>). Same rule for Today's <code>score</code>
<code>lead</code>	<code>{ level, basis, is_estimate, shown_not_scored: true, contributes_to_score: false, text, sentence, utility_inventory }</code> ; <code>null</code> for well/spring. <code>level</code> : <code>high</code> (built before 1950) · <code>elevated</code> (1950-1987) · <code>good</code> (1988 or later) · <code>no_data</code> (year unknown); <code>severe</code> is knowingly unreachable (no address-level inventory exists). <code>utility_inventory</code> : <code>{ utility, reported, lead, galvanized_requiring_replacement, non_lead, unknown, needs_replacement, address_level: false, source: { name, url, retrieved } }</code> , the utility's published service-line counts (NC DEQ, July 2025), context only, or <code>null</code> when the utility isn't in the published results
<code>action_plan[]</code>	<code>{ rank, key: water lead radon, severity, title, reason, cost, certification, renter_can_self_serve, action }</code> , only items at Elevated or above; text already written for renter or owner
<code>breakdown[]</code>	In order: water, radon, lead (public water only), one per excluded substance. Each: <code>{ key, label, severity, contributes_to_score, risk, score, confidence: full limited stale none, is_measured, is_estimate, detail, note }</code> (<code>risk</code> and <code>score</code> are numbers for the server's use; the app shows the severity word and the page's own sentences). Excluded-substance boxes add <code>health_reference_level_ug_l</code> , <code>detail[].value_ug_l</code>

D4. Journal

Endpoint	Request	Response
GET <code>/api/journal</code>	<code>from</code> , <code>to</code> (default last 90 days)	<code>{ from, to, count, entries: [{ id, entry_date, band, symptoms[], severity, note, retrospective, possibly_illness, created_at, updated_at }] }</code>
POST <code>/api/journal</code>	<code>{ entry_date, band, symptoms[], severity, note, retrospective, possibly_illness }</code> , every field, every time (a save rewrites the whole entry). <code>band</code> : one of the seven group keys or <code>household</code> . Validation: <code>entry_date</code> must be <code>YYYY-MM-DD</code> ; <code>severity</code> is <code>mild</code> , <code>moderate</code> , <code>bad</code> or <code>null</code> ; <code>symptoms</code> keeps at most 20 (each as text); <code>note</code> at most 500 characters. <code>entry_date</code> is checked for shape only: an impossible date like 2026-02-31 fails in the database as a 500, so the date picker must only offer real dates	<code>{ entry }</code> . One entry per date and group; saving again updates
DELETE <code>/api/journal</code>	<code>?id=<uuid></code> one entry · <code>?all=true</code> every entry (400 for anything else, including a non-UUID id)	<code>{ deleted: n }</code>
POST <code>/api/journal/retrospective</code>	<code>{ dates: ["YYYY-MM-DD", ...] }</code>	<code>{ ready, factor, statement, flagged_days_with_readings, flagged_avg?, month_avg?, unflagged_avg? }</code> . 400 when <code>dates</code> isn't an array or has more than 62 entries; future dates are silently dropped. Compares the flagged days' readings with the rest of that month, e.g. "On the 3 days you flagged, grass pollen averaged High. The month's average was Low." <code>ready: false</code> (with a <code>statement</code>) when too few of the dates have readings
GET <code>/api/journal/findings</code>	None	Not ready: <code>{ ready: false, flaggedDays, totalDays, needed: { flagged, total } }</code> . Ready: <code>{ ready: true, findings: [{ factor, statement, elevated_rate,</code>

		other_rate, counts: { elevated_days, other_days } }], disclaimer, flaggedDays, totalDays } (note the camelCase)
GET /api/journal/summary	season, year (default: the season that most recently ended)	{ season, year, logged_days, statement, factor?, from?, to? } . 400 for an unknown season or a year outside 2000 to 2100. E.g. "This spring you logged 22 days. On the 9 highest-pollen days, you flagged symptoms 7 times."

D5. Learn and Assistant

Endpoint	Request	Response
GET /api/learn	topic (pfas · radon · lead · air · pollen · uv · mold), locale, and context (every one optional): pfas → contaminant, value, limit, severity; radon → county, zone; lead → home_year; air → source, value, severity, pollutant; uv → value, severity, peak_start, peak_end; pollen → value, severity, category; mold → risk, humidity, precip, severity. Errors: 400 unknown topic · 404 no content · 500	{ topic, locale, title, what_it_is, why_yours, household_note, protect[], sources: [{ label, url, retrieved }] } . locale in the response is the language actually served. why_yours is null when no context was sent (except Lead and Air, which always have a sentence); household_note is null when the topic has no note for this household
GET /api/assistant	page: today · homeguard · journal · map · act. Anything else (or none) gets the default three questions	{ suggestions: [3 strings], disclaimer }
POST /api/assistant	{ question, page } (page is one of today, home, homeguard, map, journal, act, settings, learn; anything else is ignored)	Branches, see C2: declined; configured: false; grounded { answer, grounded: true, citations: [{ n, label, url, retrieved }], uses_household_data, disclaimer }; no answer { answer: null, reason: "no_source" "unavailable", message, disclaimer }; 429 { answer: null, configured: true, rateLimited: true, message } . citations can be empty when uses_household_data is true. 400 { error } for an empty question or one over 1,000 characters. Limits: 20 per household per hour, 450 per day app-wide

D6. Alerts and notifications

Endpoint	Request	Response
GET /api/alerts	None	{ count, unread, alerts: [{ id, type, severity, title, message, fired_at, read }] }, newest first, dismissed alerts excluded. type: air_quality_change · weather_advisory · new_water_results · radon_season · season_summary
POST /api/alerts	{ id } mark read · { all: true } mark all read · { id, dismissed: true } dismiss · { id, dismissed: false } undo	{ ok: true, updated } (updated: 0 on an id or dismiss call means no such alert; Mark all read returns 0 when nothing was unread). 400 for an invalid id or an unrecognized body; 501 when migration 0010 is missing (dismiss only)
GET /api/notifications · PUT	PUT: { preferences: { air_quality_change, weather_advisory, new_water_results, radon_season, season_summary } }	{ preferences: {...five booleans} }, all default true. The daily job skips types a household turned off. 400 for an unknown type or a value that isn't true or false
POST /api/push/subscribe · DELETE	POST: the browser's push subscription object (public key: NEXT_PUBLIC_VAPID_PUBLIC_KEY). DELETE: { endpoint }	{ ok: true } . 400 for an endpoint that isn't https or missing keys; 429 when changed too often. New alerts are also pushed to subscribed browsers with the payload { id, type, title, message, url } (url = the page to open)

D7. Act, Map, District, Sources

Endpoint	Request	Response
----------	---------	----------

GET /api/volunteer	causes (comma list: water, pfas, radon, air, wells, advocacy), county (defaults to the household's)	{ county, causes, count, orgs: [{ id, name, description, counties[], causes[], url, last_verified, link_checked, match_count }] }. last_verified is null on all 13 organizations today (D9)
GET /api/map?layer=water	quarter (optional, from quarters)	{ layer, count, located, quarters[], counts: { total, over_limit, hazard_index_over, by_contaminant: { X: { tested, over } } }, features: [{ pwsid, name, lat, lng, population, overall_severity, coverage, latest_sample_iso, detected_unregulated[], hazard_index: { value, severity, proposed_for_rescission } null, contaminants: { PFOA, PFOS, PFHxS, PFNA, "HFPFO-DA": { value_ppt, limit_ppt, ratio, severity, date_iso, proposed_for_rescission } { severity: "no_data", limit_ppt, proposed_for_rescission } } }], assembled_at }. Systems with only unscorable detections report overall_severity: "no_data" with coverage: "unscorable". counts describe exactly the features returned (statewide). quarters uses the form "2024Q3", oldest first, and is empty when the data has no time series. Also on the root: quarter (null unless requested), population_source, note, and prebuilt (not present on a ?quarter= response). Each feature also carries legacy status and latest_sample_date (never bind); hazard_index also has exceeds, components, basis. A ?quarter= request returns only the systems that reported in that quarter, and its assembled_at is the request time. Errors: 400 (bad layer or quarter), 503 (a layer source is down), 500
GET /api/map?layer=radon	None	{ layer, count, features: [{ county, zone, severity }], assembled_at } (county shapes are a static asset bundled with the frontend)
GET /api/map?layer=facilities	None	{ layer, count, features: [{ id, name, lat, lng, programs[], compliance_status, severity, open_violations, recent_violations, penalties, last_inspection }], assembled_at }. compliance_status: no_violation · violation · significant_violation · unknown. recent_violations = quarters in noncompliance out of the last 12 (3 years), not a count of violations; open_violations = programs currently reporting a violation; penalties = dollars over the last 5 years. Each feature also has county. The root has scope ("NC-08") and selection (a server sentence; don't display, B7). Only active major facilities and repeat violators inside NC-08
GET /api/map?layer=air	None	{ layer, count, features: [{ id, name, lat, lng, aqi, severity, dominant_pollutant, last_reading_at, agency }], assembled_at } (dominant_pollutant as in D2; the root also has source)
GET /api/district	None	{ scope: "NC-08", total_systems, systems_over_limit, affected_population, affected_population_complete, systems_tested_any, over_limit_population_unknown, by_contaminant: { X: { tested, over, limit, population_over, over_population_unknown } }, exceedance_lines: { X: string }, counties: [{ county, in_district: "whole" "partial", systems, systems_over_limit, worst_water_severity, radon_zone, radon_severity }] (always the nine NC-08 counties), state_comparison: { district_share_over, state_share_over } (fractions 0 to 1, either can be null), household_county: { county, water_rank, of, share_over } (an unranked county has water_rank and share_over null; share_over is a 0 to 1 fraction of people served), ranked_counties, nc_counties, ranking_rule, unlocated_systems (a number), boundaries: { district, counties, retrieved }, prebuilt, note, assembled_at }. affected_population is a number in practice, but if the geography lookup failed it is 0 alongside total_systems: 0 (the frontend treats that as no data, B7). Never display note or ranking_rule (B7). Exceedance lines count only systems tested for that compound. Most fields are NC-08-wide and pre-built daily (exceptions: state_comparison.state_share_over, ranked_counties, nc_counties and unlocated_systems are statewide); household_county is filled in per request from the caller's profile
GET /api/sources	None	{ sources: [{ name, provides, last_retrieved }] } (currently 10). last_retrieved is null when unknown (shown as "Retrieval date not recorded"). Feeds Settings, About and the health card's sources footer
GET /api/health	None	{ all_ok, checked_at, checks: [{ name, reachable, ok?, ms, status?, error? }] } (an unreachable check has error and no ok). For Vibhav's live checks only; the app never calls it

Map and district data are built once a day by the scheduled job and served from storage, so they load fast and can't break during a demo because a government service is slow (\$13.10, \$62.1).

Part E: Definition of Done and Build Order

E1. A page is done when LOCKED

Not when it renders (\$42). Every box, every page:

☐	All four states, loading, error, empty, populated, with different copy for error and empty
☐	Every value formatted per A5; every string from the content inventory or approved as PROPOSED , no invented copy
☐	Every severity shown as color + icon + word; missing data as the words "No data" or an explicit state, never zero or green
☐	Every estimate labeled as an estimate; every data point's source and date available to the person
☐	Bound to the exact fields in Part D; nothing computed that the server decides; no legacy <code>status</code> fields
☐	Every field on the page tested absent/null without breaking
☐	Back button correct for any sheet it opens; scroll position kept on return
☐	Keyboard navigable with visible focus; sheets trap and return focus; 44px targets; reduced motion respected
☐	Checked at 375px on a real phone, in light and dark mode
☐	No console errors; meets the performance targets (first content under 1.5s, cached readings under 200ms, tab change under 100ms, map first render under 2s)
☐	Every string goes through the translation layer; both languages present once reviewed Spanish exists
☐	Built only from Phase 0's shared components and tokens (Part P2); no page-local copy of a shared component; no hard-coded color, size or string
☐	Bound to the live API from the first commit; the <code>/<page>/preview</code> route shows every state with mock data; every server string passes through <code>displayText()</code> ; the U+2014 lint test passes
☐	Wireframe frozen and proven: the built page has zero visual difference from the approved wireframe screenshots, and the element list in the page's README is complete (A13)
☐	(Once, app-wide) Page titles, link-preview metadata, <code>theme-color</code> , favicon and home-screen icons set per A5's identity rules

E2. Build order

#	Build	Why here
0	Onboarding (done) , merged to main (c84cbb3). The reference implementation for everything below: adapter pattern, preview route, copy catalog, tokens, screenshots (B1)	It already set the button, input, chip and spacing styles and proved the design foundation
1	Phase 0: the shared foundation (Parts P2 and P3): shell, shared components, tokens moved to <code>:root</code> , sheets and toast host, data hooks and modules, copy files, fonts and icons, the U+2014 lint test. One owner, named by Yogi (D1). Merged to main before anything in row 2 starts	Three people building at once need one shared layer; otherwise each rebuilds it differently
2	Three lanes in parallel (Part P1). Lane 1 Yogi: Today, Journal, Settings, then the two Onboarding follow-ups. Lane 2 Naggi: HomeGuard + Learn, then Act + Alerts. Lane 3 Vibhav: Map + District, then Assistant. Every page follows the same four steps: wireframe (preview route, mock states), freeze (screenshots + README element list), build (live API, same component), prove (zero visual difference, live check, flow check)	Cross-page dependencies are covered by Phase 0's contracts (P5); anything not built yet is a labeled placeholder, never a stub that ships
3	Device notifications (B6, D8), Spanish (if reviewed), and polish	After the core pages

Cut order if time runs short, from the end: Assistant, Settings, Act, district panel (the blueprint's own order). Today is the first page built on a real phone with live data (the blueprint's own measurement, \$26.4): if it takes hours the plan holds; if days, cut scope now.

Two finished pages beat five half-finished ones under every judging criterion, and an unfinished page reads as broken rather than scoped (\$26.3).

Part F: Decision Log

Every decision that shapes the frontend, made during planning (September 2026). **LOCKED** = decided, build to it. **PROPOSED** = a recommendation awaiting Yogi's confirmation. A superseded decision is listed with what replaced it, so an old document or branch can't reintroduce it.

F1. Product and scope

#	Decision	Status	Why / notes
1	HALO is a web app with a mobile-shaped layout, running in a phone browser, not a native iOS or Android app	LOCKED	§3. The Capacitor config in the repo isn't part of this build
2	The blueprint's function, content, buttons, states, copy, and data rules are ground truth. Its visual and style spec is void and redesigned	LOCKED	Team decision. This document carries the function; A11 and A13 carry the new look
3	Vibhav's glassmorphic HomeGuard frontend (branch <code>vmag211/homeguard-frontend</code>) and its <code>docs/HALO-design-language.md</code> are discarded	LOCKED	With Vibhav's permission. Don't merge, copy, or follow them
4	Every page holds all the content it can; Onboarding is the deliberate exception and stays minimal	LOCKED	§10.1, every extra onboarding question costs completions
5	This specification targets the finished backend; Part D is the contract both sides build to	LOCKED	The absent-field rule (D0) covers anything not live yet
6	Build order: Onboarding (done) → Phase 0 foundation → three parallel lanes (Today, Journal, Settings / HomeGuard + Learn, Act + Alerts / Map + District, Assistant)	LOCKED	E2 and Part P. Cut order from the end: Assistant, Settings, Act, district panel
7	Frontend work is built with Codex (wireframes, visual design, code) by three page owners, after one shared Phase 0 foundation with a single owner named by Yogi. Every page follows wireframe, freeze, build, prove. Yogi supplies reference inspiration and approves each page	LOCKED	September 27 (Codex), September 29 (three owners). Process in A13; owners and lanes in Part P
8	Spanish ships only after a person reviews it; every string goes through a translation layer from day one	LOCKED	§68, machine-translated health guidance is not acceptable

F2. Design

#	Decision	Status	Why / notes
9	Six severity words, good, moderate, elevated, high, severe, <code>no_data</code> , exactly as the backend's <code>lib/severity.js</code> emits them	LOCKED	An earlier idea to drop "elevated" was superseded: the scale follows the existing code
10	Severity colors: Good green, Moderate orange, Elevated red, High a red-leaning warning purple, Severe a darker purple, No data black drawn as an outline only	LOCKED	Exact light and dark values in A11. The palette wasn't tuned for red-green color blindness, the team's choice, which is safe only because severity is always color + icon + word
11	Brand: dark blues, navy, dark teals, turquoise, with a sprinkle of light aqua; no saturated green in the brand	LOCKED	A11. Green always means Good
12	Typefaces, the "field journal" set: Fraunces (headings, hero scores), Instrument Sans (everything else, tabular digits for data), IBM Plex Mono (codes only)	LOCKED	Chosen over more generic candidates for personality without losing clarity
13	The app follows the phone's appearance setting (light or dark). There is no default mode and no in-app toggle	LOCKED	§59. Every color is a token defined with <code>light-dark()</code> . Whether to force dark only is open (D3, Part P10); the default is to keep following the phone
14	The palette swatch sheet and type specimen made during planning were previews of colors and fonts only, not page designs	LOCKED	Page design happens per page, in Codex
15	Onboarding tone: "something in between", light and calm, with a navy-to-teal brand band behind the logo on Welcome only	LOCKED	Details in B1

16	Base visual system, type scale, spacing, radius, buttons, inputs, chips, progress, pills, motion	LOCKED	Built and final in Onboarding (A13, Part P6). New pages reuse it and add nothing without a Phase 0 change
17	The blueprint's simplifications are the defaults: three button variants, two sheet heights, pill rather than colored card edge, two skeleton shapes, three motion durations	LOCKED	\$65.4 (A9 note)

F3. Data, privacy, and backend behavior the frontend depends on

#	Decision	Status	Why / notes
18	The server decides every severity, score, and household sentence; the frontend never applies a threshold	LOCKED	\$6.1. Bind <code>severity</code> , never the legacy <code>status</code>
19	401 = session definitively invalid → reset silently. 503 = couldn't verify now → retry, never reset	LOCKED	Protects a household's data during an auth outage. Merged to main (the session handling was rewritten in the Onboarding commit c84cbb3; Vibhav reviews it)
20	Route on <code>onboarding_complete</code> (not <code>onboarded</code>), backed by the local <code>halo.onboarded</code> flag	LOCKED	A1
21	The Assistant may send the relevant readings and journal symptom names and dates to its AI provider, never the free-text notes, but only when a server switch is on. It ships off by default	LOCKED	Approved September 27 for this build; off by default because free models may log prompts (D4, Yogi turns it on). The privacy statement changes with the switch (B6)
22	Stored coordinates are rounded to about 100 m; ZIP is no longer stored	LOCKED	Matches the privacy statement's "approximate coordinates"
23	Radon zones apply only to North Carolina addresses; others get <code>out_of_state</code>	LOCKED	County names repeat across states (Union County, SC)
24	The Severe lead level is knowingly unreachable, no address-level service-line inventory exists; the utility's published counts are shown as context	LOCKED	Recorded by the backend, September 27
25	Journal's first-visit comparison is computed server-side (<code>POST /api/journal/retrospective</code>)	LOCKED	The frontend never compares readings itself
26	Map layers and district figures are pre-built daily and served from storage	LOCKED	Fast, and a slow government service can't break a demo
27	Anonymous sign-in is enabled in the Supabase project	LOCKED	Confirmed by Yogi. Re-check if the project is ever recreated (\$66 item 1)
28	Settings toasts only on failure (the blueprint contradicts itself: \$49.2 vs \$54). The one exception is Clear journal	LOCKED	Quieter; one rule for all of Settings (B6)

F4. Decisions made while building (September 27 to 29)

#	Decision	Status	Why / notes
29	Onboarding is built and merged to main (c84cbb3). It is the reference implementation: every other page copies its adapter pattern, preview route, copy catalog, tokens and screenshot workflow	LOCKED	B1, Part P
30	No em dash (U+2014) in anything the app shows. A missing value reads "No data". Page titles read <code>HALO Page</code> . Server strings pass through <code>displayText()</code> ; a lint test enforces the rule	LOCKED	A5, A6, Part P2. The backend still emits em dashes in some text, so <code>displayText()</code> is required until Vibhav removes them (P9 ask 5)
31	Bind to the live API from the first commit. Mock data exists only for wireframe previews and tests. Never run a destructive call against the production database	LOCKED	The backend is merged into main; Part P4
32	Fonts are self-hosted with <code>next/font/local</code> (Fraunces, Instrument Sans; IBM Plex Mono is added in Phase 0). Body weight 500. The logo is <code>public/halo-logo-mark.png</code> , a white mask on Welcome's navy band	LOCKED	A11, A13, B1
33	Alerts badge: shows the unread count and drops as rows are read, not when the sheet opens	LOCKED	C3. Settled September 29 (was open)
34	The map time slider is built: the water data has 12 quarters, 2023Q1 to 2025Q4. Its final position is "Latest"	LOCKED	B7. Yogi confirmed the history exists and the backend notes list these quarters; the slider hides itself if <code>quarters</code> comes back

			short
35	The daily cache no longer serves an old address's reading after Change address: the server clears today's row and resets the limiter	LOCKED	Backend done for a changed location (closes a former open item with Vibhav). B6
36	Alerts never include boil-water advisories (no national source exists); weather alerts are flood and heat from the National Weather Service	LOCKED	C3, B6 switch copy

F5. Still open

- **Nine decisions with default rules (Part P10, D1 to D9).** Builders proceed on the default; Yogi changes any of them. D1 Phase 0 owner, D2 Change address household step, D3 dark-only or follow the phone, D4 Assistant household context, D5 content owner for missing copy, D6 season summary target, D7 map library, D8 device notifications timing, D9 volunteer organization verification.
- **Nine backend asks for Vibhav (Part P9).** Each has a stated frontend fallback, so none blocks a page.
- Every **PROPOSED** item in Parts A to C: the page's owner keeps or cuts each as the page is built, and Yogi approves.
- The privacy statement's incomplete "stores" line: a proposed addition awaits approval (B6).
- With Vibhav: server timeouts per source (the blueprint's §62.2 targets aren't enforced) and reloading the water data with non-detects, which would let the map say "no detection" and "tested" precisely (B7).
- Which district map the app uses (the 119th Congress lines, 2023) versus the newer 120th Congress map (B7).
- Whether Spanish ships: depends on a reviewer (§68).
- Whether the Assistant ships: it is last in the build order and its household-data mode is off (§68, D4).
- Next.js 16.2.10 has security advisories (fix in 16.3.6, outside the pinned range). Upgrade separately, never inside a page branch (Part P4).
- Which evaluation criteria the NC-08 office uses this year: one email to ask (§66 item 12).